



## OM-C HeliService USA SOP (Standard Operating Procedures)

Version 3 Revision 7  
Effective Date 01.03.2026

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| Responsibility | Name            |
|----------------|-----------------|
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## Record of Revision (RoR)

| Version | Date                | Entered by      | Highlight                                                                                                        |
|---------|---------------------|-----------------|------------------------------------------------------------------------------------------------------------------|
| V3.7    | 01.03.2026<br>09:49 | Zachary Leggett | Sorry forgot a bit                                                                                               |
| V3.6    | 01.03.2026<br>08:43 | Zachary Leggett |                                                                                                                  |
| V3.5    | 03.02.2026<br>12:55 | Zachary Leggett | Sea State Defined<br>Addition of Virginia SOPs Addition of Martha's Vineyard SOPs Sabine<br>Helipad Restrictions |
| V3.4    | 07.01.2026<br>14:55 | Zachary Leggett |                                                                                                                  |
| V3.3    | 04.01.2026<br>17:07 | Zachary Leggett |                                                                                                                  |
| V3.2    | 17.12.2025<br>15:19 | Zachary Leggett | Ops Memo 25-11<br>Addition of Premier SOPs                                                                       |
| V3.1    | 03.12.2025<br>16:23 | Zachary Leggett | Addition of 3.5.2.6                                                                                              |
| V3.0    | 02.12.2025<br>15:25 | Zachary Leggett | Addition of Premier Health                                                                                       |
| V2.1    | 14.11.2025<br>10:52 | Zachary Leggett | HHO Duty Time and Rest<br>AW 169 Single Pilot ops                                                                |
| V2.0    | 08.10.2025<br>09:17 | Zachary Leggett |                                                                                                                  |
| V1.7    | 05.08.2025<br>16:05 | Zachary Leggett |                                                                                                                  |
| V1.6    | 22.07.2025<br>16:27 | Zachary Leggett |                                                                                                                  |
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| V1.4    | 21.07.2025<br>13:16 | Zachary Leggett |                                                                                                                  |
| V1.3    | 08.06.2025<br>16:55 | Zachary Leggett |                                                                                                                  |
| V1.2    | 30.04.2025<br>09:18 | Zachary Leggett |                                                                                                                  |
| V1.1    | 29.04.2025<br>12:33 | Zachary Leggett |                                                                                                                  |



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| V1.0 | 29.04.2025<br>07:23 | Zachary<br>Leggett |  |
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## List of effective Articles

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| NEW | 6.1.4 Charybdis          |             | 01.03.2026  |
| NEW | 6.1.8 MV HOS Riverbend   |             | 01.03.2026  |



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## 0 Purpose

The purpose of the Standard Operating Procedures (SOP) manual is to provide clear, consistent guidance for conducting specific operations safely and efficiently. The SOP manual outlines standardized procedures for flight, maintenance, emergency actions, customer requirements, and crew coordination, ensuring all personnel operate in compliance with regulatory requirements and best practices. It promotes operational consistency, enhances safety, reduces risk, and supports training and decision-making by establishing a shared understanding of how tasks should be performed under various conditions.



## 1 Operational SOPs

### 1.1 HeliService USA Operations

#### 1.1.1 HeliService USA – Pilot Scheduling Procedure

##### 1. Schedule Development

The **Chief Pilot**, **Lead Pilot**, or their **designee** is responsible for creating the pilot flight schedule. The scheduling process shall adhere to the following criteria:

- **FAA Regulations:** Including, but not limited to, 14 CFR Part 135.267 (Flight Time Limitations and Rest Requirements) and applicable duty/rest guidelines from OM-A and OM-D.
- **Customer Requirements:** Such as mission timelines, airframe-specific qualifications, and special mission readiness.
- **Fatigue Risk Management:** Fatigue mitigation strategies will be employed per SMS guidance (ref: SMS N 8900.700) to ensure crew fitness for duty and prevent cumulative fatigue.

##### 2. Schedule Approval

Once the draft schedule is complete, it will be reviewed and approved by the **Director of Operations** and/or **Chief Pilot**. Only upon their documented approval will the schedule be finalized and distributed to flight crew.

##### 3. Pilot Sign-In and Flight Readiness Confirmation

Upon signing into the **Flight and Duty Time Tracking Software**, each pilot shall:

- **Review assigned flight duties**
- **Confirm personal eligibility and qualifications** (currency, recency, and required checks per OM-D §2.3 and §3.8)
- **Acknowledge compliance** with duty/rest limitations and self-certify they are fit for duty

Pilots who cannot meet qualification or fitness criteria must immediately notify the Chief Pilot or designee, and will not be released for duty until resolved.

#### 1.1.2 Hoist Operator Duty and Rest Requirements (Non-Crew Designation)

**Responsible Authority:** Director of Operations / Chief Pilot

**Applicability:** All hoist operators participating in HeliService USA operations under 14 CFR Part 133 or Part 135.



#### 1.1.2.1 Purpose

To ensure operational readiness and fatigue mitigation for hoist operators conducting mission-critical flight support tasks, while maintaining compliance with FAA, SMS, and OSHA standards for non-crew personnel.

#### 1.1.2.2 Duty Day Limitations

- **Maximum Duty Period:** 24 hours per calendar day.
- **Standby Requirement (14+ hours):**  
If a hoist operator is scheduled or extended beyond 14 hours of duty, the period beyond 14 hours must be designated as **standby only**, where the operator:
  - Remains in a rest-capable, alert posture (e.g., base quarters or hotel).
  - May not conduct physical hoist operations without a pre-shift fatigue evaluation.
  - Must be relieved after activation from standby beyond 3 hours of active involvement.
- **Maximum Active Duty (Working Time):** No more than 14 cumulative hours of **active tasking** within any 24-hour duty period.

#### 1.1.2.3 Fatigue Risk Mitigation

- Operators must self-monitor and report fatigue symptoms without fear of reprisal.
- Any operator showing signs of fatigue or having been activated during standby for more than 2 hours must be evaluated before further assignment.
- Supervisors are authorized to suspend or reassign duties at their discretion based on fatigue risk factors.

#### 1.1.2.4 Scheduling and Compliance

- Duty and standby time must be scheduled and logged in company duty tracking software.
- Duty beyond 14 hours requires standby classification and pre-approval by the Director of Operations or Designee.
- The Safety Department will review any standby activations and extensions quarterly to evaluate schedule effectiveness.



## 1.1.3 Adverse Weather Policy

### 1.1.3.1 Purpose

This policy defines the operational limits and procedural framework for offshore helicopter flights conducted in adverse weather conditions, to align with recommended practices, ICAO Annexes, and relevant national aviation regulations. The goal is to ensure weather-related risks are managed to As Low As Reasonably Practicable (ALARP) levels, prioritizing the safety of personnel, aircraft, and equipment.

### 1.1.3.2 Scope

This policy applies to all offshore helicopter operations, including passenger transport, Helicopter Hoist Operations (HHO), sling load operations, and emergency response missions. It is applicable to all contracted aircraft operators and subcontractors performing offshore flights on behalf of the company, regardless of region.

### 1.1.3.3 Policy Statement

- Helicopter operations will only be conducted when actual and forecasted meteorological conditions meet or exceed the defined minimum weather criteria.
- The Pilot-in-Command (PIC) retains the ultimate authority to delay, reroute, or cancel flights when weather conditions present a safety concern.
- Decision-making is to prioritize safety over any operational, commercial, or scheduling pressures.
- Operations may proceed beyond adverse weather conditions specified only after:
  - Conducting an operational risk assessment.
  - Documenting mitigation measures.
  - Obtaining authorization from the Chief Pilot/Director of Operations or accountable Operational control designee (when practicable).

### 1.1.3.4 Definitions

- **Adverse Weather:** Any weather conditions that may adversely affect the safe conduct of helicopter operations, including but not limited to low visibility, low cloud ceiling, high wind speeds, gusting winds, icing conditions, thunderstorms, and severe sea states.

- **Decision Point:** A scheduled evaluation before or during the flight to reassess operational decisions based on updated weather conditions.

#### 1.1.3.5 Minimum Weather Criteria



| Operation Type                     | Visibility (Statute Miles)  | Cloud Base (Feet AGL)                                                                                                                          | Winds over Helideck (knots) | Max Gust Spread (Knots) | Sea State (Significant Wave Height)                    |
|------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------|--------------------------------------------------------|
| Day VFR Passenger Transport        | 1sm Onshore<br>2sm Offshore | 300ft                                                                                                                                          | 60 KIAS                     | 20 KIAS                 | Does not exceed the aircrafts floatation system rating |
| Night VFR Passenger Transport      | 2sm Onshore<br>3sm Offshore | 500ft Onshore<br><br>1000ft Offshore 10+ NM of a HSUSA base of Operations<br><br>700ft Offshore Within ten 10 NM of a HSUSA base of operations | 60 KIAS                     | 20 KIAS                 | Does not exceed the aircrafts floatation system rating |
| IFR Operations                     | As Per IFR Approach Minima  | As Per IFR Approach Minima                                                                                                                     | 60 KIAS                     | 20 KIAS                 | Does not exceed the aircrafts floatation system rating |
| Hoisting (HHO) or Sling Operations | 3 sm                        | 600ft                                                                                                                                          | 43 KIAS                     | 20 KIAS                 | Does not exceed the aircrafts floatation system rating |



**Note:**

Customer Specific Weather minimums may be utilized / Required by specific customers. Check applicable customer requirements and or OM-C.

**Note:**

For operations in known or forecast icing conditions, the aircraft must be certified for such conditions. Gust spreads greater than 10 knots should be carefully evaluated, especially for hoisting and deck landing operations.

#### **1.1.3.6 Decision Making and Planning**

- Pre-flight weather assessments are required using validated meteorological sources (such as METAR/TAF, offshore platform reports, and satellite data).
- Final weather confirmation must occur immediately prior to engine start.
- Weather conditions along the route, at destination, and at designated alternates must be reviewed and documented.
- A formal risk assessment must accompany any operational decision near weather minima.

#### **1.1.3.7 In-Flight Monitoring**

- Continuous monitoring of enroute weather conditions through onboard systems and communications with air traffic services is mandatory.
- If actual weather conditions deteriorate below minimum limits, the PIC must follow pre-briefed contingency actions, including diversion or mission abort.

#### **1.1.3.8 Exceptions for Emergency Operations**

- Emergency operations, such as Search and Rescue (SAR) or Medevac, may proceed under adverse weather conditions only after:
  - Conducting an operational risk assessment.
  - Documenting mitigation measures.
  - Obtaining authorization from the accountable manager or designated authority (when practicable).



#### 1.1.3.9 Training and Competency

- All flight crew members must receive initial and recurrent training on adverse weather recognition, weather decision-making, and associated contingency planning.
- Specific training will cover regional weather phenomena relevant to the areas of operation, including frontal systems, icing conditions, turbulence, and fog formation.

#### 1.1.3.10 Continuous Improvement

- All weather-related incidents, near-misses, and operational challenges must be reported through the Safety Management System (SMS).
- This policy and associated minima will be reviewed annually or after any significant weather-related event, with updates incorporated as needed.

### 1.1.4 Bird Avoidance Policy

#### 1.1.4.1 Flight Planning

Pilots shall review known bird activity areas using FAA NOTAMs, advisory publications, and local hazard briefings during preflight planning.

When scheduling flights during peak bird activity hours—typically one hour after sunrise and one hour before sunset pilots will consider flying at or above 2000 feet.

#### 1.1.4.2 Airspeed Restrictions Below 1000Ft

Within 3 nm of the shoreline and overland, all HeliService USA pilots shall maintain an airspeed at or below 100 knots indicated airspeed (KIAS) when operating at altitudes below 1000 feet above ground level (AGL) , except when:

1. Conducting emergency procedures
2. Required by ATC instructions
3. Operating in areas where terrain or obstacles dictate a higher speed for safe flight.



#### **1.1.4.3 Altitude Considerations**

Whenever operationally feasible and safe, pilots should maintain altitudes above 1000 feet AGL when transiting areas known for high bird activity

#### **1.1.4.4 In Flight Avoidance**

Maintain vigilant visual scanning for bird activity, especially near bodies of water, coastal areas, landfills, or agricultural fields. Use anti-collision lights and landing lights in high-risk areas to increase aircraft visibility.

Report all bird sightings and strikes to the Director of Safety and enter events into the Fleetplan SMS reporting tool.

#### **1.1.4.5 Training and Awareness**

All flight crews shall receive annual training on bird strike hazards and mitigation strategies. Bird strike events shall be reviewed in Safety Review Board (SRB) meetings as part of safety assurance and continuous improvement.

#### **1.1.4.6 Reporting Requirements**

Any bird strike incident, regardless of damage, must be reported per HeliService USA's SMS protocols and to the FAA using Form 5200-7.

#### **1.1.4.7 Local Area Bird Policies**

For local bird policies please see HeliService OM-C Chapter 2



## 1.1.5 AW169 Single Pilot Operations Policy

### 1.1.5.1 Purpose

To establish minimum qualification standards for single-pilot operations in the Leonardo AW169, tailored for both local area operations and cross-country operations. This policy ensures operational safety while recognizing pilot experience specific to the aircraft and operational environment.

### 1.1.5.2 Definitions

- **Local Operations:** Flights within a 50 NM radius of the home base (e.g., North Kingstown), in familiar terrain, with established infrastructure and short mission durations.
- **Cross-Country Operations:** Any flight exceeding 50 NM from base or involving unfamiliar airspace, international borders, mountainous terrain, or offshore operations.

### 1.1.5.3 General Minimum Experience Requirements

All single-pilot AW169 operations must meet the following FAA baseline per Part 135 and internal OM-D training requirements:

- **Rotorcraft-Helicopter ATP or Commercial Certificate**
- **HeliService USA's Basic Indoc course**
- **Any of the requirements below may be waived by the Chief Pilot or Director of Operations**

#### 1.1.5.3.1 Local Operations Pilot Experience (Minimums)

- **Total Helicopter Time:** 500 hours
- **AW169 Time:** 150 hours
- **VFR Conditions**
- **Time in Local Area (50 NM radius):** 100 hours
- **Recent Flight Experience in AW169:** At least 10 hours within the past 90 days

*Rationale:* The familiarity with terrain, ATC procedures, and company routes mitigates risk and allows for slightly lower total time provided AW169 proficiency is strong.



#### 1.1.5.3.2 Cross Country Operations Pilot Experience (Minimums)

- **Total Helicopter Time:** 2,000 hours
- **AW169 Time:** 50 hours (25 may be in approved simulator)
- **VFR Conditions**
- **Instrument Time (if IFR capable route):** 75 hours total, 10 hours last 6 months
- **Cross-Country Helicopter Time:** 250 hours
- **Recent Flight Experience in AW169:** 20 hours within the past 90 days

*Rationale:* Greater exposure to complex airspace, weather variability, and navigation demands justifies a more robust experience threshold.

#### 1.1.5.4 Exceptions and Considerations

- **Currency Waivers:** May be granted by the Chief Pilot or Director of Operations for pilots with documented recent military or Part 91/135 experience in AW169 or equivalent aircraft.
- **Initial Operating Experience (IOE):** All single-pilot AW169 pilots must complete IOE with a company Check Pilot, Instructor Pilot, Line Training Captain or approved by Director of Operations or Chief Pilot.

### 1.1.6 Sea State Limitations

When determining Sea State HSUSA utilizes the Beaufort Scale of measurement. Which measures wind, wave, and swell to determine the sea state. HSUSA defines Sea state as the Significant wave height (Hs). This is defined as the average height of the highest one-third waves”.

## 1.2 Night Vision Goggles

### 1.2.1 Maintenance and Inspection Standards

- **180-Day Professional Inspection:** Every set of NVGs must be inspected by a qualified facility every 180 days. Certified technicians must verify tube performance, alignment, and calibration. HeliService USA (HSUSA) verifies that vendors performing these inspections utilize manufacturer-approved data.
- **Daily Pre-Flight Checks:** Pilots must conduct a pre-flight inspection of NVGs before each flight operation using the manufacturer’s operating manual. This includes:
  - **Tracking and Documentation :** When using NVGs, sign out the goggles on the form for tracking and documentation.



- Verifying fresh batteries and having spares readily available.
- Ensuring no visible physical damage to the unit.
- Confirming all controls function properly.
- **Cleaning Protocols:**
  - Remove loose dust with a soft brush or air blower.
  - Clean optics using only lint-free lens paper and approved cleaning solutions; harsh chemicals that strip protective coatings are prohibited.
  - Keep battery contacts free of corrosion and remove any leaking batteries immediately.

### 1.2.2 Storage and Security Protocols

- **Controlled Environment:** NVGs must be stored in cool, dry, and dark locations to prevent damage from temperature extremes or humidity. Avoid storage in hot vehicles or humid basements. Use desiccant or silica gel packs in storage cases to absorb moisture and prevent internal fogging.
- **Protective Housing:** Always store goggles in their padded protective case when not in use.
- **Light Protection:** Objective lens caps must be placed on the device before storage to prevent accidental light exposure.
- **Battery Management:** Remove batteries if the device will be stored for more than a few days to prevent leakage and corrosion.
- **Security of Units:** When NVGs are not in personal possession, goggles must be locked away.
- **Tracking and Documentation:** When goggles are no longer in use, sign in the goggles on the appropriate form.

### 1.2.3 Operational Security and Handling

- **Light Discipline:** Always use objective lens caps during any daylight testing. Never point active goggles at bright light sources, as intense exposure accelerates photocathode wear and ages components.
- **Physical Protection:** Use secure helmet mounts and handle units with the same care as expensive camera equipment. Never press on lenses or yank battery packs.
- **Acclimation:** When moving between temperature zones, allow the device to acclimate inside its case to prevent condensation.
- **Unauthorized Repairs:** Personnel must never attempt home or field repairs beyond basic cleaning. Opening housings voids warranties and is dangerous due to high-voltage internal components.



## 1.2.4 Discrepancy Reporting and ALSE Notification

- **Immediate Notification:** Any discrepancies, damage, or irregularities discovered during pre-flight or post-flight inspections must be reported immediately.
- **ALSE Personnel Reporting:** Pilots must notify **ALSE (Aviation Life Support Equipment) personnel** immediately upon identifying a fault.
- **Digital Entry:** All equipment discrepancies must be entered into the aircraft discrepancy section of the computerized maintenance tracking program (Digital AirWare) to ensure a formal record is created and the unit is appropriately "red-tagged" or removed from the active inventory.

## 1.3 Training

### 1.3.1 LOFT Training Program

#### 1.3.1.1 Purpose

The LOFT program provides scenario-based training to enhance crew resource management (CRM), decision-making, and operational proficiency in a realistic environment. It aligns with FAA AC 120-35 (LOFT) and incorporates HeliService USA's OM-A, OM-D, and RLCFM procedures.

#### 1.3.1.2 Objectives

- Reinforce standard operating procedures (SOPs) and emergency procedures.
- Improve communication and teamwork using CRM techniques.
- Train crews to manage realistic operational scenarios including Part 133 (external load) and Part 135 (air taxi) environments.
- Assess decision-making under time-critical and abnormal conditions.

#### 1.3.1.3 Scope

Applicable to:

- PIC and SIC under Part 135.
- Hoist Operators (HHO) and Task Specialists during hoist operations.



- Ground crew integration for Part 133 external load missions.

#### 1.3.1.4 Individual Modules

##### Module 1: Pre-Flight and Briefing

- Conduct **realistic dispatch briefing** including weather, MEL status, load configuration, and customer-specific requirements.
- Review **risk assessment** and operational risk management (ORM).

##### Module 2: Normal Operations

- Offshore/onshore operations with **standard crew coordination**.
- Simulate routine tasks: hoist site approach, load handling, external load release, and post-mission checks.

##### Module 3: Abnormal Scenarios

- Simulate failures and unusual situations (without jeopardizing safety):
  - Single engine failure during hoist.
  - Hoist motor runaway / cable entanglement.
  - Lost comms with ground crew.
  - Hazardous weather enroute to offshore site.
- Emphasize crew communication, checklist usage, and adherence to emergency procedures.

##### Module 4: CRM Integration

- Crew evaluates leadership, workload management, decision-making.
- Debrief with instructor focusing on CRM effectiveness.

##### Module 5: Post-Flight Review

- Crew self-assessment.
- Instructor feedback aligned with Heliservice SMS reporting and continuous improvement.

#### 1.3.1.4.1 Offshore / IFR Operations



- 1.3.1.4.2 Hoist / External Load Operations
- 1.3.1.4.3 HAZMAT and SMS Events
- 1.3.1.4.4 Emergency Response Scenarios
- 1.3.1.4.5 Additional Offshore and SAR Challenges



## **2 Local Base Procedures**

### **2.1 Quonset**

#### **2.1.1 Daily Work Flow**

- Arrive with sufficient time to brief, chat, preflight to be ready for a late shift flight request
- Duty in to complete flight, check currency.
- Check digital Airware
  - ◦ Check all aircraft at your base
  - ◦ Next due calendar/Hobbs
  - ◦ Check hoist next due calendar/Hobbs/cycles
  - ◦ Check open MEL/Discrepancies
  - ◦ Check Logbook has been opened at previous night shift/New one completed at current night shift.
  - ◦ Confirm there are X hours/cycles/etc. Available for each shift.
- Confirm Flights/missions on fleetplan
- Check Weather
- Conduct Daily Brief with Crew
- Preflight Aircraft/Fuel Samples
- Preflight iPads, download new data, confirm custom content is loaded and updated.
- Inspect and prepare NVGs for night shift
- Rinse Engines/Full runup for night shift.
- Clean aircraft exterior and interior on X day.
- Confirm crew has Duty Phones and aircraft Keys
- Test Dispatch System

#### **2.1.2 Operational Readiness**

- Every effort should be made to depart on time for scheduled operations and as quickly as possible for HAA missions.
- Pre-Fill a FRAT/Manifest during each briefing session. This will allow each member of the flight crew to take part.
- Establish and keep all aircraft fueled to a base minimum allowing for quick dispatch for local calls and some scheduled flights without the need to refuel.
- Configure the aircraft where applicable
- Pre-stage and prepare all crew gear (suits, helmets, NVGs, Medical equipment, etc.
- Confirm tow cart and battery cart are charged and located with the HAA Helicopter for ease of



use.

- Crew briefs each members role and responsibility for the shift, launching, during the mission, etc.
- Scheduled flights are to DEPART at a given time. Passengers should be called for 10 minutes prior to that time. Crews should be starting the aircraft with sufficient time to allow for an on time departure.

### 2.1.3 Passenger Briefings

Before taxiing, the following briefing shall be given to the passengers after they are all settled in and the cabin door is closed:

"Good (morning/afternoon/evening), welcome to HeliService USA flight to the "Vessel name", confirming that is everyone's destination, if not please speak up or tap one of us on the shoulder. Our flight time to the "Vessel name" is XX minutes. Weather en route today is \_\_\_\_\_. Please remain in your seats with your seat belts fastened until the HLO has directed you otherwise. Enjoy the flight and thank you for flying HeliService USA."

Before departing from an offshore helideck, the following briefing shall be given to the passengers after they are all settled in and the cabin door is closed:

"Good (morning/afternoon/evening), welcome to HeliService USA flight back to Quonset. Our flight time to Quonset is XX minutes. Weather en route today is \_\_\_\_\_. Please remain in your seats with your seat belts fastened until our ground staff has directed you otherwise. Enjoy the flight and thank you for flying HeliService USA."

### 2.1.4 Hoist Operations

### 2.1.5 HAA Operations

### 2.1.6 Wind Turbine Hoisting Operations

#### 2.1.6.1 Hours of Operations

Non HAA hoist operations (unless NVG's are used) shall be accomplished between civil twilight hours only, which is generally 30 minutes prior to sunrise and 30 minutes after sunset. For more precise times, Pilots shall use [suncalc.net](https://www.suncalc.net), or another similar site:

[SunCalc - sun position, sunlight phases, sunrise, sunset, dusk and dawn times calculator](https://www.suncalc.net)



Dispatch shall not schedule pick up times later than 15 minutes before the end of civil twilight.

## 2.1.7 VHF Frequencies



## U.S. VHF Channel Information

| New Channel Number | Old Channel Number | Ship Transmit MHz | Ship Receive MHz | Use                                                                                                                                                                                                          |
|--------------------|--------------------|-------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1001               | 01A                | 156.050           | 156.050          | Port Operations and Commercial, VTS. Available only in New Orleans / Lower Mississippi area.                                                                                                                 |
| 1005               | 05A                | 156.250           | 156.250          | Port Operations or VTS in the Houston, New Orleans and Seattle areas.                                                                                                                                        |
| 06                 | 06                 | 156.300           | 156.300          | Intership Safety                                                                                                                                                                                             |
| 1007               | 07A                | 156.350           | 156.350          | Commercial. VDSMS                                                                                                                                                                                            |
| 08                 | 08                 | 156.400           | 156.400          | Commercial (Intership only). VDSMS                                                                                                                                                                           |
| 09                 | 09                 | 156.450           | 156.450          | Boater Calling. Commercial and Non-Commercial. VDSMS                                                                                                                                                         |
| 10                 | 10                 | 156.500           | 156.500          | Commercial. VDSMS                                                                                                                                                                                            |
| 11                 | 11                 | 156.550           | 156.550          | Commercial. VTS in selected areas. VDSMS                                                                                                                                                                     |
| 12                 | 12                 | 156.600           | 156.600          | Port Operations. VTS in selected areas.                                                                                                                                                                      |
| 13                 | 13                 | 156.650           | 156.650          | Intership Navigation Safety (Bridge-to-bridge). Ships >20m length maintain a listening watch on this channel in US waters.                                                                                   |
| 14                 | 14                 | 156.700           | 156.700          | Port Operations. VTS in selected areas.                                                                                                                                                                      |
| 15                 | 15                 | --                | 156.750          | Environmental (Receive only). Used by Class C EPIRBs.                                                                                                                                                        |
| 16                 | 16                 | 156.800           | 156.800          | International Distress, Safety and Calling. Ships required to carry radio, USCG, and most coast stations maintain a listening watch on this channel. See our <a href="#">Watchkeeping Regulations page</a> . |
| 17                 | 17                 | 156.850           | 156.850          | State & local govt maritime control                                                                                                                                                                          |
| 1018               | 18A                | 156.900           | 156.900          | Commercial. VDSMS                                                                                                                                                                                            |
| 1019               | 19A                | 156.950           | 156.950          | Commercial. VDSMS                                                                                                                                                                                            |
| 20                 | 20                 | 157.000           | 161.600          | Port Operations (duplex)                                                                                                                                                                                     |
| 1020               | 20A                | 157.000           | 157.000          | Port Operations                                                                                                                                                                                              |
| 1021               | 21A                | 157.050           | 157.050          | U.S. Coast Guard only                                                                                                                                                                                        |
| 1022               | 22A                | 157.100           | 157.100          | Coast Guard Liaison and Maritime Safety Information Broadcasts. Broadcasts announced on channel 16.                                                                                                          |
| 1023               | 23A                | 157.150           | 157.150          | U.S. Coast Guard only                                                                                                                                                                                        |
| 24                 | 24                 | 157.200           | 161.800          | Public Correspondence (Marine Operator). VDSMS                                                                                                                                                               |
| 25                 | 25                 | 157.250           | 161.850          | Public Correspondence (Marine Operator). VDSMS                                                                                                                                                               |
| 26                 | 26                 | 157.300           | 161.900          | Public Correspondence (Marine Operator). VDSMS                                                                                                                                                               |
| 27                 | 27                 | 157.350           | 161.950          | Public Correspondence (Marine Operator). VDSMS                                                                                                                                                               |



|       |       |         |         |                                                                                                          |
|-------|-------|---------|---------|----------------------------------------------------------------------------------------------------------|
| 28    | 28    | 157.400 | 162.000 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 1063  | 63A   | 156.175 | 156.175 | Port Operations and Commercial, VTS. Available <b>only</b> in New Orleans / Lower Mississippi area.      |
| 1065  | 65A   | 156.275 | 156.275 | Port Operations                                                                                          |
| 1066  | 66A   | 156.325 | 156.325 | Port Operations                                                                                          |
| 67    | 67    | 156.375 | 156.375 | Commercial. Used for Bridge-to-bridge communications in lower Mississippi River. Intership <b>only</b> . |
| 68    | 68    | 156.425 | 156.425 | Non-Commercial. VDSMS                                                                                    |
| 69    | 69    | 156.475 | 156.475 | Non-Commercial. VDSMS                                                                                    |
| 70    | 70    | 156.525 | 156.525 | Digital Selective Calling (voice communications not allowed)                                             |
| 71    | 71    | 156.575 | 156.575 | Non-Commercial. VDSMS                                                                                    |
| 72    | 72    | 156.625 | 156.625 | Non-Commercial (Intership <b>only</b> ). VDSMS                                                           |
| 73    | 73    | 156.675 | 156.675 | Port Operations                                                                                          |
| 74    | 74    | 156.725 | 156.725 | Port Operations                                                                                          |
| 77    | 77    | 156.875 | 156.875 | Port Operations (Intership <b>only</b> )                                                                 |
| 1078  | 78A   | 156.925 | 156.925 | Non-Commercial. VDSMS                                                                                    |
| 1079  | 79A   | 156.975 | 156.975 | Commercial. Non-Commercial in Great Lakes <b>only</b> . VDSMS                                            |
| 1080  | 80A   | 157.025 | 157.025 | Commercial. Non-Commercial in Great Lakes <b>only</b> . VDSMS                                            |
| 1081  | 81A   | 157.075 | 157.075 | U.S. Government <b>only</b> - Environmental protection operations.                                       |
| 1082  | 82A   | 157.125 | 157.125 | U.S. Government <b>only</b>                                                                              |
| 1083  | 83A   | 157.175 | 157.175 | U.S. Coast Guard <b>only</b>                                                                             |
| 84    | 84    | 157.225 | 161.825 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 85    | 85    | 157.275 | 161.875 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 86    | 86    | 157.325 | 161.925 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 87    | 87    | 157.375 | 157.375 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 88    | 88    | 157.425 | 157.425 | Commercial, Intership <b>only</b> . VDSMS                                                                |
| AIS 1 | AIS 1 | 161.975 | 161.975 | <a href="#">Automatic Identification System (AIS)</a>                                                    |
| AIS 2 | AIS 2 | 162.025 | 162.025 | <a href="#">Automatic Identification System (AIS)</a>                                                    |

**Notes:** VDSMS (VHF Digital Small Message Services). Transmissions of short digital messages in accordance with RTCM Standard 12301.1 is allowed.

## Four-digit channel numbers

Recommendation ITU-R M.1084-5 Annex 5, adding the prefixes "10" and "20" to the existing channel number instead of the suffixes "A" and "B", respectively.



Four-digit VHF maritime channel numbers were established by the International Telecommunications Union World Radio Conference meeting in Geneva, Switzerland, 2-27 November 2015. Its purpose was to improve spectrum efficiency, making allowances for countries making simplex use of duplex channels, just as the US had done beginning in 1965. US channel 05A is now designated and recognized internationally by the channel 1005, and the US Coast Guard channel 22A is now designated and recognized internationally by the channel "1022". These new channel number designations are being displayed on new models of VHF marine radios.

See ITU Radio Regulations Appendix 18, [summarized here](#).

Frequencies are in MHz. Modulation is 16KF3E or 16KG3E.

Boaters should normally use channels listed as Non-Commercial. Channel 16 is used for calling other stations or for distress alerting. Channel 13 should be used to contact a ship when there is danger of collision. All ships of length 20m or greater are required to guard VHF channel 13, in addition to VHF channel 16, when operating within U.S. territorial waters. Users may be fined by the FCC for improper use of these channels. See [Radio Watchkeeping regulations](#) for further information.

## VHF Maritime Spectrum Chart

### NOAA Weather Radio Frequencies

| Channel | Frequency (MHz) |
|---------|-----------------|
| WX1     | 162.550         |
| WX2     | 162.400         |
| WX3     | 162.475         |
| WX4     | 162.425         |
| WX5     | 162.450         |
| WX6     | 162.500         |
| WX7     | 162.525         |

Channel numbers, e.g. (WX1, WX2) etc. have no special significance but are often designated this way in consumer equipment. Other channel numbering schemes are also prevalent.

The order of channels shown is the order they were established and is slowly becoming less "popular" over time than a numerical ordering of channels.

See [NATIONAL WEATHER SERVICE MARINE PRODUCTS VIA NOAA WEATHER RADIO](#) and the [NOAA Weather Radio Homepage](#) for more information.

Additional Information, Frequencies, & Charts

Also available are [International VHF Maritime Radio Channels and Frequencies](#), [Narrowband VHF Maritime Channels and Frequencies](#), [Radio Information for Boaters](#), and [U.S. Coast Guard VHF Distress and Safety Coverage Charts](#).

## 2.1.8 Preferred Routes/Noise Abatement

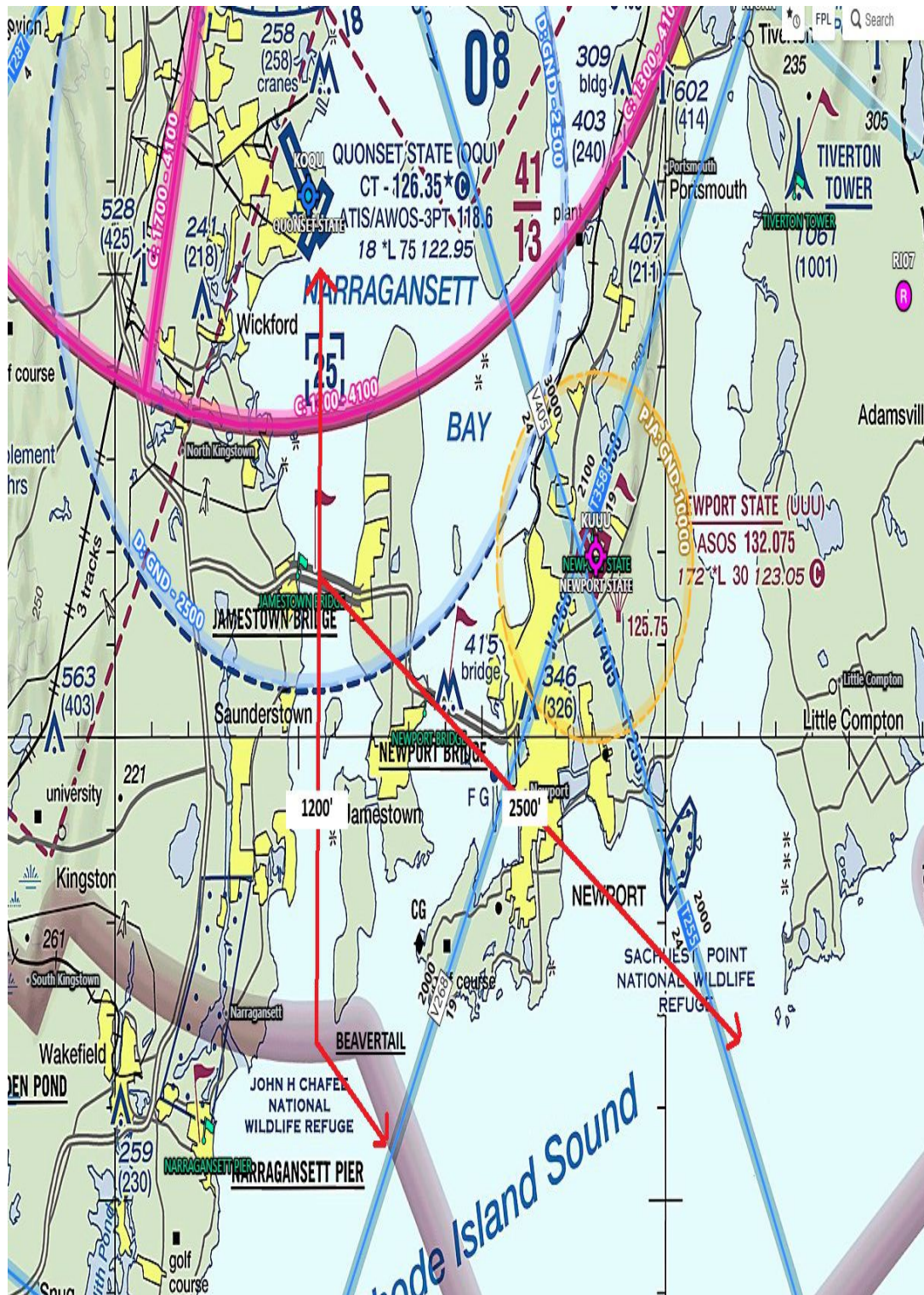


The following course rules apply for OQU flight operations. The map below show the courses and altitudes that are to be flown to the greatest extent possible. As always, safety of flight, traffic avoidance, and ATC instructions take precedence. These course rules are established for neighborly flying/noise abatement and safety.

When weather conditions do not allow for the prescribed route altitudes, lower altitudes are permissible. Ensure that proper cloud clearance is maintained for the class of airspace you are in.

**Preferred Routes to/from KOQU, Vineyard Wind, and Sunrise Wind.**





### Preferred Routing to and From KMVY and KOQU

Below is a standard departure and arrival profile for operations between KOQU and KMVY utilizing direct (straight-line) routing.



### Departure – KOQU

- Depart KOQU and track toward the southern tip of Prudence Island until activating a fresh "direct to" in the FMS.
- Climb and maintain 1,200 feet MSL until clear of KPVD Class C airspace.
- Once clear, continue climb to 2,500 feet MSL and proceed direct KMVY.

### Arrival – KOQU

- Descend to 1,200 feet MSL to remain below the KPVD Class C shelf.
- Maintain 1,200 feet MSL until reaching the shoreline.
- Descend as required to enter the airport environment and conduct the approach.



### Note:

The flight path passes within approximately 2 NM of the departure end of KUUU Runway 04; therefore, the climb to 2,500 feet MSL is initiated once clear of KPVD airspace. Heightened vigilance should be maintained when in close proximity to the pattern traffic at KUUU, as well as parachute operations when active. Within 4 NM of the route, the highest obstacle is Tiverton Tower, located approximately 3 NM north of the direct track, with a height of 1,061 feet MSL.

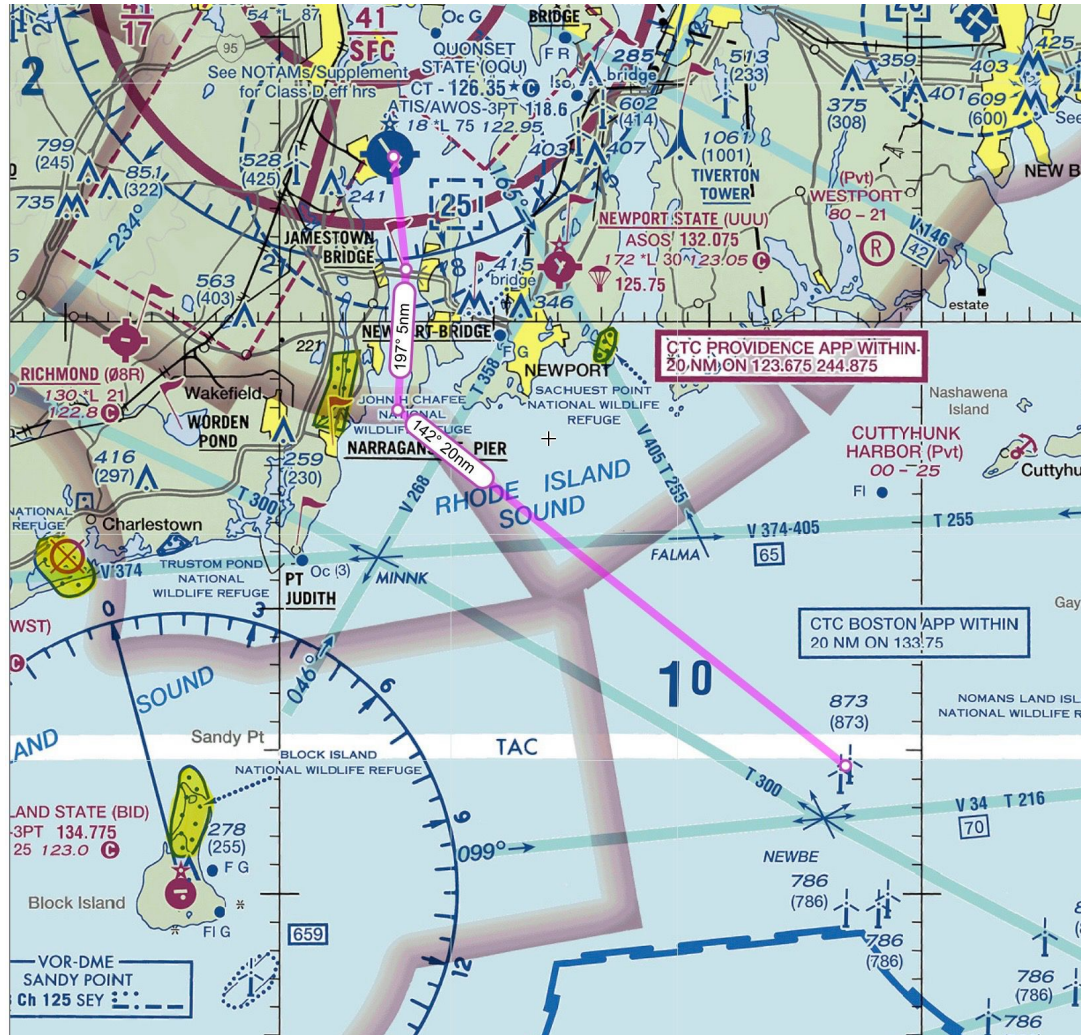
This profile complies with the 1,000-foot minimum altitude requirement for bird strike avoidance in accordance with OM-C 1.1.4.3, as well as neighborly flying guidance outlined in OM-A 6.4.3.2.

## 2.1.9 Bird Avoidance KOQU to Vineyard Wind





## 2.1.10 Bird Avoidance KOQU to South Fork and Revolution Wind



## 2.2 Martha's Vineyard

### 2.2.1 Daily Work Flow

#### Evening Before the Shift

- Send group text. The HHO or PIC sends a joint text to the boat captain, PIC, SIC, and HHO. Include the meet-up time at the boat (1 hour and 15 minutes prior to the scheduled lift time) and note any special details or instructions.

#### Day-of: Pre-Departure at the Boat

- Meet at the boat at the designated time.
- Discuss safety and sea conditions. Go over safety concerns and review anticipated sea conditions



for the ride to Martha's Vineyard.

- Transit to the island.

#### **Arrival at Vineyard Haven**

- Secure the vessel.
- Use the company vehicle to drive to the GE hangar at Martha's Vineyard Airport.

#### **At the GE Hangar**

- Preflight planning and checks. Pilots start preflight planning while the HHO performs a preflight check on the aircraft.
- Move aircraft. When ready, the HHO requests wing walkers to move the helicopter out of the hangar and onto the ramp.
- Check-in and manifests. The boat captain initiates check-in and finalizes manifests, weighing crew and equipment if needed.
- Monitor messages. Monitor message chats for updates on departure times and turbine status.
- Conduct briefing. Once technicians are geared up and a green or yellow light is obtained, conduct a full briefing covering the flight request, aircraft operations, items on the HHO briefing sheet, and turbine information provided by GEMC.

#### **Preparing for Flight**

- Load gear and passengers into the helicopter.
- Perform a complete walk-around of the aircraft.
- Begin the engine start sequence.

#### **Flight Operations**

- Fly neighborly, avoiding noise-sensitive areas as depicted on ForeFlight.
- Contact Vineyard MCC. Prior to entering and exiting the Vineyard Wind Park, contact Vineyard MCC on frequency 122.9.
- Stand down flight watch. Notify Vineyard MCC of positive communications with the tower and confirm that they can stand down on flight watch.
- Notify FBO. On the leg back into the airport, call Vineyard Operations (FBO) on 122.95 to notify ground operations of an upcoming fuel request.

#### **Turnaround Between Hoist Operations**

- APU mode on return. After returning from a hoist operation (if another group is scheduled shortly), bring the helicopter to APU mode.
- Check messages for turbine status and refuel as necessary.
- Load bags into the helicopter.
- Brief next group. PIC and HHO brief the next group of technicians inside the hangar, using the same briefing points as the previous group.
- Rotor start. Once a green or yellow light is obtained, load passengers and begin the rotor start checklist.
- Reposition if needed. If additional flights or heavier maintenance are required, reposition the



helicopter to KOQU.

#### **After Morning Flight Completion**

- Once morning flights are done, pilots complete flight and operational paperwork.
- Maintenance. The mechanic begins necessary maintenance on the aircraft.
- Clean up. After paperwork and maintenance, chores begin: clean the helicopter and tidy the hangar.

#### **Afternoon/Evening Hoist Pickups**

- Receive readiness notification. Receive notification that technicians are ready with a yellow or green light for approach to the turbine.
- Transport technicians back. Fly the technicians back to the GE hangar and switch to APU mode.
- Offload and refuel. Offload baggage, refuel if necessary, perform a walk-around, and check messages for additional green/yellow lights for subsequent pickups.
- Shut down after final pickup. After the final pickup of the day, shut down to main mode for off-loading.
- Estimate next day's fuel. Obtain an estimated fuel load for the next day's anticipated flight schedule.

#### **End-of-Day Wrap-Up**

- Move aircraft into hangar. HHO moves the aircraft into the hangar with the help of wing walkers; the pilot completes end-of-day paperwork.
- End-of-day maintenance. HHO performs end-of-day maintenance, checks, and paperwork, and provides pass-down information about that day's flights.
- Return to dock. Once duties are completed, drive the company vehicle back to the Vineyard Haven dock.
- Transit back to Falmouth Harbor. The boat captain conducts a briefing, and the team transits back to Falmouth Harbor. Upon arrival, help secure the boat.
- Plan next day. Plan the next day's arrival time based on the anticipated flight schedule and confirm who will make up the following day's crew.
- Send group text. Later in the evening, send a group text to the next day's crew detailing the boat departure time and any pertinent instructions.

#### **Important Frequencies, Phone Numbers, and Codes**

- KMOV Tower – 121.4 MHz (508-693-1170)
- KMOV ATIS – 126.5 MHz (508-693-7685)
- KMOV ASOS – 126.5 MHz (508-696-6988)
- Vineyard Operations (FBO) – 122.95 MHz (508-693-7022)
- KMOV Ground – 124.35 MHz
- Boston APP/DEP – 133.75 MHz
- Gate Code – 1717
- Front Door Key Lockboxes (2) by trash cans code - 2016



- Lockbox on south side of building code - 2016
- Falmouth HeliHouse Lock Box code - 0169

## 2.2.2 Wind Turbine Hoist Operations

### 2.2.2.1 Hours of Operations

Non HAA hoist operations (unless NVG's are used) shall be accomplished between civil twilight hours only, which is generally 30 minutes prior to sunrise and 30 minutes after sunset. For more precise times, Pilots shall use [suncalc.net](https://www.suncalc.net), or another similar site:

[SunCalc - sun position, sunlight phases, sunrise, sunset, dusk and dawn times calculator](https://www.suncalc.net)

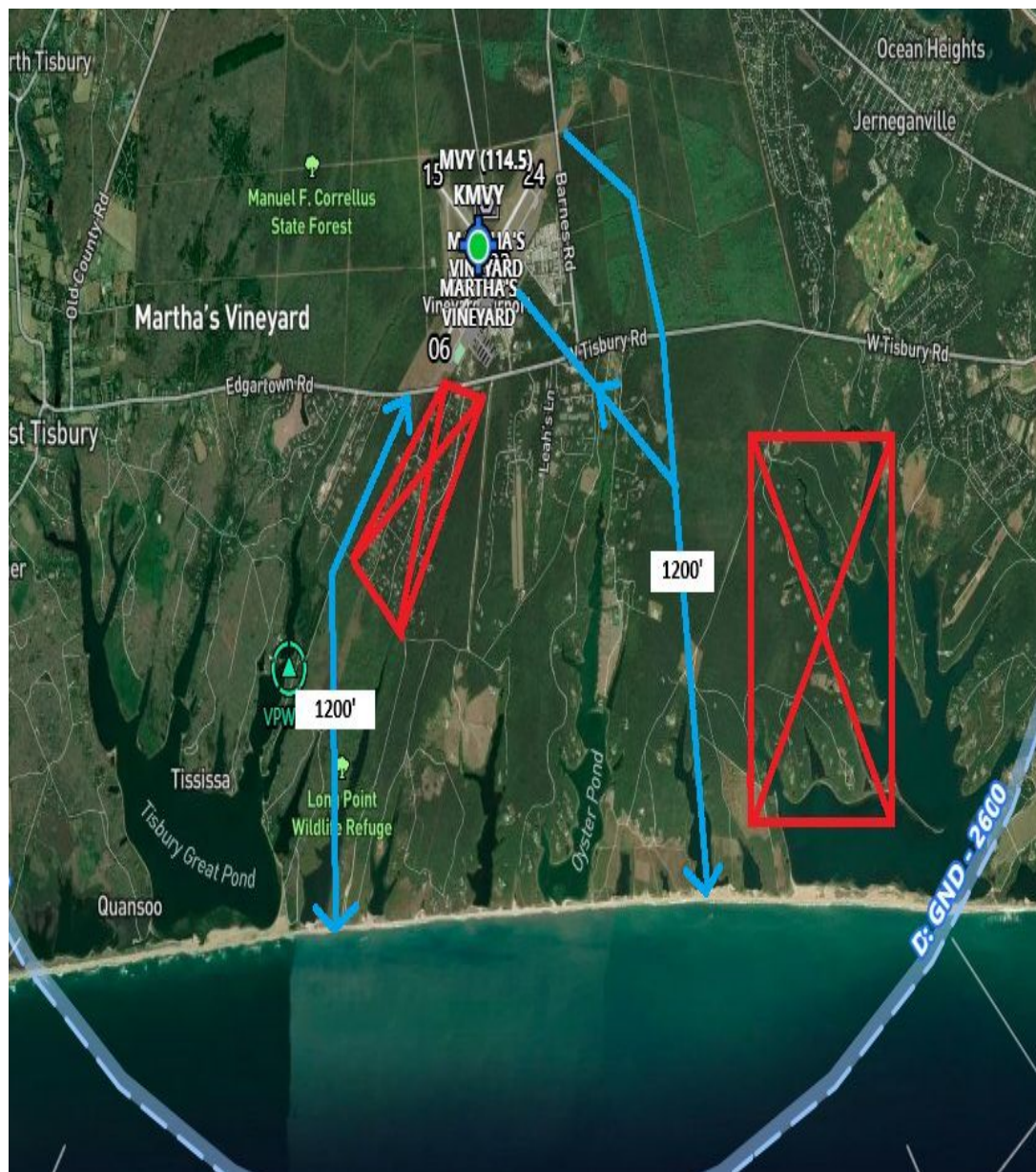
Dispatch shall not schedule pick up times later than 15 minutes before the end of civil twilight.

## 2.2.3 Preferred Routes/Noise Abatement

The following course rules apply for MVY flight operations. The map below show the courses and altitudes that are to be flown to the greatest extent possible. Do not overfly the red X boxes unless it is unavoidable. As always, safety of flight, traffic avoidance, and ATC instructions take precedence. These course rules are established for neighborly flying/noise abatement and safety.

When weather conditions do not allow for the prescribed route altitudes, lower altitudes are permissible. Ensure that proper cloud clearance is maintained for the class of airspace you are in.





## 2.2.4 VHF Frequencies

## U.S. VHF Channel Information

| New Channel Number | Old Channel Number | Ship Transmit MHz | Ship Receive MHz | Use                                                                                                                                                                                                          |
|--------------------|--------------------|-------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1001               | 01A                | 156.050           | 156.050          | Port Operations and Commercial, VTS. Available only in New Orleans / Lower Mississippi area.                                                                                                                 |
| 1005               | 05A                | 156.250           | 156.250          | Port Operations or VTS in the Houston, New Orleans and Seattle areas.                                                                                                                                        |
| 06                 | 06                 | 156.300           | 156.300          | Intership Safety                                                                                                                                                                                             |
| 1007               | 07A                | 156.350           | 156.350          | Commercial. VDSMS                                                                                                                                                                                            |
| 08                 | 08                 | 156.400           | 156.400          | Commercial (Intership only). VDSMS                                                                                                                                                                           |
| 09                 | 09                 | 156.450           | 156.450          | Boater Calling. Commercial and Non-Commercial. VDSMS                                                                                                                                                         |
| 10                 | 10                 | 156.500           | 156.500          | Commercial. VDSMS                                                                                                                                                                                            |
| 11                 | 11                 | 156.550           | 156.550          | Commercial. VTS in selected areas. VDSMS                                                                                                                                                                     |
| 12                 | 12                 | 156.600           | 156.600          | Port Operations. VTS in selected areas.                                                                                                                                                                      |
| 13                 | 13                 | 156.650           | 156.650          | Intership Navigation Safety (Bridge-to-bridge). Ships >20m length maintain a listening watch on this channel in US waters.                                                                                   |
| 14                 | 14                 | 156.700           | 156.700          | Port Operations. VTS in selected areas.                                                                                                                                                                      |
| 15                 | 15                 | --                | 156.750          | Environmental (Receive only). Used by Class C EPIRBs.                                                                                                                                                        |
| 16                 | 16                 | 156.800           | 156.800          | International Distress, Safety and Calling. Ships required to carry radio, USCG, and most coast stations maintain a listening watch on this channel. See our <a href="#">Watchkeeping Regulations page</a> . |
| 17                 | 17                 | 156.850           | 156.850          | State & local govt maritime control                                                                                                                                                                          |
| 1018               | 18A                | 156.900           | 156.900          | Commercial. VDSMS                                                                                                                                                                                            |
| 1019               | 19A                | 156.950           | 156.950          | Commercial. VDSMS                                                                                                                                                                                            |
| 20                 | 20                 | 157.000           | 161.600          | Port Operations (duplex)                                                                                                                                                                                     |
| 1020               | 20A                | 157.000           | 157.000          | Port Operations                                                                                                                                                                                              |
| 1021               | 21A                | 157.050           | 157.050          | U.S. Coast Guard only                                                                                                                                                                                        |
| 1022               | 22A                | 157.100           | 157.100          | Coast Guard Liaison and Maritime Safety Information Broadcasts. Broadcasts announced on channel 16.                                                                                                          |
| 1023               | 23A                | 157.150           | 157.150          | U.S. Coast Guard only                                                                                                                                                                                        |
| 24                 | 24                 | 157.200           | 161.800          | Public Correspondence (Marine Operator). VDSMS                                                                                                                                                               |
| 25                 | 25                 | 157.250           | 161.850          | Public Correspondence (Marine Operator). VDSMS                                                                                                                                                               |
| 26                 | 26                 | 157.300           | 161.900          | Public Correspondence (Marine Operator). VDSMS                                                                                                                                                               |
| 27                 | 27                 | 157.350           | 161.950          | Public Correspondence (Marine Operator). VDSMS                                                                                                                                                               |



|       |       |         |         |                                                                                                          |
|-------|-------|---------|---------|----------------------------------------------------------------------------------------------------------|
| 28    | 28    | 157.400 | 162.000 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 1063  | 63A   | 156.175 | 156.175 | Port Operations and Commercial, VTS. Available <b>only</b> in New Orleans / Lower Mississippi area.      |
| 1065  | 65A   | 156.275 | 156.275 | Port Operations                                                                                          |
| 1066  | 66A   | 156.325 | 156.325 | Port Operations                                                                                          |
| 67    | 67    | 156.375 | 156.375 | Commercial. Used for Bridge-to-bridge communications in lower Mississippi River. Intership <b>only</b> . |
| 68    | 68    | 156.425 | 156.425 | Non-Commercial. VDSMS                                                                                    |
| 69    | 69    | 156.475 | 156.475 | Non-Commercial. VDSMS                                                                                    |
| 70    | 70    | 156.525 | 156.525 | Digital Selective Calling (voice communications not allowed)                                             |
| 71    | 71    | 156.575 | 156.575 | Non-Commercial. VDSMS                                                                                    |
| 72    | 72    | 156.625 | 156.625 | Non-Commercial (Intership <b>only</b> ). VDSMS                                                           |
| 73    | 73    | 156.675 | 156.675 | Port Operations                                                                                          |
| 74    | 74    | 156.725 | 156.725 | Port Operations                                                                                          |
| 77    | 77    | 156.875 | 156.875 | Port Operations (Intership <b>only</b> )                                                                 |
| 1078  | 78A   | 156.925 | 156.925 | Non-Commercial. VDSMS                                                                                    |
| 1079  | 79A   | 156.975 | 156.975 | Commercial. Non-Commercial in Great Lakes <b>only</b> . VDSMS                                            |
| 1080  | 80A   | 157.025 | 157.025 | Commercial. Non-Commercial in Great Lakes <b>only</b> . VDSMS                                            |
| 1081  | 81A   | 157.075 | 157.075 | U.S. Government <b>only</b> - Environmental protection operations.                                       |
| 1082  | 82A   | 157.125 | 157.125 | U.S. Government <b>only</b>                                                                              |
| 1083  | 83A   | 157.175 | 157.175 | U.S. Coast Guard <b>only</b>                                                                             |
| 84    | 84    | 157.225 | 161.825 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 85    | 85    | 157.275 | 161.875 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 86    | 86    | 157.325 | 161.925 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 87    | 87    | 157.375 | 157.375 | Public Correspondence (Marine Operator). VDSMS                                                           |
| 88    | 88    | 157.425 | 157.425 | Commercial, Intership <b>only</b> . VDSMS                                                                |
| AIS 1 | AIS 1 | 161.975 | 161.975 | <a href="#">Automatic Identification System (AIS)</a>                                                    |
| AIS 2 | AIS 2 | 162.025 | 162.025 | <a href="#">Automatic Identification System (AIS)</a>                                                    |

**Notes:** VDSMS (VHF Digital Small Message Services). Transmissions of short digital messages in accordance with RTCM Standard 12301.1 is allowed.

## Four-digit channel numbers

Recommendation ITU-R M.1084-5 Annex 5, adding the prefixes "10" and "20" to the existing channel number instead of the suffixes "A" and "B", respectively.



Four-digit VHF maritime channel numbers were established by the International Telecommunications Union World Radio Conference meeting in Geneva, Switzerland, 2-27 November 2015. Its purpose was to improve spectrum efficiency, making allowances for countries making simplex use of duplex channels, just as the US had done beginning in 1965. US channel 05A is now designated and recognized internationally by the channel 1005, and the US Coast Guard channel 22A is now designated and recognized internationally by the channel "1022". These new channel number designations are being displayed on new models of VHF marine radios.

See ITU Radio Regulations Appendix 18, [summarized here](#).

Frequencies are in MHz. Modulation is 16KF3E or 16KG3E.

Boaters should normally use channels listed as Non-Commercial. Channel 16 is used for calling other stations or for distress alerting. Channel 13 should be used to contact a ship when there is danger of collision. All ships of length 20m or greater are required to guard VHF channel 13, in addition to VHF channel 16, when operating within U.S. territorial waters. Users may be fined by the FCC for improper use of these channels. See [Radio Watchkeeping regulations](#) for further information.

## VHF Maritime Spectrum Chart

### NOAA Weather Radio Frequencies

| Channel | Frequency (MHz) |
|---------|-----------------|
| WX1     | 162.550         |
| WX2     | 162.400         |
| WX3     | 162.475         |
| WX4     | 162.425         |
| WX5     | 162.450         |
| WX6     | 162.500         |
| WX7     | 162.525         |

Channel numbers, e.g. (WX1, WX2) etc. have no special significance but are often designated this way in consumer equipment. Other channel numbering schemes are also prevalent.

The order of channels shown is the order they were established and is slowly becoming less "popular" over time than a numerical ordering of channels.

See [NATIONAL WEATHER SERVICE MARINE PRODUCTS VIA NOAA WEATHER RADIO](#) and the [NOAA Weather Radio Homepage](#) for more information.

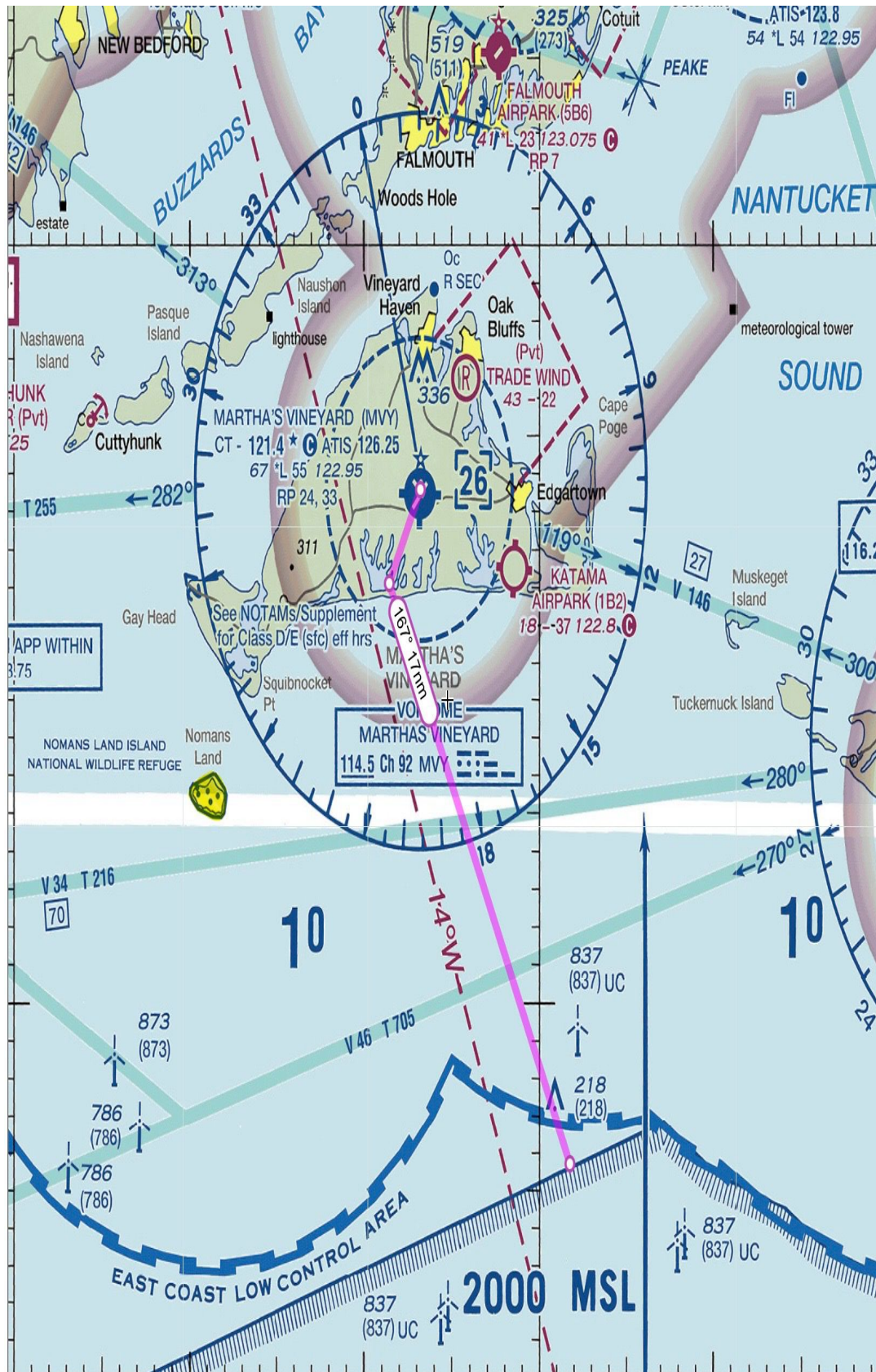
Additional Information, Frequencies, & Charts

Also available are [International VHF Maritime Radio Channels and Frequencies](#), [Narrowband VHF Maritime Channels and Frequencies](#), [Radio Information for Boaters](#), and [U.S. Coast Guard VHF Distress and Safety Coverage Charts](#).



## 2.2.5 Bird Avoidance KMOV to Vineyard Wind





## 2.3 Sabine Pass

### 2.3.1 Base Standards

#### 2.3.1.1 Employee Expectations

- Crew members must stay within 20 minutes of the aircraft while on duty
- Crew members are expected to respond to a flight within 30 minutes of request from the customer

#### 2.3.1.2 Aircraft Expectations

- Primary aircraft will be ready to fly at all times
- Secondary aircraft, if available, will be ready for flight at all times
- This includes maintenance logs, preflight inspections, daily checks and fuel onboard

#### 2.3.1.3 Shift Change

- Crew members will conduct a formal shift change/pass down using the Sabine Shift Change Briefing form located in the Sabine Ops folder on Teams at the beginning of each duty period
- All crew members will be present for a shift change/pass down brief at the change of each duty period
- If crews hot-swap and a shift change brief is not conducted face-to-face, the outgoing PIC will debrief with the outgoing team and leave applicable pass down notes for the oncoming crew
- Day PIC will submit a daily report to the Sabine Pilots at shift start

#### 2.3.1.4 Duty Phone

- Duty phones will be handed off at each shift change
- Duty PIC is to keep duty phone on hand while on shift
- Do not "call forward" this phone to personal phones
- Monitor Sabine Pilots Teams chat

#### 2.3.1.5 Ground Handling Aircraft

- Aircraft handling will be the responsibility of the duty mechanic/hoist operator
- When towing the aircraft you must have wing walkers to insure clearance of obstacles

#### 2.3.1.6 Maintenance

- The Operator shall notify the Customer immediately of any unplanned Aircraft On Ground (AOG) occurrences and expected out of service (OOS) duration.
- Planned OOS events for the Primary Aircraft shall provide notices of not less than 72 hours and planned back-up support plan provided at that time. When possible, the Operator will work with



the Customer to conduct scheduled maintenance during times least intrusive to the Customers operation.

Routine maintenance shall be scheduled, and notification made to the Customer, at least 72 hours in advance

#### **2.3.1.7 Manifest Instructions**

- Flight legs shall be logged from landing to landing, i.e. KBPT-00TA with ship name noted in the comment line. Harbor pilot(s) on/off deck time should be noted by non-flying pilot and reported to Sabine Operations
- Both pilots should confirm accuracy of each manifest leg prior to closing manifests
- When conducting a flight with a different customer, i.e. maintenance/training, close the current manifest and start a new manifest

#### **2.3.1.8 DFL Instructions**

- Prior to submitting a DFL, ALL crew members will review the DFL for accuracy by using the DFL Checklist in the Teams Sabine Ops folder

#### **2.3.1.9 Fueling**

##### **KBPT**

- Fuel can be requested by calling 409-719-4950
- Aircraft must be shut down to APU mode or both engines off
- Place fuel receipt in the basket on PIC desk for collection by KUSA

##### **00TA**

- Fuel truck located at Sabine pilot office
- Conduct daily fuel sump and record on log located in truck
- After fueling, record the amount taken into the fuel log

#### **2.3.1.10 Contacts List**

- A list of contacts can be found in the Sabine Ops Channel located in the Teams App

#### **2.3.1.11 Day PIC Duties**

- Submit a daily status report to the customers. The report file can be found in the Sabine Ops Channel in the Teams App
- Conduct daily fuel sump and record on log located in fuel truck
- Submit a fuel order request for the fuel truck when lower then 1000 gallons

#### **2.3.1.12 Night PIC Duties**



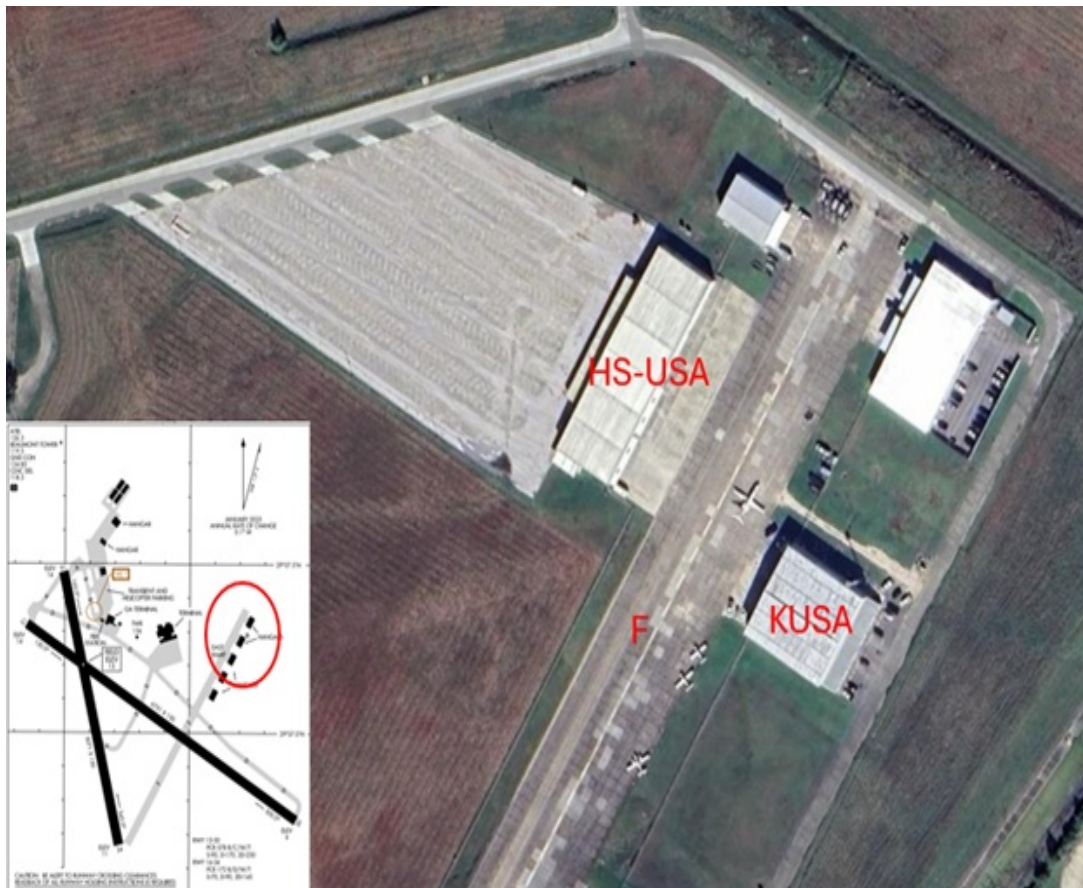
- Complete daily flight checks on duty helicopter

## 2.3.2 Local Area

### 2.3.2.1 Airports

KBPT - Jack Brooks Regional/Beaumont

- - TOWER - 119.3
  - GND CON - 124.85
  - CLNC DEL - 118.3
  - ATIS - 126.3



Exercise caution when transitioning on taxiway Foxtrot with parked aircraft. If able, ground taxi past parked aircraft in the vicinity of the KUSA hangar.

KORG - Orange County

- CTAF - 122.8
- AWOS - 118.975

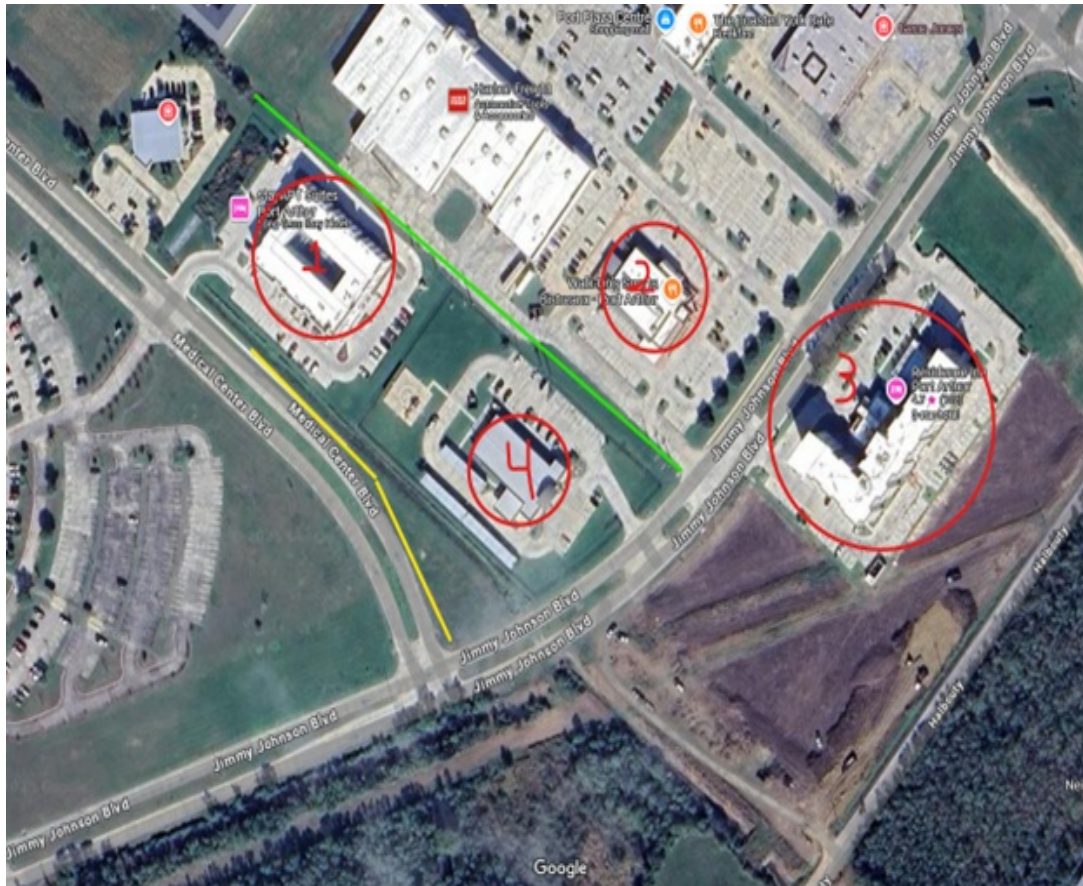
KBMT - Beaumont Municipal

- - CTAF - 123.0



- AWOS - 118.425

### 2.3.2.2 Sabine Pilot Helipad (00TA)



1. StayAPT Suites – 3 story building
2. Walk-Ons Sports Bar – outside seating area – Busy
3. Residence Inn Port Arthur – 5 story building? - Busy
4. Sabine Pilots Office – 2 Story building
5. Green Line – Power lines
6. Yellow Line – Street lamp poles

#### Sabine Helipad Safety

- Customers do not approach without approval from flight crew
- 15ft Hover Height is recommended for any pedal turns at the Heliport for obstacle clearance.
- Pilot and HHO will conduct a safety brief with harbor pilots prior to each flight to confirm ship, review safety procedures, and discuss any areas of concern
- FOD check of the helipad will be completed before take-off
- Start cart must be plugged in while not in use.
- Install sunshade/nose covers during hot weather when the aircraft will be shut down for an extended period



### 2.3.2.3 Helipad Limitations

- No pad operations with **gust spread greater than 25 kts**.
- No pad operations with **temperatures above 40°C**.
- No pad operations with winds greater than **30 kts from 290°–360°**.
- No pad operations with **aircraft gross weight exceeding 4,600 kg** when temperatures are above 30°C.
- No takeoff or landing with **crosswind greater than 20 kts** (CAT A limit).

### 2.3.2.4 Hazardous Areas



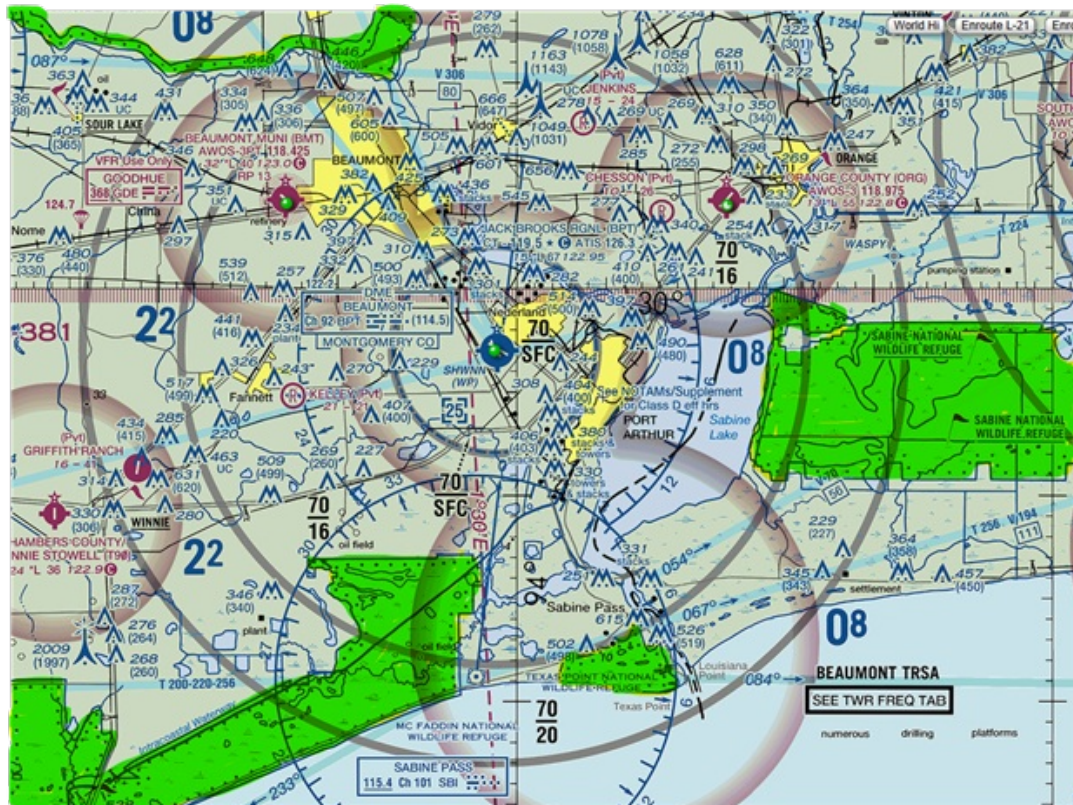
1. Heavy use hunting area (year around)
2. Motiva Oil Depot
3. Motiva & Valero Refinery
4. Golden Pass LNG & Cheniere LNG Refinery
5. Sunoco Logistics Refinery
6. Motiva Refinery
7. Total Refinery
8. Duck Hunting Area
9. Blue box - KBPT Airport
10. Green Box - Sabine Pilot Helipad (00TA)
11. Yellow Box – Sabine Pilot Boat Station and Helipad



### 2.3.2.5 Noise Abatement

- Maintain at least 1000ft AGL over urban areas
- Avoid over flying houses east of taxiway Foxtrot and north of taxiway Delta

### 2.3.2.6 Bird Areas (Sanctuary)



- Pilots will avoid flying over National Wildlife Refuges.
- If needed to fly near or over a Wildlife Refuge, follow HSUSA's company Bird Avoidance Policy OM-C Section 1.1.3.
- Known Wildlife Refuges in the operational area.
  - Sabine National Wildlife Refuge
  - Texas Point National Wildlife Refuge
  - McFaddin National Wildlife Refuge
  - Big Thicket National Preserve

## 2.3.3 Pre-Flight operations

### 2.3.3.1 Weather

- Flights with forecasted or known lighting within 10 miles of the route to be flown should take with extreme caution.

Links to weather tracking resources



- Flight Service Weather
- Windy
- Lightning Tracker
- Coastaware

#### 2.3.3.1.1 VFR Weather minimums

- Day VFR
  - 600 ft ceiling – 2 mile visibility
- Night VFR
  - 800 ft ceilings – 3 mile visibility

#### 2.3.3.2 Weight and Ballance

- Pilots will compute a weight and balance using the appropriate performance data found in the AW169 RFM prior to each flight

#### 2.3.3.3 Take-Off Times

- **Outbound Traffic** - Depart airport when outbound vessels are at buoy's 13/14, reference SeaIQ
- **Inbound Traffic** - Depart airport 25 minutes before Sabine pilot onboard time located on the Sabine Pilot Dispatch portal for inbound vessels
- **00TA Departure** - use times above
- **KBPT Departure** - Leave 10 minutes early for inbound traffic to pick up Sabine Pilots from 00TA



| Outbound Transit Times    |                    |       | Outbound Transit Times   |                    |       |
|---------------------------|--------------------|-------|--------------------------|--------------------|-------|
| Terminal                  | Transit Time (Hrs) | Miles | Terminal                 | Transit Time (Hrs) | Miles |
| Below Texaco Is           |                    |       | Motiva PN #2             | 5.25               | 45.8  |
| Port City Partners        | 2                  | 22    | TPC                      | 4.75               | 46    |
| Sabine Pass LNG West      | 2.5                | 23    | Phillips 66              |                    |       |
| Sabine Pass LNG East      | 2.5                | 23    | P66 #1                   | 6                  | 48.5  |
| Sabine Pass LNG #3        | 2.5                | 23    | P66 #5                   | 6                  | 48.5  |
| Golden Pass LNG #1        | 3                  | 27    | P66 #2                   | 6                  | 48.5  |
| Golden Pass LNG #2        | 3                  | 27    | P66 #4                   | 6                  | 48.5  |
| Port Arthur LNG #1        | 3.25               | 28.8  | Sun Oil                  |                    |       |
| Port Arthur LNG #2        | 3.25               | 28.8  | Sun #1                   | 6                  | 48.8  |
| Valero Pleasure Island    | 3.25               | 30.5  | Sun #2                   | 6                  | 48.8  |
| Texaco Is Intersection    | 3.25               | 30.5  | Sun #3                   | 6                  | 48.8  |
| Behind Texaco Is          |                    |       | Sun #4                   | 6.5                | 49.1  |
| Motiva PA #2              | 3.5                | 30.6  | Sun #5                   | 6.5                | 49.1  |
| Motiva PA #6              | 3.5                | 30.6  | Sun #6                   | 6.5                | 49.1  |
| Motiva PA #7              | 3.5                | 30.6  | Lower Sun Anchorage      | 6.5                | 49.1  |
| Oxbow Load                | 3.5                | 31    | Upper Sun Anchorage      | 6.5                | 49.1  |
| Oxbow Discharge           | 3.5                | 31    | OCI (Dupont/Lucite)      | 6.5                | 51    |
| GT Logistics #1           | 3.5                | 31.1  | Enterprise               |                    |       |
| GT Logistics #2           | 3.5                | 31.1  | BMT #1                   | 6.75               | 51.7  |
| Valero #1                 | 4                  | 31.3  | BMT #2                   | 6.75               | 51.7  |
| Valero #3                 | 4                  | 31.3  | BMT North                | 6.75               | 51.7  |
| Valero #5                 | 4                  | 31.3  | BMT South                | 6.75               | 51.7  |
| Port of Port Arthur       |                    |       | BMT East                 | 6.75               | 51.7  |
| PoPA #1                   | 3.5                | 33.2  | BMT Ethane               | 6.75               | 51.7  |
| PoPA #2                   | 3.5                | 33.2  | Martin Gas Sulphur Basin | 6.75               | 52.5  |
| PoPA #3                   | 3.5                | 33.2  | Neches Industrial Park   | 6.75               | 53    |
| PoPA #4                   | 3.5                | 33.2  | ExxonMobil               |                    |       |
| PoPA #5                   | 3.5                | 33.2  | XOM #4                   | 7                  | 55.5  |
| PoPA #6                   | 3.5                | 33.2  | XOM #5                   | 7                  | 55.5  |
| Port Arthur Turning Basin | 3.5                | 33.2  | XOM Coke/BLADE           | 7                  | 55.5  |
| Gulf Copper               | 4                  | 37.5  | XOM Chemical             | 7                  | 55.5  |
| Pabtex                    | 4.25               | 38.5  | Port of Beaumont         |                    |       |
| Hydra/Old River           | 4.25               | 39    | Carrol St Wharf          | 7                  | 56.5  |
| Neches River              | 4.25               | 39    | Orange County Wharf #1   | 7                  | 56.5  |
| Total                     |                    |       | Orange County Wharf #2   | 7                  | 56.5  |
| Total #2 Crude Dock       | 5                  | 42.5  | Harbor Is A              | 7                  | 56.5  |
| Rainbow Bulk              | 5                  | 43    | Harbor Is B              | 7                  | 56.5  |
| Lower Fina Anchorage      | 5                  | 44    | Harbor Is C              | 7                  | 56.5  |
| Upper Fina Anchorage      | 4.5                | 45    | PoB #4                   | 7                  | 56.5  |
| Huntsman                  | 4.5                | 45.6  | PoB #5                   | 7                  | 56.5  |
| Motiva PN                 |                    |       | PoB #7                   | 7                  | 56.5  |
| Motiva PN #1              | 5.25               | 45.8  |                          |                    |       |

### 2.3.3.4 Risk Assessment

- Pilots shall complete risk assessment located on manifest in Complete Flight prior to flight

### 2.3.3.5 Vessel Information

- All crew members shall gather information on vessels where operations will be conducted
  - Locate viable landing or hoisting areas
  - Identify potential hazards
  - Length and width of vessel
- Information can be found on the following sites
  - Vessel Finder
  - Marine Traffic
- 

## 2.3.4 In-Flight Operations

### 2.3.4.1 Routes

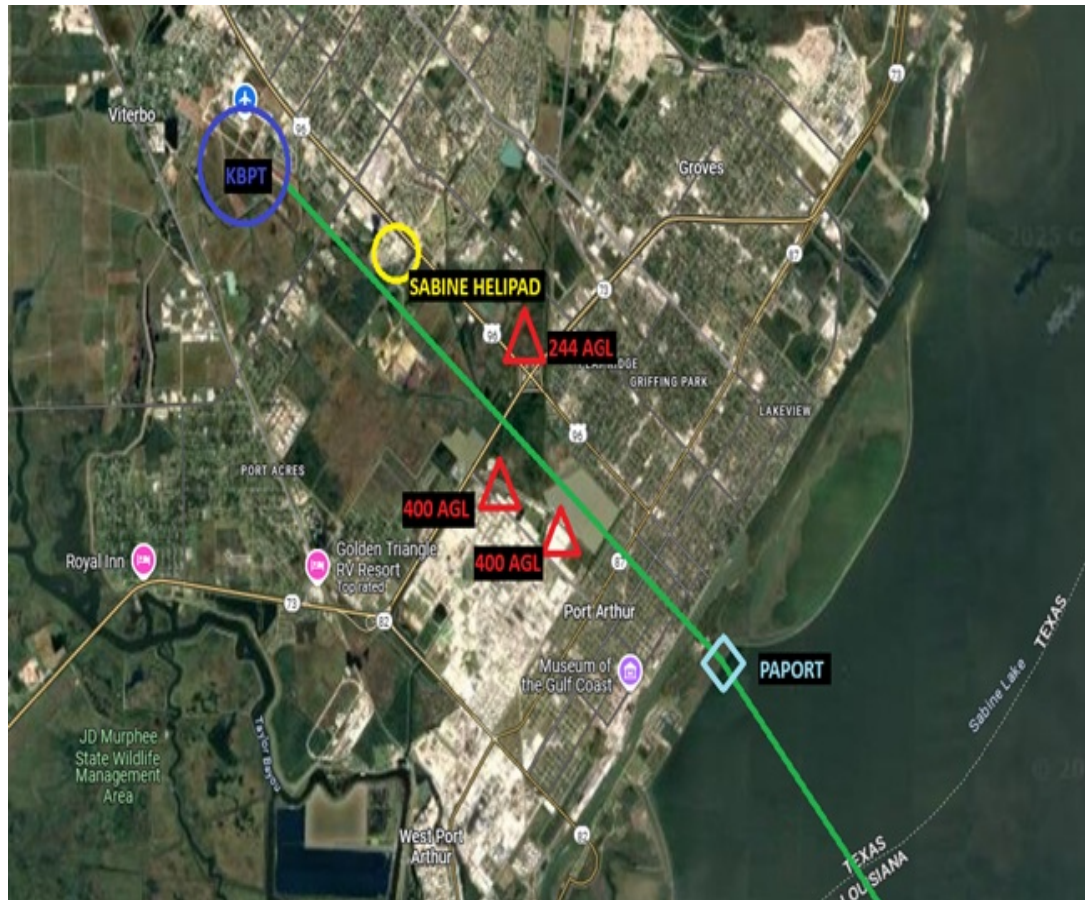


### 2.3.4.1.1 VFR Routes

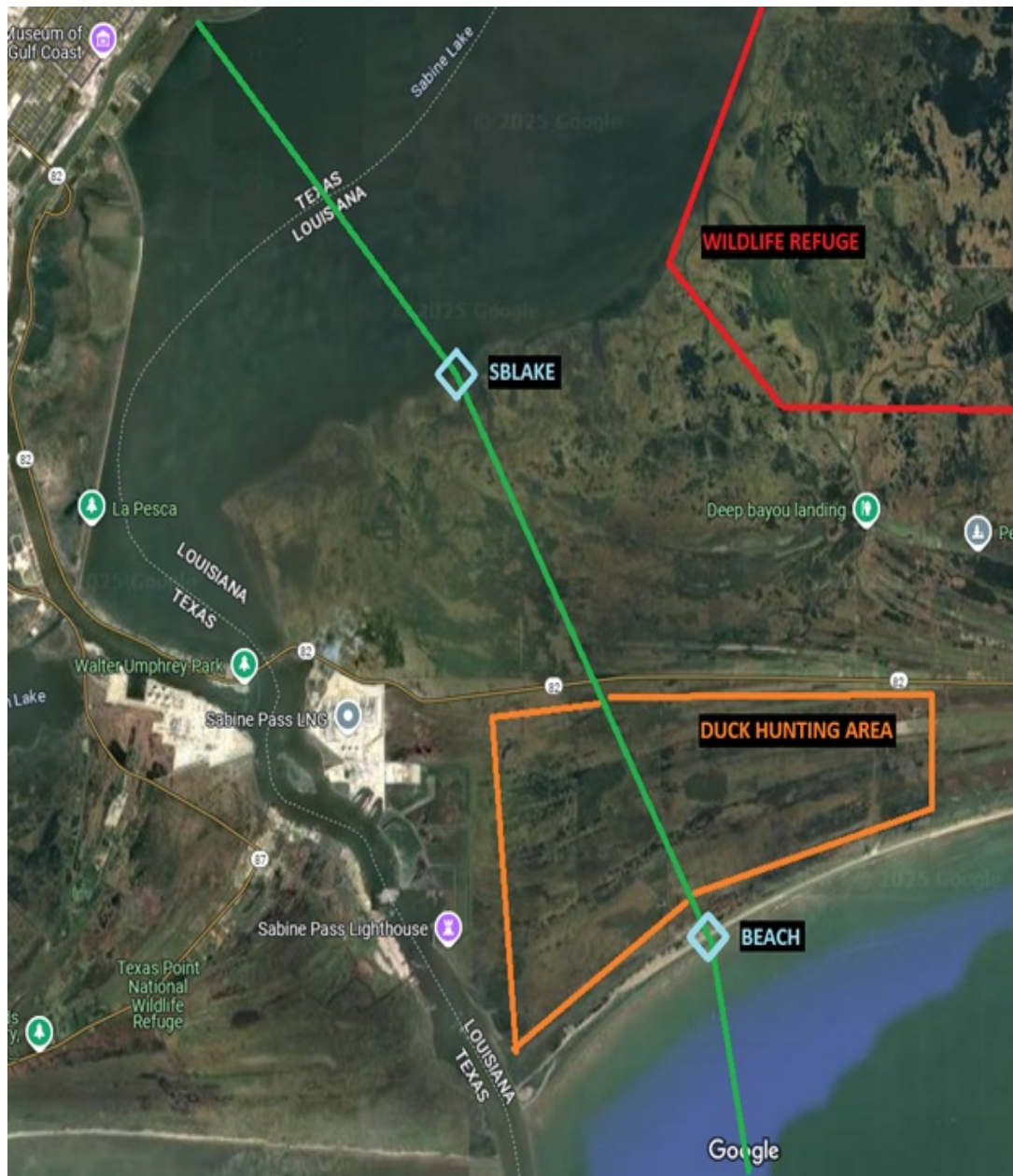


- Navigation Points
  - KBPT
  - 1 – PAPORT (N29.52.02, W093.54.88)
  - 2 – SBLAKE (N29.49.07, W093.52.28)
  - 3 – BEACH (N29.42.85, W093.48.95)
  - 4 – SBBUOY (N29.25.00, W093.40.00)
- Standard Route
  - KBPT-PAPORT-SBLAKE-BEACH-SBBUOY-BEACH-SBLAKE-PAPORT-KBPT
  - This route is saved as SBBUOY in the navigation page of EDCU





- Depart KBPT or 00TA, climb to a minimum of 1000ft to PAPORT waypoint. As conditions allow, maintain a minimum of 1000ft over the city of Port Arthur for noise abatement.



- From PAPORT fly over the Sabine Lake to SBLAKE waypoint. Maintain a minimum of 1000ft to BEACH waypoint. Be cautious of ducks in this area during hunting seasons (October-January).
- From BEACH continue to SBBUOY at a minimum of 1000ft.

#### 2.3.4.1.2 IFR Routes

Outbound IFR routes may be filed direct to the applicable HEDA location.

HEDA approaches may be conducted at the following locations:

SPEAST: N29 30.00 W093 35.00

SPWEST: N29 24.00 W093 47.00



Inbound IFR flights should be filed direct to KBPT from SBBUOY: N29 25.00 W093 40.00

#### 2.3.4.2 Communications

All communications go through marine channel 09. Utilizing ROIP systems at the hangar, the Sabine helicopter Office and the two lead pilot laptops. With ROIP Quonset dispatchers monitor the marine channel 09 from dispatch. All marine transfer operations and communications take place on channel 9. HeliService Ops, Sabine Dispatch, Helicopter crew, Sabine boat crew.

Pre-Flight communications may be on channel 09 or by phone.

In-Flight communication will be on channel 09.

- 2 hours prior to flight call Sabine Boat Ops:
  - Brief weather conditions.
  - Brief ship maneuvering and placement for optimal pickup performance.
  - Will pilots be flown from/to KBPT/00TA.
  - Discuss Go/No-Go for helicopter ops:
    - If No-Go is planned/possible, notify Sabine Dispatch to get in contact with Sabine Pilot assigned to vessel.
    - Confirm with assigned Sabine Pilot if they would like to commit to the boat transfer or wait for a last call from Helicopter.
- 1 Hour 15 minutes to flight:
  - Last call for weather cancels to not impede vessel traffic.
  - Notify Sabine Boat Ops/Sabine Dispatch of decision.
- 30 minutes prior to flight:
  - Notify Sabine Boat ops prior to run up.
    - Requested inbound vessel heading and location for pilot drop off.
  - Notify outbound vessel with pick up information.
    - Requested hoist location.
    - Requested vessel heading for performance.
  - After departure:
    - Report "OFF DECK" to HeliService Ops/Sabine Dispatch:
      - Off deck from KBPT/00TA
      - Souls on board
      - Fuel Endurance
      - Number of vessels to visit
    - If no departure i.e. maintenance or weather issue notify Boat Ops/Outbound Vessel immediately
  - BEACH waypoint outbound:
    - Report "BEACH OUT" to HeliService Ops/Sabine Dispatch.
    - Contact Vessels for final direction if needed.
      - Ships heading for performance.



- Location of hoisting position.
- ETA.
- **On Approach to Vessel:**
  - Report "**OPS NORMAL APPROACH #**" to HeliService Ops/Sabine Dispatch
- **After Hoist Complete:**
  - Report "**Pilot off/on [time]**" to HeliService Ops/Sabine Dispatch
- Next vessel if applicable:
  - Contact vessel for setup information
  - Follow steps for "**On Approach to Vessel**" and "**After Hoist Complete**"
- Return KBPT/TA00
- BEACH Navigation Point Inbound:
  - Report "**BEACH IN**" to HeliService ops/Sabine Dispatch
- Before Landing KBPT/00TA:
  - Report "**LANDING ASSURED**" to HeliService Ops/Sabine Dispatch

#### Frequencies to Monitor:

Channel 09 Ship Operations Communications.

- HeliService USA Dispatch (ROIP)
- Sabine HeliService Ops
- Sabine Boat Ops
- Sabine Dispatch

Channel 16 Emergency Marine Traffic Channel.

- All marine traffic
- Possible man overboard calls

Harbor Pilots may also monitor Channel 14 while transitioning to/from port.

Aviation Channel 130.325.

- HeliService USA Ops Dispatch (ROIP)
- Sabine HeliService Ops

#### 2.3.4.3 Approach to Vessel

- 10 miles out descend to 300 RHT
- Approximately 5 minutes from vessel, conduct Before Hoist Checklist
- 3 miles out with ship in sight, descend to 200 RHT
- 2 miles out, begin decelerating to 50 GSPD
- Make a normal descent at a 45-degree relative angle to the vessel into the intended hoisting



position

- Hoist at 30 RHT above intended hoisting position, or at a suitable height to maintain visual hover references and obstacle avoidance. While the PIC is the final authority, all crew members should be in mutual agreement on hoist location and height prior to commencing hoist operations
- Complete hoist

#### 2.3.4.4 Hoisting

##### Standard Hoisting Locations

- Port side or starboard side mid-ship hoist shall be standard if vessel configuration allows.
- Ship hoist or landing areas will also be used if the pilot has adequate visual reference to complete a safe hoist.
- Bridge wings are allowed but will be on a case-by-case basis. Bridge wings that meet the following criteria will not be authorized:
  - Bridge wings with large antennas, flags, radar equipment, or any other dangerous items located beyond the structure of the bridge house. Care should be taken to avoid excessive downwash on sensitive ship equipment
  - Cages/awnings located on the bridge wings.

##### Hoisting Location Hazards

- Loose FOD on ship main deck
- Ship cranes, stanchions
- Guy-wires/guy-lines

#### 2.3.4.5 Departure from Vessel

- Hoist complete.
- Move left away from vessel at 90 degree angle.
- Ascend to 100-150 RHT.
- Complete end of hoist checks.
- Make standard ITO (2.5°-5° nose down and 95% PI) away from vessel.
- Climb to cruising altitude, continue navigation to next intended waypoint.

### 2.3.5 Post Flight Operations

#### 2.3.5.1 Manifest

- Submit completed manifest in Complete Flight at end of shift
- When changing customers, submit current manifest and open a new manifest
- Complete manifest using reference material located in the Ops Training folder under Sabine Ops Channel in Teams



### 2.3.5.2 Digital Flight log

- Submit a Digital Flight Log in Digital AirWare at the end of each shift or upon request from maintenance personnel
- Complete DFL using the reference material located in the Ops Training folder under Sabine Ops channel in Teams

## 2.4 Premier Health

### 2.4.1 Introduction CareFlight Program

These procedures sets requirements for Pilots assigned to the Premier Health CareFlight program, that is not covered by other company publications. The intent of this document is to give a logical sequence of a pilot's daily duties and specific information for the Premier Health, CareFlight Program.

### 2.4.2 Pre-Mission

The following section contains information and task required prior to accepting a flight.

#### 2.4.2.1 Base Duty times

The following are the Pilot shift times for each base. CareFlight 3 is intentionally staggered to avoid all three bases shift changing at the same time.

Location Pilot shift times:

| Location                                      | Pilot Shift Times                         |
|-----------------------------------------------|-------------------------------------------|
| CareFlight 1 (CF1) Reserved                   | Reserved                                  |
| CareFlight 2(CF2) Warren County Airport (I68) | Day Shift 0700-1900 Night Shift 1900-0700 |
| CareFlight 3(CF3) Grimes Field (I74)          | Day Shift 0600-1800 Night Shift 1800-0600 |
| CareFlight 4(CF4) Darke County Airport (KVES) | Day Shift 0700-1900 Night Shift 1900-0700 |

#### 2.4.2.2 Pre-Flight / Aircraft Logbook

Heliservice USA (HSUSA) uses a digital logbook. Each aircraft will have one iPad Mini and one 11-inch iPad. Digital Airware <https://lb.digitalairware.com/DAW20/> may be accessed from either iPad. Each



aircraft will have a binder containing fuel cards and weight & balance forms.

After their Pre-Flight the Day Pilot or Mechanic will open a Digital Logbook and sign off the PreFlight. The Night Pilot at the end of their shift will close the Digital Logbook.

At the end of every Pilot's shift, if an engine start or flight took place. The DFL will be generated and closed via Digital Airware.

#### **2.4.2.3 Full Aircraft Run up**

The aircraft will be started every morning using a battery cart. All systems will be powered up and tested IAW the QRH, RFM or local checklist. This will be accomplished by the day pilot. This will allow for all subsequent starts to be done using the "Quick Start" procedures.

#### **2.4.2.4 Complete Flight**

Prior to the first flight of the day and any subsequent flights. A manifest must be prefilled with the pilot, current crew and fuel load. A Pre-flight risk assessment (PFRA) will also be filled out and adjusted accordingly based off the flight request.

#### **2.4.2.5 Medical Crew Briefing**

Prior to the first flight of the day, the PIC will conduct a thorough crew briefing using the 24hr shift change briefing sheet located in the base product folder on Microsoft teams.

#### **2.4.2.6 Base Lead Daily Briefing**

Every day at 0730 and 1930 the Base Lead or Manager on Call will conduct a briefing via phone and Microsoft Teams. During the briefing the PIC from each base will brief the following:

- Call Sign
- Tail Number
- Crew for the day
- Mechanic on call
- Maintenance HRs until next inspection
- Number of flights in the last 12hrs
- Forecasted weather for the day

#### **2.4.2.7 Activated for a flight**

Upon being activated the PIC will determine if the flight can be completed within their remaining duty day, perform a final check of WX, determine the highest Obstacle along the route and sign the PFRA. If the pilot does not have enough duty day remaining they must immediately inform the COM center so another aircraft can be dispatched.



If the flight request happens during the Medical Crew shift change. It will be at the discretion of the PIC to take the current Medical Crew on duty or allow for the oncoming crew to take the flight, only if this does not cause a delay in taking the flight.

Prior to starting the engines, the Pilot must use the "Pre-Start Confirmation check" located in Appendix A. This checklist should be placed in the aircraft and visible to the pilot or a digital copy may be read from the iPad.

#### **2.4.2.8 Prior to lifting**

Once the aircraft is started and ready for take-off, the Pre-takeoff Confirmation Check located in Appendix A, must be verbalized to the crew and include the phrase "Crew, patient, mission equipment secured?" The Med crew must respond with they are secure and ready for takeoff. After the above is completed a call to the COM center must be made prior to lift for the initial and all subsequent legs. The call must include: Call sign, SOB, risk score, fuel in hours and minutes. Example: "Carecom, CF3, three SOB, risk of 15.2+30 on the fuel. Departing"

### **2.4.3 During the Mission**

During the mission, the crew will adhere to the following procedures.

#### **2.4.3.1 Communications**

All aircraft in the CareFlight program will use the following call signs and transponder codes for each base:

These call signs may be utilized when obtaining an IFR Clearance. However, ATC will issue a discrete transponder code.

| Location      | Call Sign          | Transponder Code |
|---------------|--------------------|------------------|
| Warren County | CareFlight 2 (CF2) | 5342             |
| Urbana        | CareFlight 3 (CF3) | 5343             |
| Versailles    | CareFlight 4 (CF4) | 5344             |

#### **2.4.3.2 Weather Monitoring**

During the flight, the PIC will continuously monitor weather using all available resources and react accordingly if there is any change in weather. Such as return, divert, land, or go IFR.

#### **2.4.3.3 Deteriorating Weather Conditions Enroute**



If, while conducting VFR operations, deteriorating weather conditions are encountered, one of the following procedures is to be followed.

- Divert to an alternate airport, heliport, or other suitable area where the patients can be transferred to other means of transportation, if applicable.
- Return to the departure point, if practical.
- Land and notify dispatch or hospital personnel of the situation, and make arrangements for care of the patients, if applicable.
- If weather conditions and regulations permit, continue the flight under IFR.
- Execute the IIMC procedure in Section 4.3 Inadvertent Instrument Meteorological Conditions (IIMC) Recovery, if the preceding options are not available.

#### **2.4.3.4 Before Landing**

Prior to landing the PIC will conduct a before landing check to include verbalizing the following: "Gear down, three in the green" followed by "Crew, patient, mission equipment check" The Med crew must respond with if they are secure and ready for landing. The PIC will also let the COM center know they are "Three in the green"

#### **2.4.4 Post Mission**

The following actions will be completed after every flight.

##### **2.4.4.1 Post flight Debrief**

The PIC will debrief the flight with the crew, giving everyone an opportunity to share any observations, insights or lessons learned. Additionally, the pilot will assist the crew with any information needed for the Post-transport debrief in Protean.

##### **2.4.4.2 Closeout the manifest**

After every completed mission, the PIC will close the current manifest out and create a new one IAW Paragraph 3.1.4 of this SOP.

##### **2.4.4.3 Create and Close a DFL**

After every flight or engine start the PIC will create and close a DFL.

#### **2.4.5 Refuel**

In addition to the requirements in the OM-A, RFM and OPSPEC. The following additional requirements apply to the CareFlight program.

##### **2.4.5.1 Miami Valley Hospital (MVH) Fuel Point**



Whenever practical, Premier aircraft will utilize the fuel point located on the South pad. Prior to getting fuel the PIC will confirm with the COM center that fuel is in service. The first PIC to land after 0001, will conduct the daily fuel point inspection utilizing the Fuel Storage Tank Checks form found in Appendix A of this document.

#### **2.4.5.2 MVH Fuel Credit Card**

When getting fuel at other airports, use the Premier fuel card. When using the card ensure you inform the FBO of the tax-free status. Ensure you keep all fuel receipts. The Base Leads will send all fuel receipts to Kristen Wright [krwright@PremierHealth.com](mailto:krwright@PremierHealth.com)

#### **2.4.5.3 KVES Fuel Point**

KVES has a fuel point located on the parking pad. Directions for operating the point are located inside the hangar where the power switches are. After refueling, the amount of fuel in gallons must be logged on the fuel log sheet provided by Midmark. The fuel point is maintained by Midmark. Any issues with the fuel point should be directed to the airport manager, Scott Naas (937)467-9360.

### **2.4.6 Quarterly IIMC Recovery Drills**

All pilots assigned to the CareFlight program are required to complete quarterly practice of an Inadvertent Instrument meteorological Conditions (IIMC) recovery with an emphasis on AMRM.

The drill will be a team-oriented exercise that will include a preflight briefing as well as a training flight with a full instrument approach recovery. IIMC Recovery Template found in Appendix A should be used.

The use of a view limiting device is not authorized for the purposes of this section.

When possible, the drill will take place on the return to base leg after the completion of a patient transport or reposition flight. Some base locations may not present this opportunity due to airport locations and instrument approach availability. For these bases it will be necessary to initiate a flight for the sole purpose of flying to the nearest location where a practice instrument approach procedure can be performed. In all circumstances this will be accomplished while remaining in-service and available for any patient transport request. Pilots will contact the Program Lead Pilot prior to departing on a training flight.

It is important that there be an equal amount of emphasis placed on avoiding IIMC as well as the recovery. Each recovery scenario and practice approach shall be coupled with a discussion around the recognition and avoidance of the conditions that may lead to such an encounter.



## 2.4.7 Additional Training

At any time, a pilot may request additional training. This request shall be coordinated through the Program Lead Pilot.

If a PIC has not flown in the last 60 days, they will, as a minimum, accomplish one start and three takeoffs and landings. If a PIC has not flown in the last 90 days, they will consult with aviation management to determine if any training is needed for the pilot to perform competently and safely in that aircraft. It is essential for the pilot to be knowledgeable of aircraft lighting and proficient with avionics and aircraft systems. This flight shall be coordinated through the Program Lead Pilot

A PIC who has not flown over a route and into an airport or heliport within the preceding 90 days will:

1. Study the appropriate IFR enroute or VFR aeronautical charts.
2. Study the destination airport and heliport diagrams, including alternate airports and heliports.
3. Study the appropriate IFR approach charts for destination and alternate, if applicable.

## 2.4.8 Miscellaneous

This section contains various Items required for the Program

### 2.4.8.1 Adverse Weather Plan

Appendix A of this document contains the Adverse Weather Plan for each base. The Plan must be posted somewhere to be visible at all times. The base Lead will review and periodically update as needed.

### 2.4.8.2 IIMC Recovery Template

Appendix A of this document has the IIMC Recovery Plan for each base. The Plan should be posted on the Hazard Map and a copy should always be in the aircraft, in an area accessible to the crew. The base Lead will review Quarterly, annotate the date reviewed and update as needed.

### 2.4.8.3 Inadvertent Instrument Meteorological Conditions (IIMC) Recovery

This procedure will be posted at the respective base and available to the pilot in the aircraft. The pilot shall become knowledgeable about the procedure. The procedure will include at least the following information.

- The priority and most important aspect of an IIMC procedure is to maintain aircraft control. Positive control of the aircraft with reference to instruments must take priority over all other duties.
- Attitude (level)
- Heading (turn only to avoid known obstacles)
- Power (adjust to climb power)
- Airspeed (adjust to climb airspeed)



- Climb to the minimum safe altitude to clear all obstacles in the area.
- De-goggle when safely able (if appropriate).
- Make all turns no greater than standard rate.
- Contact appropriate Approach Control/Center and declare an emergency. Squawk 7700 when able. Report location, altitude, heading, fuel status, and number of persons on board. State that you are inadvertent IMC and request assistance.

In order of preference below, the pilot shall inform ATC as to how they wish to proceed:

1. Radar vectors to VMC conditions
2. Vectors to the closest approach

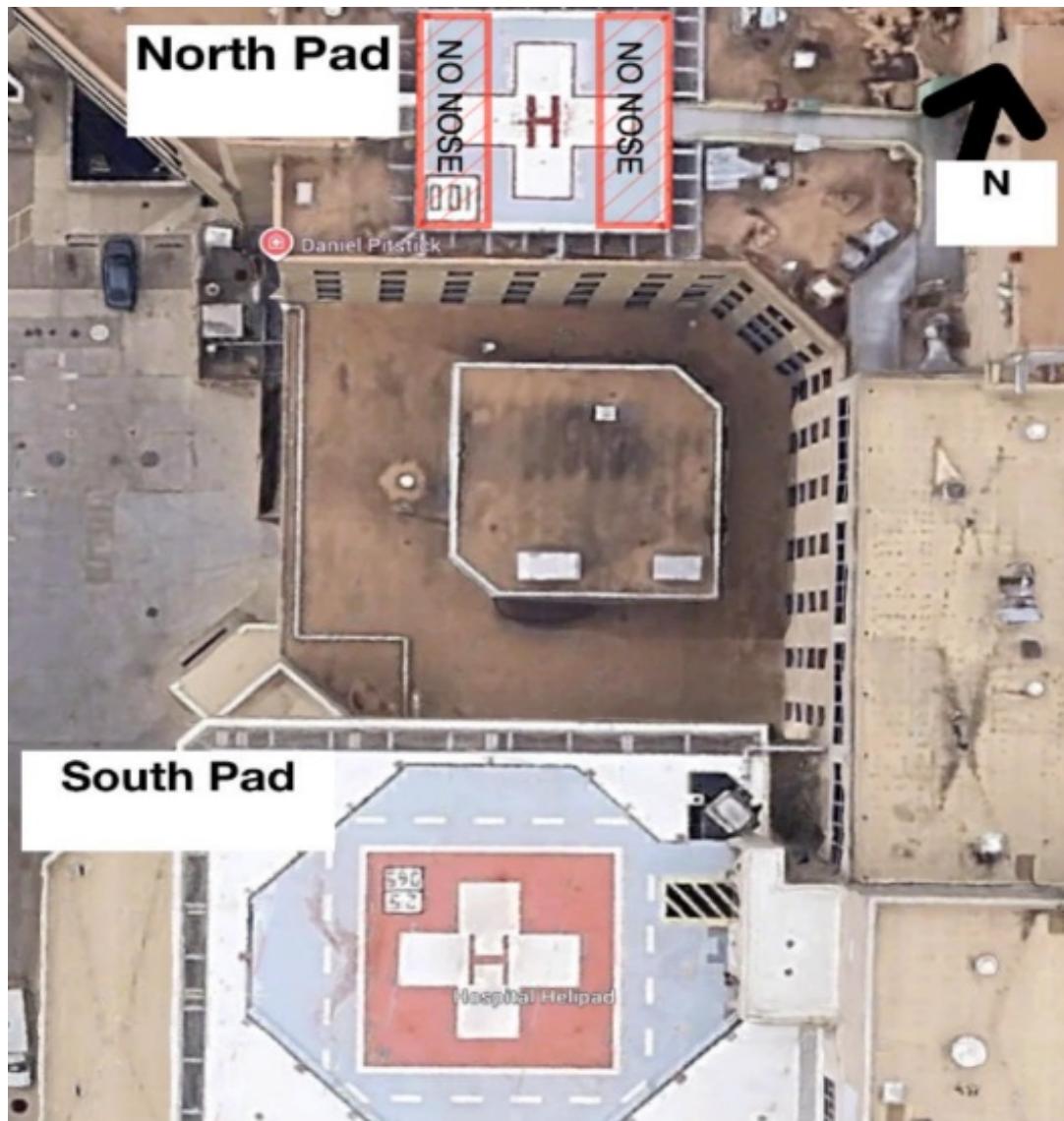
Complete the approach procedure to landing or entering VMC conditions. Maintain VMC, and land

## **2.4.9 Hospital Helipads**

### **2.4.9.1 Miami Valley North Pad**

The South pad has a MGWT of 25,000lbs The North pad has a MGWT of 10,000lbs and is significantly smaller than the south pad. As such the North pad has the below landing restrictions.





The North Pads MGWT is 10,000LBS or 4535KGs.

When utilizing the North Pad Aircraft, **Max allowable GW is 4500KG.**

All Aircraft must land with their nose pointing directly to the north or south, with the aircraft centered on the pad as much as possible.



## 2.4.10 Night Vision Goggle (NVG) Maintenance and Inspection Program

Base leads will be responsible for managing and maintaining the NVGs assigned to their base. NVGs are required to be inspected every 180 days. Each base will have a minimum of two sets on hand. This allows for Base leads to rotate their NVGs accordingly. Base leads will keep the Careflight NVG Tracker spreadsheet located in the Premier Teams NVG INFO folder, up to date with inspection dates. Additionally, Base leads will update their assigned NVGs in Digital Airware (DAW)

NVGs will be shipped to NightFlight Concepts utilizing the NVG shipping form located in Appendix A and the Premier Teams NVG INFO folder:

<https://heliserv.sharepoint.com/:f:/r/sites/HeliServiceUSALLCPremier/Shared%20Documents/Premier/NVG%20INFO/csf=1&web=1&e=aYFeuy>

NVGs will be shipped Via UPS or FedEx using the following account numbers:

**UPS Account Number = E74B78**

**FedEx Account Number = 201 610 626**

Users / Contacts are Ken Porter



The above information along with links are also in the NVG INFO folder.

## **2.4.11 Maintenance**

### **2.4.11.1 Procedures for taking an Aircraft Out of Service (OOS) for Maintenance**

Whenever a Pilot or Mechanic takes the aircraft OOS for scheduled or unscheduled maintenance. The following procedures will be followed:

1. Call the CareFlight Communication Center at (937)208-3838. Let them know you are OOS and why. Give them an Estimated Return to Service time (ERTS).
2. Call the Mechanic on Call.
3. Send an OOS email using the established DISTRO or at a minimum email the COM Center, the CareFlight Base Leader for the day, and the Program Lead Pilot. The email should include the affected base, short explanation why and ERTS time.

### **2.4.11.2 Pilot servicing of Aircraft LOX Requirements**

HSUSA Pilots assigned to the CareFlight program are authorized to service the AW169 LOX system provided the following requirements are met:

1. The pilot has received "hands on" training with one of the mechanics on how to properly remove, fill and reinstall the LOX bottle.
2. Has watched the CareFlight LOX training video located in the Maintenance folder on the Premiere Share Point.
3. The pilot must have an entry in their formTF-07 stating they have been trained on LOX servicing prior to being authorized to service the LOX

## **2.4.12 Appendix A Sample Forms**

### **2.4.12.1 Fuel Storage Tank Checks**



#### 2.4.12.2 Adverse Weather Plans

#### 2.4.12.2.1 Lebanon

Purpose: To devise an Adverse Weather Plan for the Heliservice USA, Ohio Base located at the Warren County Airport, Lebanon, OH.

Actions by the Pilot-in-Command (PIC):

1. The on-duty PIC shall retain responsibility for maintaining an accurate and continuous watch on weather conditions. He or she shall exercise this policy and procedures when conditions warrant no later than 30-minutes prior to the onset of adverse weather conditions.
2. The PIC shall keep the CareFlight Program crew leader or manager-on-call, the communications specialists and the on-duty medical crew informed of adverse weather and potential relocation of the aircraft, if appropriate.
3. The Lebanon, Ohio aircraft shall be located on the helipad of base when adverse weather conditions do not warrant hangaring the aircraft.
4. The Lebanon, Ohio aircraft shall be moved into the hangar when adverse weather conditions warrant hangaring the aircraft. These conditions are listed in paragraphs 5b through 8 below.
5. Wind:
  - a. The aircraft shall have its Main Rotor Blades and Tail Rotor Blades (if applicable) tied down and the rotor brake engaged when: § at pilot discretion, the wind causes the main rotor blades to flex up and down in excess.
  - b. The aircraft shall be relocated no later than 30-minutes prior to the occurrence of winds forecasted to be in excess of 35 knots (40 MPH).
6. Precipitation: The aircraft shall be moved into the hangar no later than 30 - minutes prior to the occurrence of forecasted hail, freezing rain, or snow conditions.
7. Temperature: The aircraft shall be moved inside the Lebanon hangar when the outside air temperature drops below 32 degrees Fahrenheit (0 degrees Celsius) as measured at the I68 AWOS (513)934-5500.
8. Proximity/Movement: The aircraft shall be moved into the hangar as soon as possible when a weather system's movement or proximity of the hazardous weather requires an exercise of caution to protect the aircraft from potential damage.
9. The PIC shall pull the aircraft out of the hangar once the adverse weather conditions have diminished to the point that there is no foreseeable negative impact on the aircraft or flight operations.

#### 2.4.12.2.2 Urbana

Purpose: To devise an Adverse Weather Plan for the Heliservice USA, Ohio Base located at Grimes Field, Urbana, OH.

Actions by the Pilot-in-Command (PIC):



1. The on-duty PIC shall retain responsibility for maintaining an accurate and continuous watch on weather conditions. He or she shall exercise this policy and procedures when conditions warrant no later than 30-minutes prior to the onset of adverse weather conditions.
2. The PIC shall keep the CareFlight Program crew leader or manager-on-call, the communications specialists and the on-duty medical crew informed of adverse weather and potential relocation of the aircraft, if appropriate.
3. The Urbana, Ohio aircraft shall be located on the helipad of base when adverse weather conditions do not warrant hangaring the aircraft.
4. The Urbana, Ohio aircraft shall be moved into the hangar when adverse weather conditions warrant hangaring the aircraft. These conditions are listed in paragraphs 5b through 8 below.
5. Wind:
  - a. The aircraft shall have its Main Rotor Blades and Tail Rotor Blades (if applicable) tied down and the rotor brake engaged when: § at pilot discretion, the wind causes the main rotor blades to flex up and down in excess.
  - b. The aircraft shall be relocated no later than 30-minutes prior to the occurrence of winds forecasted to be in excess of 35 knots (40 MPH).
  - c. The Urbana Hangar door is limited to/rated at 26 knots (30 MPH)
6. Precipitation: The aircraft shall be moved into the hangar no later than 30- minutes prior to the occurrence of forecasted hail, freezing rain, or snow conditions.
7. Temperature: The aircraft shall be moved inside the Urbana hangar when the outside air temperature drops below 32 degrees Fahrenheit (0 degrees Celsius) as measured at the I74 AWOS (937) 484-5863.
8. Proximity/Movement: The aircraft shall be moved into the hangar as soon as possible when a weather system's movement or proximity of the hazardous weather requires an exercise of caution to protect the aircraft from potential damage.
9. The PIC shall pull the aircraft out of the hangar once the adverse weather conditions have diminished to the point that there is no foreseeable negative impact on the aircraft or flight operations.

#### 2.4.12.2.3 Versailles

Purpose: To devise an Adverse Weather Plan for the Heliservice USA-Versailles, Ohio Base (VEROH) located at the Darke County Airport (VES).

Actions by the Pilot-in-Command (PIC):

1. The on-duty PIC shall retain responsibility for maintaining an accurate and continuous watch on weather conditions. He or she shall exercise this policy and procedures when conditions warrant prior to the onset of adverse weather conditions.
2. The PIC shall keep the CareFlight Program crew leader or manager-on-call, the communications specialists and the on-duty medical crew informed of adverse weather and potential hangaring of the aircraft, if appropriate.
3. The Versailles, Ohio aircraft shall be located on the CareFlight helipad at VES



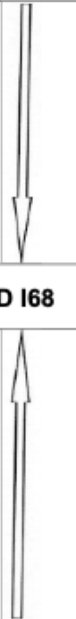
when adverse weather conditions do not warrant hangaring the aircraft.

4. The Versailles, Ohio aircraft shall be relocated to the CareFlight Hangar when adverse weather conditions warrant hangaring the aircraft. These conditions are listed in paragraphs 5b through 8 below.
5. Wind:
  - a. The aircraft shall be hangared when:
    - i. at pilot discretion, the wind causes the main rotor blades to flex up and down in excess.
  - b. The aircraft shall be hangared prior to the occurrence of winds forecasted to be in excess of 35 knots (40 MPH).
6. Precipitation: The aircraft shall be hangared prior to the occurrence of forecasted hail, freezing rain, or snow conditions.
7. Temperature: The aircraft will be placed inside the CareFlight hangar when the outside air temperature drops below 32 degrees Fahrenheit (0 degrees Celsius) as measured at VES AWOS (937)526-3154. Furthermore, the aircraft will be hangared when conditions for frost exist or are imminent.
8. Proximity/Movement: The aircraft shall be hangared as soon as possible when a weather system's movement or proximity of the hazardous weather requires an exercise of caution to protect the aircraft from potential damage.
9. The PIC shall return the aircraft to the outside helipad once the adverse weather conditions have diminished to the point that there is no foreseeable negative impact on the aircraft or flight operations.

#### 2.4.12.3 IIMC Procedure Template

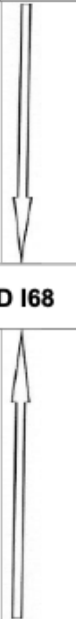


## IIMC Procedure Template

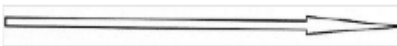
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                    |                                                     |                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------|
| <b>Location: Lebanon, OH (I68)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                    | <b>IIMC Procedure for: 30 NM Orientation Mag. N</b> |                              |
| <b>Date: 1/11/2025</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                    |                                                     |                              |
| <b>APP FREQ: 134.45</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                    | <b>APP FREQ: 134.45</b>                             |                              |
| Minimum Safe Altitude: 3,500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                                                     | Minimum Safe Altitude: 3,500 |
| Recovery ARPT KDAY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                    |                                                     | Recovery ARPT KDAY           |
| Recovery ARPT KLUK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                    |                                                     | Recovery ARPT KILN           |
| Minimum Safe Altitude: 3,500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                    |                                                     | Minimum Safe Altitude: 3,500 |
| <b>APP FREQ: 121.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                    | <b>APP FREQ: 118.85</b>                             |                              |
| <ul style="list-style-type: none"> <li>Altitude (Level)</li> <li>Heading (Turn only to avoid known obstacles)</li> <li>Power (Adjust to climb power)</li> <li>Airspeed (Adjust to climb airspeed)</li> </ul> <p>Climb to the minimum safe altitude to clear all obstacles in the area.<br/>           Make all turns no greater than standard rate.<br/>           Contact appropriate Approach Control/Center and Declare an Emergency, Squawk 7700 when able.<br/>           Report location, altitude, heading, fuel status, and number of persons on board. State that you are inadvertent IMC and request assistance.</p> <p>The following procedures shall be utilized in the following order to preference;</p> <ol style="list-style-type: none"> <li>Radar vectors to VMC conditions.</li> <li>Vectors to the closest approach.</li> </ol> |                                                                                    |                                                     |                              |



## IIMC Procedure Template

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                    |                                                     |                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------|
| <b>Location: Lebanon, OH (I68)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                    | <b>IIMC Procedure for: 30 NM Orientation Mag. N</b> |                              |
| <b>Date: 1/11/2025</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                    |                                                     |                              |
| <b>APP FREQ: 134.45</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                    | <b>APP FREQ: 134.45</b>                             |                              |
| Minimum Safe Altitude: 3,500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                                                     | Minimum Safe Altitude: 3,500 |
| Recovery ARPT KDAY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                    |                                                     | Recovery ARPT KDAY           |
| Recovery ARPT KLUK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                    |                                                     | Recovery ARPT KILN           |
| Minimum Safe Altitude: 3,500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                    |                                                     | Minimum Safe Altitude: 3,500 |
| <b>APP FREQ: 121.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                    | <b>APP FREQ: 118.85</b>                             |                              |
| <ul style="list-style-type: none"> <li>Altitude (Level)</li> <li>Heading (Turn only to avoid known obstacles)</li> <li>Power (Adjust to climb power)</li> <li>Airspeed (Adjust to climb airspeed)</li> </ul> <p>Climb to the minimum safe altitude to clear all obstacles in the area.<br/>           Make all turns no greater than standard rate.<br/>           Contact appropriate Approach Control/Center and Declare an Emergency, Squawk 7700 when able.<br/>           Report location, altitude, heading, fuel status, and number of persons on board. State that you are inadvertent IMC and request assistance.</p> <p>The following procedures shall be utilized in the following order to preference;</p> <ol style="list-style-type: none"> <li>Radar vectors to VMC conditions.</li> <li>Vectors to the closest approach.</li> </ol> |                                                                                    |                                                     |                              |



| <b>IIMC Procedure Template</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                    |                                                     |                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Location: Versailles, OH (KVES)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                    | <b>IIMC Procedure for: 30 NM Orientation Mag. N</b> |                                                                                    |
| <b>Date: 1/11/2025</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                    |                                                     |                                                                                    |
| <b>APP FREQ: 132.15</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                    | <b>APP FREQ: 134.45</b>                             |                                                                                    |
| Minimum Safe Altitude: 3,100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   | Minimum Safe Altitude: 3,100                        |                                                                                    |
| Recovery ARPT KPLD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   | Recovery ARPT KAXV                                  |  |
| <b>ID KVES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                    |                                                     |                                                                                    |
| Recovery ARPT KDAY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | Recovery ARPT KDAY                                  |                                                                                    |
| Minimum Safe Altitude: 3,100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                    | Minimum Safe Altitude: 3,100                        |                                                                                    |
| <b>APP FREQ: 134.45</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                    | <b>APP FREQ: 134.45</b>                             |                                                                                    |
| <ul style="list-style-type: none"> <li>Altitude (Level)</li> <li>Heading (Turn only to avoid known obstacles)</li> <li>Power (Adjust to climb power)</li> <li>Airspeed (Adjust to climb airspeed)</li> </ul> <p>Climb to the minimum safe altitude to clear all obstacles in the area.<br/>           Make all turns no greater than standard rate.<br/>           Contact appropriate Approach Control/Center and Declare an Emergency, Squawk 7700 when able.<br/>           Report location, altitude, heading, fuel status, and number of persons on board. State that you are inadvertent IMC and request assistance.</p> <p>The following procedures shall be utilized in the following order to preference;</p> <ol style="list-style-type: none"> <li>Radar vectors to VMC conditions.</li> <li>Vectors to the closest approach.</li> </ol> |                                                                                    |                                                     |                                                                                    |

#### 2.4.12.4 Pre-Start Confirmation Check



### **Pre-Start Confirmation Check**

Unauthorized PEDs – Off (Crew Response Required)

PFRA Approved (Crew Response Required)

**Before Start Checklist Complete**

Clear To Start (Crew Response Required)

### **Pre-Takeoff Confirmation Check**

Engine(s) – Fly

**Before Takeoff Checklist Complete**

Ready for Takeoff (Crew Response Required)

2.4.12.5 NVG Shipping Form



**SHIP TO:**  
7805 Karl May Drive  
Waco, TX 76708



The following are shipping instructions for night vision imaging systems being shipped to Night Flight Concepts for NVG inspections, service, and repair.

**PLEASE INCLUDE MOUNT & BATTERY PACK**

**Return to Customer:**

Company: \_\_\_\_\_  
Name: \_\_\_\_\_  
Street Address: \_\_\_\_\_  
City/State/Zip: \_\_\_\_\_

**Bill to Customer:**

Company: \_\_\_\_\_  
Name: \_\_\_\_\_  
Street Address: \_\_\_\_\_  
City/State/Zip: \_\_\_\_\_

**Contact Information**

Fill in the contact information in the event a technician needs to discuss any un-scheduled work to be performed.

Contact Name: \_\_\_\_\_  
Work Phone: \_\_\_\_\_  
Cell Phone: \_\_\_\_\_  
Email: \_\_\_\_\_

**Inventory**

Inventory each night vision goggle system and record the model and serial number where indicated. See example below.

|            | Example | 1 | 2 | 3 | 4 |
|------------|---------|---|---|---|---|
| Make/Model | F4949   |   |   |   |   |
| NVG S/N    | 10532   |   |   |   |   |

**Customer Shipping Information**

Fill in your return shipping information, otherwise, packages will be sent back UPS-Ground and billed to the customer.

| Purchase Order # | Shipment Option<br>FedEx / UPS / DHL | Delivery Option<br>Overnight / 2-day / Ground | Shipping Account # |
|------------------|--------------------------------------|-----------------------------------------------|--------------------|
|                  |                                      |                                               |                    |

**MUST INDICATE ALL PARTS NVG'S WILL OPERATE UNDER**

| Part 61 | Part 91 | Part 121 | Part 125 | Part 135 | Part 141 |
|---------|---------|----------|----------|----------|----------|
|         |         |          |          |          |          |

admin@nightflightconcepts.com

1-888-632-8764

www.nightflightconcepts.com

NFC FORM 1141

April 2020

1/12/25 Rev 1



## 2.5 Norfolk, VA

### 2.5.1 Base Standards

#### Team Members

1. All team members are the face of HeliService USA when at work and will be professional and courteous at all times. Take initiative and constantly strive to improve your work environment.
2. Primary Duties: Perform all duties as described in the OM-A. Continuously strive to be a professional in your field and embody the relentless pursuit of excellence.
3. Additional Duties: Perform additional duties, whether assigned or volunteered, with the same commitment and motivation as primary duties.

### 2.5.2 Aircraft

#### Aircraft

1. Ground Handling: Signature Aviation will be responsible for moving the aircraft in and out of the hangar. HeliService USA team members will be present to assist as needed. Signature Aviation will also provide a GPU for starts when requested.
2. When requesting Signature Aviation support, send an email the day prior to the flight to [ORF@signatureaviation.com](mailto:ORF@signatureaviation.com) (757-857-3463) and include:
3. Tail number and date in the subject line: N693HS 26FEB2026
4. Example Request:
  - At 0700 tug out helicopter N693HS from hangar 8 to run-up location. Refuel with 75 gallons of Jet A (-) and provide a GPU for a 0800 crank.
  - Helicopter's first departure will be at 0830 and will conduct numerous flights throughout the day with refueling of Jet A (-) requested between legs.
  - At approximately 1700 when flights are complete, tug helicopter back into hangar 8.
  - Aircraft Status: The Flight Engineer will do their best to ensure the aircraft is always in a flyable status and proactively forecast scheduled maintenance and inspections at times that will not impact customer support.
  - Aircraft Unattended: When the aircraft is left unattended in the hangar or outside on the parking ramp, wheel chocks will be in place. If outside, all doors will be closed and blade tie-downs and covers will be installed if left overnight.
  - Fueling: Aircraft will be fueled by Signature Aviation (757-857-3463).

•

### 2.5.3 Work Days

#### 1. Flight Workdays:

1. a. Ground Ops: Show two hours prior to the first departure of the day and be



prepared to receive and process customers.

- b. Flight Engineer: Show two hours prior to the first departure of the day and get the aircraft and logbook prepared for flight. Coordinate with Signature Aviation to move the aircraft from the hangar to its runup location. Coordinate for Signature Aviation to refuel the aircraft and provide a GPU as necessary.
  - c. Pilots: Show two hours prior to the first departure of the day and duty in, conduct all planning, ADIZ coordination, weight and balance calculation and equipment preparation. Relay to Quonset Dispatch max payload for the flight. Relay to the Flight Engineer gallons of fuel that need to be add
  - d. All Team Members: After all flight operations are completed for the day, work together to conduct all post flight tasks. Put the aircraft back in the hangar, complete any maintenance or inspections, complete paperwork and reset all equipment for the next day.
- 
1. **Non-flight Workdays:** All team members will be required to be available during normal duty hours of their duty days and are expected to be at work if there is work to be done. If all office and hangar tasks are completed or the work is digital and can be conducted remotely, team members may leave but are expected to report to the office within two hours of notification if a need comes up. Any phone calls, messages or emails will be answered and responded to during your normal workday.
  
  1. **Duty Day and Crew Rest:** Pilots and Flight Engineers are expected to abide by established crew rest policies and speak up when scheduled flights and timelines will conflict with this. All crewmembers and mechanics working on aircraft will be well rested, alert and ready to conduct flight operations safely.
  -

#### 2.5.4 Vehicle Operations on Flight Line

1. Towing Aircraft: Aircraft will be towed by Signature Aviation and monitored by HeliService personnel.
2. Driving Vehicles: HeliService vehicles will only be driven on the Signature Aviation ramp by team members that have completed the required safety training. While driving on the parking ramp, drivers will remain aware of taxiing aircraft and will give the right of way to them.

#### 2.5.5 Quonset Based Dispatch

1. Customer Coordination: Dispatch operations will be conducted remotely from Quonset. All coordination and direct communication with CVOW vessels will be handled by Quonset Dispatch and communicated to Virginia team members via



Teams messaging to the CVOW group and email.

2. Manifest: Quonset Dispatch will provide an initial manifest to Virginia Ground Ops for all flights. Virginia Ground Ops will update the manifest with passenger and cargo weights. Quonset Dispatch will send final manifests to pilot and aircraft emails.
3. Flight Delays: Any flight delays originating from CVOW (late passenger, flight changes, etc.) will be communicated from Quonset Dispatch to Virginia team via Teams messages. Any flight delays originating from Virginia team (weather impacts, maintenance issues, etc.) will be communicated to Quonset Dispatch via Teams messages.

## 2.5.6 Communication and Data Management

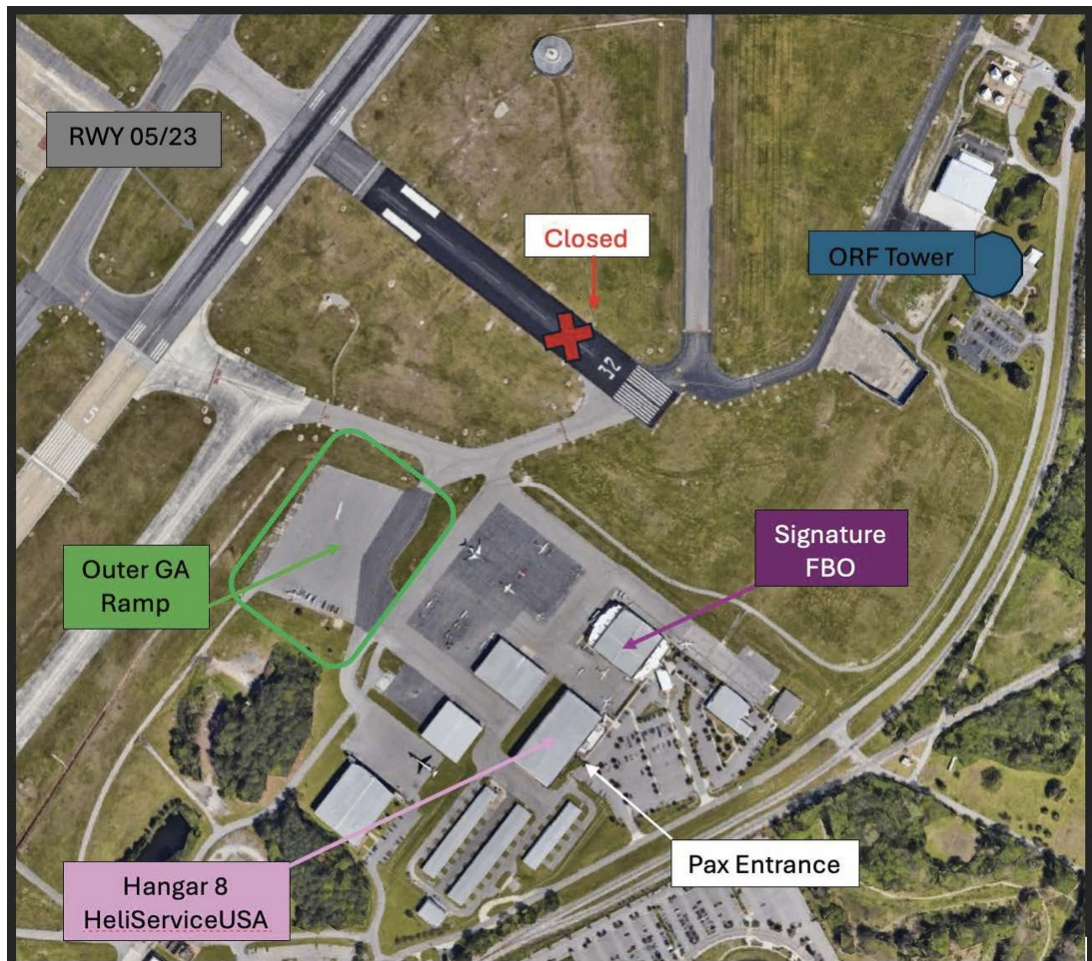
1. Messaging: Teams CVOW group message will be the primary means of communication for day-to-day things. HeliService email will be used for more extensive communication.
2. Phone Calls: Team members are expected to be available for phone calls during their normal workday if not at the office.
3. Location for Saving Files: The Virginia SharePoint folder will be the primary location for all files. This folder will be kept organized and old or outdated files placed in an archived folder.

## 2.5.7 Local Area

### 2.5.7.1 Airports

1. Norfolk International Airport (ORF)
  - HeliService Virginia is based at ORF on the general aviation ramp in Signature Aviation hangar 8. General aviation is located at the southern end of the airport and the scheduled airline flight terminal is located at the northern end of the airport. KORF is a Class C airport with many instrument approaches to choose from.
  -





2. Oceana Naval Air Station (NTU)
  - NTU is a military Class D airport located 9 nautical miles southeast of ORF.
3. Fort Story
  - Fort Story is an uncontrolled military training site located 9 nautical miles northeast of ORF. Low level UAV and parachute operations occur here with training times posted in NOTAMS.
4. Norfolk Naval Station Chambers Field (NGU)
  - NGU is a military Class D airport located 5 nautical miles from ORF.
5. There are many other civilian and military airports located within 20 nautical miles of ORF but should not impact flights from ORF to the wind park.
6. Norfolk International Airport (ORF)
  - HeliService Virginia is based at ORF on the general aviation ramp in Signature Aviation hangar 8. General aviation is located at the southern end of the airport and the scheduled airline flight terminal is located at the northern end of the airport. KORF is a Class C airport with many instrument approaches to choose from.

## 2.5.8 Coastal Virginia Offshore Wind Park (CVOW)

CVOW park consists of 176 offshore wind turbines and 3 offshore substations at various stages of



completion. The closest edge of the park is 34 nautical miles east of ORF. The park is 14 nautical miles from east to west and 10 nautical miles from north to south.

### **2.5.9 Hazardous Areas**

1. The CVOW is bracketed by Warning Areas (W-386 and W-72A) that encompass military aviation ordnance ranges. These ranges are active multiple hours of the day and status must be checked via radio with VACAPES callsign "Giantkiller" prior to entry to the windfarm.
2. There are additional Warning Areas to the North and South of CVOW routing that border the coastline. These ranges are active various times of the day. Any routing or passage through the areas shall be coordinated through VACAPES callsign "Giantkiller".
3. Fort Story UAS/Parachute area. The Fort Story UAS area is defined as a 3.0 NM radius of NTU025007.2 and surface to 700' AGL. NOTAMs will identify when this ceiling is raised for parachute operations.

### **2.5.10 Noise Abatement**

While there are no specific noise abatement areas, fly neighborly and avoid built up areas while operating below 1000 feet AGL.

### **2.5.11 Local Bird Avoidance**

### **2.5.12 Pre-Flight**

#### **2.5.12.1 Showtime**

Showtimes for everybody are at least 30 min prior to the first flight of the day.

#### **2.5.12.2 Weather**

Weather information will be derived from local airfield weather reports along with vessel and offshore substation weather reports.

#### **2.5.12.3 Aircraft Status**

Aircrew will check Digital Airware to obtain aircraft status.

During crew brief the status of the aircraft will be briefed by the Flight Engineer. Any inspections or maintenance coming due and any limitations to the aircraft will be relayed to the pilots prior to the first flight of the day. Any tests or checks that will be conducted by the pilots during the flight day will also be briefed.



If AMT is not available the crew will open a logbook and sign off the pre-flight IAW HSUSA OM-A.

#### **2.5.12.4 Aircrew Reading File**

All crewmembers are expected to be current on required reading file items prior to conducting any flight operations. It is the responsibility of the PIC to confirm this prior to the first flight of the day.

#### **2.5.12.5 Weight and Balance**

Pilots will confirm that the aircraft will remain within max gross weight for the conditions and within CG limits for the duration of all flights. IAW HSUSA approved methods.

#### **2.5.12.6 Fuel Required and Payload Available**

Fuel planning requirements will be conducted by pilots for each flight and include a mandatory 30-minute reserve. Virginia pilots will use a minimum of 150kg as their planned landing fuel. Max load for passengers and cargo will be relayed to Quonset Dispatch and Virginia Ground Ops for each leg of the flights.

#### **2.5.12.7 Vessel Information**

1. Charybdis
  - Jack Up vessel with a port-bow helideck with a D Value of 22.2. The height of the helideck is 58 feet above the water when underway and higher when jacked.
2. HOS Riverbend
  - Floating vessel with a bow helideck with a D Value of 21. The height of the helideck is 67 feet above the water.
3. Orion
  - Floating vessel with a port-bow helideck with a D Value of 22.2. The height of the helideck is 58 feet above the water.
4. Paul Candies
  - Floating vessel with a bow helideck with a D value of 21. The height of the helideck is 125 feet above the water.

#### **2.5.12.8 ADIZ Operations**

1. The FAA established a Certificate of Waiver or Authorization (COA) and Letter of Agreement (LOA) with Heliservice-USA and local ATC facilities allowing Heliservice-USA aircraft to operate in the offshore environment with a discrete transponder code. Heliservice-USA pilots shall maintain required communications with ATC services when crossing the ADIZ.
2. The COA/LOA waives the requirement for Heliservice-USA to file a Defensive Visual Flight Rules (DVFR) flight plan to cross the ADIZ. An IFR flight plan is still required if Heliservice-USA pilots operate in IFR conditions.



#### **2.5.12.9 Manifest**

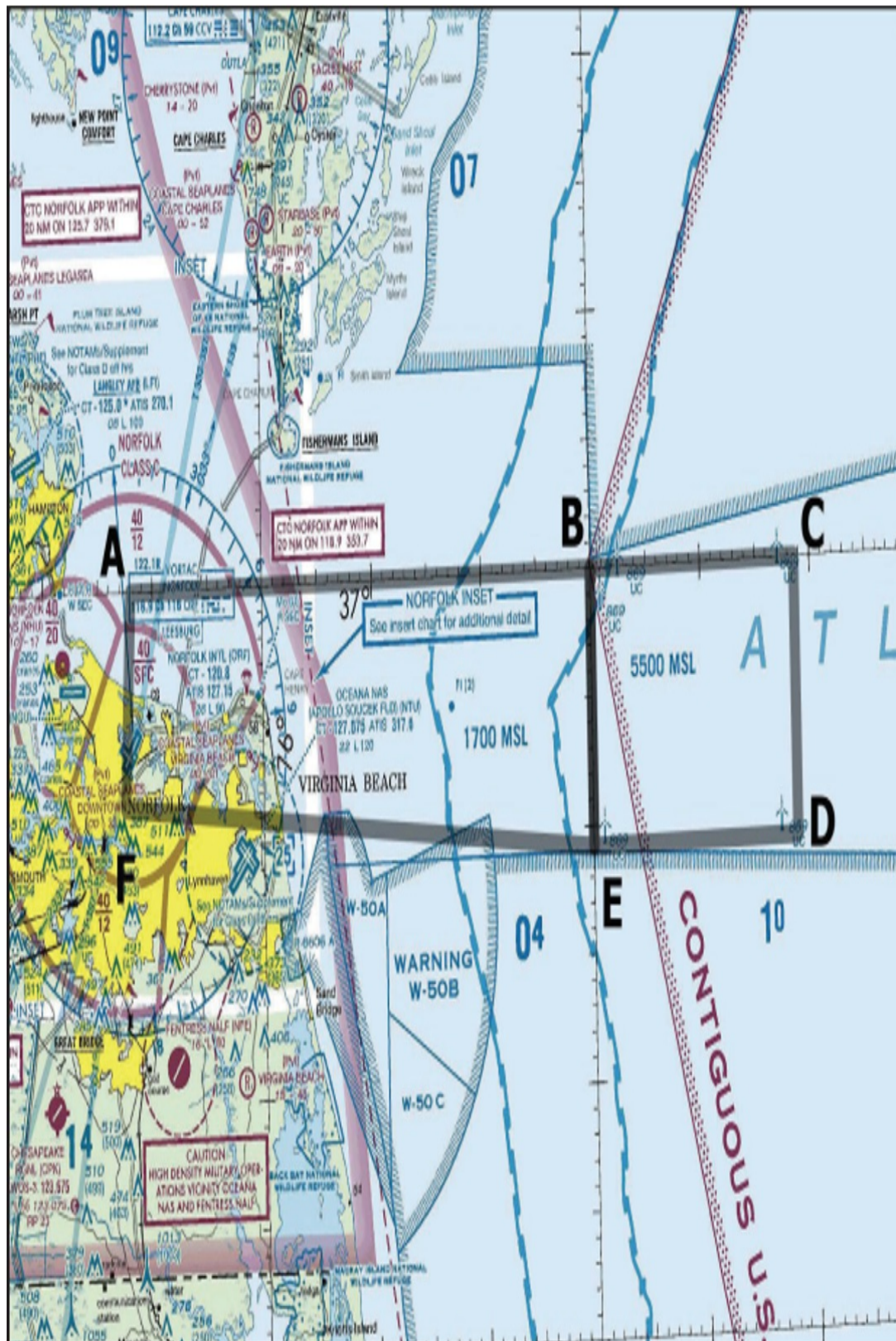
Flight manifests will be created by Quonset Dispatch and modified with passenger and cargo weights by Virginia Ground Ops. Final completed manifest will be sent to aircraft and pilots emails by Quonset Dispatch.

#### **2.5.12.10 Crew Brief**

At the beginning of each flight day, the PIC will lead a brief covering the operations that will be conducted that day. This brief will include pilots, Flight Engineer and Ground Ops and will use the established briefing format. Ground Ops will brief passenger manifest information. Flight Engineers will brief aircraft status, restrictions and flight hours remaining until the next scheduled event. SIC will brief weather, NOTAMS, aircraft performance planning numbers, fuel required for planned legs and weight available for passengers and cargo. PIC will run the brief flow, brief the flight routes used with Foreflight displayed on TV and lead a discussion on an aviation related topic.

#### **2.5.13 In-Flight**

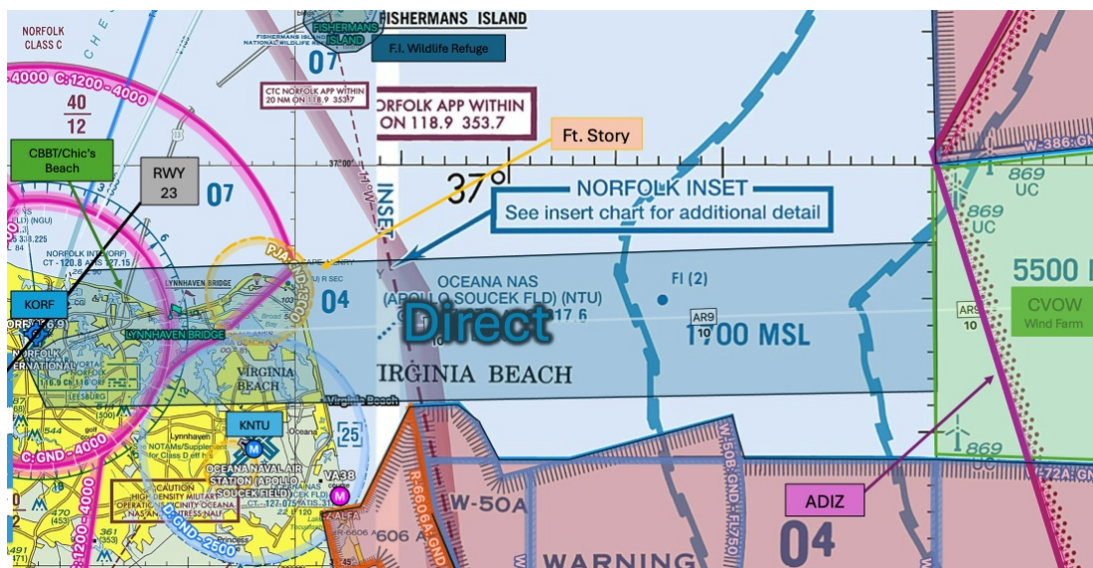
The FAA has established the following transit corridor and area for the CVOW:



| TRANSIT CORRIDOR |               | COASTAL VIRGINIA OFFSHORE WINDFARM (CVOW) |                |
|------------------|---------------|-------------------------------------------|----------------|
| Point            | Lat/Long      | Point                                     | Lat/Long       |
| A                | N37.00 W76.20 | B                                         | N37.00 W075.50 |
| B                | N37.00 W075.5 | C                                         | N36.82 W075.19 |
| E                | N36.82 W075.5 | D                                         | N37.00 W075.19 |
| F                | N36.86 W76.21 | E                                         | N36.82 W075.50 |

## 2.5.14 Routes

1. All planned routing should avoid ORF's runway 05/23 extended centerline and remain east of this at all times. This does not preclude PICs from requesting the runway on days that environmental conditions require a clear area takeoff or landing.
2. Be aware of parachute and UAV operations at Fort Story and check NOTAMS for operating times.
3. 1200' MSL is the standard routing altitude for both routes and allows for clearance from UAS operations (< 700' AGL). Pilots may deviate from this altitude for comms or as directed by ATC.
4. Direct
  - After departing ORF, fly due east toward the wind park.
  - The flight path cuts through either Fort Story or Oceana's airspace.
    - If you are cutting through Oceana's surface Class D, request transition through from Oceana Approach/Tower.
    - If overflying Ft. Story UAS, ensure transit at altitudes > 700' AGL. If parachute operations are in effect, avoid overflight of Ft. Story area.
  - The return route procedure is the same routing in reverse.
  -



## 5. Around Fort Story

- After departing ORF, fly northeast toward the intersection of Chick's Beach and Highway 13 bridge tunnel.
- Continue Northeast to waypoint FTSWAY
- Turn East and continue to CVOW.
- The return route procedure is the same routing in reverse.
- 



## 2.5.15 Transponder Code Requirements

1. Heliservice-USA must continuously squawk the discrete beacon code listed in Table 1 while conducting CVOW operations in accordance with the ZDC LOA
2. The discrete beacon code must only be used during VFR operations in accordance with the LOA and not during any other flights.



**Table 1, Beacon Codes**

| <b>Aircraft Registration Number</b> | <b>Radio Call Sign</b> | <b>Beacon Code</b> | <b>Aircraft Type</b> |
|-------------------------------------|------------------------|--------------------|----------------------|
| N691HS                              | Offshore 1             | 4331               | A169                 |
| N692HS                              | Offshore 2             | 4332               | A169                 |
| N693HS                              | Offshore 3             | 4333               | A169                 |
| N694HS                              | Offshore 4             | 4334               | A169                 |
| N697HS                              | Offshore 7             | 4335               | A169                 |

#### 2.5.16 Communication

1. When departing ORF, the communication sequence will be:
  - Norfolk Clearance Delivery
  - Norfolk Tower
  - Norfolk Approach (as required)
  - Oceana Tower (as required)
  - Oceana Approach
  - Giantkiller
  - Virginia MCC/vessel.
2. The return flight from the wind park will reverse the sequence but will include a call to Signature Operations for fuel and ground support.
3. Heliservice-USA must establish and maintain radio communications with the applicable air traffic facility whenever possible. Contact must be established on departure into the ADIZ and as soon as possible upon return
4. Pilots and ATC will use the applicable radio callsign in Table 1 during CVOW operations.

#### 2.5.17 Vessel Operations

1. Heliservice-USA crews will verify the name of the vessel and receive a “green deck” call from the HLO prior to landing on any vessel. This two-step verification mitigates wrong deck landings.
2. All approaches, takeoffs, and landings to vessels and windmills shall be CAT A profiles per the OM-D, Appendix A to the max extent possible.



3. When on vessels for crew changes, one pilot will get out of the aircraft and one pilot will remain on the controls and communicate with the HLO.
4. The pilot not at controls will:
  - Unlock the baggage door and supervise the offloading and loading of cargo and passengers.
  - Be responsible for the safe operations of all personnel around the aircraft and will intervene if an unsafe condition occurs.
  - Lock the baggage door and do a walkaround of the front and sides of the aircraft.
5. When on floating vessels, at least one set of wheel chocks will be used.

## 2.5.18 Post-Flight

### 2.5.18.1 Aircraft

At the end of the flight day, the aircraft will be tugged back into the hangar by Signature Aviation with HeliService USA supervising. All post-flight maintenance, inspections and paperwork will be completed by the Flight Engineer with the assistance of the pilots when their paperwork and tasks are finished.

If the aircraft needs to be washed the crew will make sure this task is completed

### 2.5.18.2 Digital Paperwork

Pilots and Flight Engineer will complete all required paperwork immediately after flight operations are completed for the day. It is up to the PIC to ensure that all paperwork is completed accurately and completely. After all tasks are completed, pilots will duty out for the day.

## 2.5.19 Points of Contact

- 1 Points of Contact Table

- The Norfolk base interacts with several military and civilian agencies in the area for deconfliction and operational requirements. Below is a small list of contacts for local area services:

| Company                          | Contact            | Phone Number   | Email                                                                              |
|----------------------------------|--------------------|----------------|------------------------------------------------------------------------------------|
| <b>Norfolk Airport Authority</b> |                    |                |                                                                                    |
| Norfolk Airport Auth.            | Melinda Montgomery | (757) 857-3351 | <a href="mailto:Mmontgomery@norfolkairport.com">Mmontgomery@norfolkairport.com</a> |



|                                                |                  |                                  |                                                                                                    |
|------------------------------------------------|------------------|----------------------------------|----------------------------------------------------------------------------------------------------|
| ORF Manager                                    | Anthony Rondeau  | (757) 857-3344                   |                                                                                                    |
| <b>Norfolk FAA</b>                             |                  |                                  |                                                                                                    |
| ORF Tower                                      | Chjumbo James    |                                  | <a href="mailto:Chumbo.k.james@FAA.gov">Chumbo.k.james@FAA.gov</a>                                 |
| ORF Approach                                   | Christopher Lane |                                  | <a href="mailto:Christopher.lane@FAA.gov">Christopher.lane@FAA.gov</a>                             |
| ORF ASOS                                       |                  | (276) 200-0952                   |                                                                                                    |
| ORF ATIS                                       |                  | (757) 464-1390                   |                                                                                                    |
| Washington ARTCC –<br>Airspace &<br>Procedures | Paul Clark       | (703) 779-3971<br>(703) 771-3530 | <a href="mailto:Paul.L.Clark@FAA.gov">Paul.L.Clark@FAA.gov</a>                                     |
| Washington ARTCC –<br>Watch Desk               |                  | (703) 771-3470                   |                                                                                                    |
| <b>NAVY POCs</b>                               |                  |                                  |                                                                                                    |
| VACAPES watch floor                            | Watch Supervisor | (757) 433-1230                   |                                                                                                    |
| VACAPES manager                                | Tamera Palmer    | (757) 433-1336                   | <a href="mailto:Tamera.l.palmer.civ@us.navy.mil">Tamera.l.palmer.civ@us.navy.mil</a>               |
| Oceana Tower                                   | PO1 Samuel Welch |                                  | <a href="mailto:Samuel.a.welch2.mil@us.navy.mil">Samuel.a.welch2.mil@us.navy.mil</a>               |
| HSCWL Safety Office                            | CDR Nik Bochette | 757-444-7540                     | <a href="mailto:nikolaus.j.bochette.mil@us.navy.mil">nikolaus.j.bochette.mil@us.navy.mil</a>       |
| NAS TRACON                                     |                  | (757) 433-2968                   |                                                                                                    |
| <b>Signature Aviation</b>                      |                  |                                  |                                                                                                    |
| Signature(ORF) GM                              | Mike Robinson    | (251)-459-2208<br>(757) 857-3274 | <a href="mailto:Micheal.robinson@signatureaviation.com">Micheal.robinson@signatureaviation.com</a> |



|                                |               |                |                                                                                          |
|--------------------------------|---------------|----------------|------------------------------------------------------------------------------------------|
| Signature(ORF)<br>Assistant GM | Cecille Gulez | (757)-323-8409 |                                                                                          |
| Operations and<br>Facilities   | Danny         | (757)-670-5844 |                                                                                          |
| Badging and Security           | Ray Knichel   | (757)-377-5194 | <a href="mailto:Ray.knichel@signatureaviation.com">Ray.knichel@signatureaviation.com</a> |
| <b>Miscellaneous</b>           |               |                |                                                                                          |
| Fish Spotter                   | Forrest Brann | 757-319-2177   | <a href="mailto:Flying4fish@verizon.net">Flying4fish@verizon.net</a>                     |
| Allied IT Systems              | Tommy Fauth   | (512) 676-5803 | <a href="mailto:Tommy@allieditsystems.com">Tommy@allieditsystems.com</a>                 |



### 3 Customer Specific Requirements

#### 3.1 Orsted

##### 3.1.1 Pilot Experience Requirements

Table 6 – Single Pilot / Commander Multi Pilot / Offshore Hoisting

| Topic                                 | Minimum                                                    | Recommended                                                | Extensive                                    |
|---------------------------------------|------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------|
| Total Hours (Helicopter)              | 1,000                                                      | 1,000                                                      | 2,000                                        |
| Total Hours in Multi Engine           | 100, or 25 hours under instruction on type                 | 500                                                        | 500                                          |
| Total Hours in Command                | 1,000 or 1,000 as co-pilot in HHO, of which 200 as PICUS** | 1,000 or 1,000 as co-pilot in HHO, of which 200 as PICUS** | 1,000                                        |
| Total Hours in Command / Multi Engine | N/A                                                        | 200                                                        | 500                                          |
| Total Hours on contract type          | 25                                                         | 50                                                         | 100                                          |
| Offshore Flying Hours                 | 0                                                          | 50                                                         | 500                                          |
| Hoist Cycles (1 down & up)            | 50 (of which 20 at night if ops conducted)                 | 100 (of which 50 at night if ops conducted)                | 150 (of which 100 at night if ops conducted) |

\*\* Pilot In Command Under Supervision

Table 7 – Multi-Pilot Co-Pilot / Offshore

| Topic                                                  | Minimum                                     | Recommended | Extensive                                  |
|--------------------------------------------------------|---------------------------------------------|-------------|--------------------------------------------|
| Total Hours (Helicopter)                               | 200                                         | 300         | 500                                        |
| Total Hours in Multi Engine                            | 15                                          | 15          | 250                                        |
| Total Hours in Command                                 | 25                                          | 50          | 100                                        |
| Total Hours in Command / Multi Engine                  | N/A                                         | N/A         | N/A                                        |
| Total Hours on contract type                           | 15                                          | 15          | 50                                         |
| Offshore Flying Hours                                  | 0                                           | 10          | 50                                         |
| Hoist Cycles (one down and up) [If hoisting conducted] | Operator Hoist Training Programme Completed | 50          | 50 (of which 20 at night if ops conducted) |

\*\* Pilot In Command Under Supervision

#### 3.1.2 Training

##### 3.1.2.1 Orsted Site Induction

Shall be completed prior to flying for any Orsted site.

#### 3.2 Vineyard Wind

#### 3.3 General Electric (GE)

#### 3.4 Sabine



## 4 Maintenance Procedures

### 4.1 Acceptance / inspection of aircraft components and material from suppliers

#### 4.1.1 Classification and Definitions

1. **Serviceable components:** A component is serviceable if it conforms with its approved design, is fit for its intended use and is accompanied by proper certification.
2. **Unserviceable components:** A component is unserviceable if it does not conform with its approved design and is not fit for its intended use but can be made serviceable.
3. **Standard parts:** are parts manufactured in complete compliance with an established industry, Agency, competent authority, or other Government specification which includes design, manufacturing, test and acceptance criteria, and uniform identification requirements. The specification should include all information necessary to produce and verify conformity of the part. It should be published so that any party may manufacture the part. Examples of specifications are National Aerospace Standards (NAS), Army-Navy Aeronautical Standard (AN), Society of Automotive Engineers (SAE), SAE Sematec, Joint Electron Device Engineering Council, Joint Electron Tube Engineering Council, and American National Standards Institute (ANSI), EN Specifications etc...Standard parts may be used on an aircraft, engine, propeller or other aircraft component when specified in the manufacturer's illustrated parts catalogue and/or the maintenance data. A conformity statement issued by the part supplier must accompany the parts.
4. **Raw and consumable material:** Material, both raw and consumable, used in the course of maintenance when the organization is satisfied that the material meets the required specification and has appropriate traceability. All material shall be accompanied by documentation clearly relating to the particular material and containing a conformity to specification statement as well as the manufacturing and supplier source.
5. **Unsalvageable components:** The following types of components should typically be classified as unsalvageable:
  - a. components with non-repairable defects, whether visible or not to the naked eye;
  - b. components that do not meet design specifications, and cannot be brought into conformity with such specifications;
  - c. components subjected to unacceptable modification or rework that is irreversible;
  - d. parts with mandatory life limitations that have reached or exceeded these limitations, or have missing or incomplete records;
  - e. components whose airworthy condition cannot be restored due to exposure to extreme forces, heat or adverse environmental conditions;
  - f. components for which conformity with an applicable airworthiness directive cannot be



- accomplished;
- g. components for which conformity with an applicable airworthiness directive cannot be accomplished.

#### 4.1.2 Component/Material certification

In this chapter the release documents to be expected/accepted for each type of part/material depending from their status (new/used) is identified. This table listing all the cases is developed for easy reference to receiving inspection personnel.

| SOURCE                              |                                                              |                                                              |                                                                                                         |                                                                                         |
|-------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
|                                     | EU Manufacturer                                              | EASA Part 145, EU CAO                                        | Non-EU Manufacturer                                                                                     | Non-EU Maintenance Organisation, CAO                                                    |
| <b>New Aircraft Component/Part</b>  | EASA-Form 1<br>JAA-Form 1<br>LBA-Formblatt 1<br>LBA Form One |                                                              | FAA Form 8130-3<br>ANAC F-100-01/<br>SEGVOO003<br>TCCA Form One<br>/ TCCA Form 24-0078<br>UK CAA Form 1 |                                                                                         |
| <b>Used Aircraft Component/Part</b> |                                                              | EASA-Form 1<br>JAA-Form 1<br>LBA-Formblatt 1<br>LBA Form One |                                                                                                         | FAA Form 8130-3<br>ANAC F-100-01 /<br>SEGVOO003<br>TCCA Form One<br>/ TCCA Form 24-0078 |
| <b>Standard Part</b>                | Certificate of Conformity (CoC)                              |                                                              | Certificate of Conformity (CoC)                                                                         |                                                                                         |
| <b>Consumable</b>                   | Certificate of Conformity (CoC)                              |                                                              | Certificate of Conformity (CoC)                                                                         |                                                                                         |
| <b>Raw Material</b>                 | Certificate of Conformity (CoC)                              |                                                              | Certificate of Conformity (CoC)                                                                         |                                                                                         |

Note: Used components maintained by a CAO appropriately approved for component maintenance and released on an EASA Form 1 cannot be installed on complex motor-powered aircraft or aircraft used by licensed air carriers

**Note:** According to "Merkblatt zum Thema Wareneingang in der Instandhaltung" only New Parts can be accepted with a UK CAA Form 1 approval.

#### PART MANUFACTURER APPROVAL (PMA) PARTS

PMA (Parts Manufacturer Approval) - Parts are components that have their own official approval from the FAA. It is not a prerequisite that the holder of this approval must have the original design specifications



of the aircraft manufacturer. PMA Parts always come from the FAA area but are approved by EASA under the following conditions.

For a PMA part that is to be installed in a product that has been certified or validated by EASA, one of the following entries must be made in the comment block (block 12: "Remarks") of the FAA Form 8130-3:

- a. "This PMA part is not a critical component."
- b. "Produced under licensing agreement by the holder of [INSERT TC or STC NUMBER]."
- c. "Produced by the holder of the EASA STC number [INSERT THE FULL REFERENCE OF THE EASA STC INCORPORATING THE PMA]."

### 4.1.3 Receiving inspection procedure

#### 4.1.3.1 Incoming Inspection

The procedures for acceptance of components, standard parts and materials have the objective of ensuring that the components, standard parts and materials are in satisfactory condition and meet HeliService USA requirements.

**INCOMING INSPECTION:** Involves -

1. Review of accompanying documentation and data;
2. Verifying the general condition of the packaging to identify damages that could affect the components;
3. Verifying that items are received in the appropriate package in respect of the type of component: e.g. correct ATA 300 or electrostatic sensitive devices packaging, when necessary;
4. Physical inspection of components, standard parts and/or materials;
5. Checking compliance with purchase order requirements;
6. Verifying that the shelf life of the component has not expired, record storage limitation/ life limits;
7. Verifying that the component has all plugs and caps appropriately installed to prevent damage or internal contamination. Care shall be taken when tape is used to cover electrical connections or fluid fittings/openings because adhesive residues can insulate electrical connections and contaminate hydraulic or fuel units;
8. Ensuring traceability of materials/standard parts received in batches (e.g. split of batches): Items (fasteners, etc.) purchased in batches should be supplied in a package. The packaging shall state the applicable specification/standard, part number, batch number and the quantity of the items. The documentation accompanying the material shall contain the applicable specification/standard, part number, batch number, supplied quantity, and the manufacturing sources. If the material is acquired from different batches, acceptance documentation for each batch shall be provided;
9. Confirming modification Standard and AD compliance if applicable;
10. Checking conformity with company requirements (e.g. type of release requested, Sources).

The traceability of parts and materials to the related documentation (e.g. internal tracking number) will be supported by the receiving inspection records.

Any fault, however small, must be brought to the attention of the Director of Maintenance or delegated



and the item labelled as unserviceable and placed in the Quarantine Store pending warranty action. An Incoming Inspection Finding must be recorded and reported to the company in which part was ordered from.

After receiving, inspection items will be logged into the parts database (Digital Airware).

Receiving inspection of components from internal sources if applicable will be handled as detailed above. **On receipt of all ESD sensitive devices the requirements of Company Procedures shall be followed in order to reduce the possibility of damage to an item due to electrostatic discharge.**

#### 4.1.3.2 Acceptance of components from Internal Sources

Components maintained by HeliService USA in an approved workshop may be accepted into stores with a Yellow Status Label.

Items for delivery to external customers must be issued with an FAA Form 8130 by an appropriately authorized licensed engineer.

#### 4.1.3.3 Components Removed Serviceable from Aircraft

To allow the appropriate removal and movement of parts fitted to a serviceable aircraft the following procedures should be followed as appropriate. This includes Customer (any applicable Customer procedures take priority over the following) and HeliService USA aircraft. For control/history/status/life of parts the relevant department must be contacted. An FAA Form 8130 must be issued when parts move from one AOC to another or are supplied by HeliService USA to an external customer.

When parts are moved internally for HeliService aircraft use, the issue of an FAA Form 8130 is not necessary only a Yellow component status tag.

**Special attention should be given to the Aircraft registration.**

#### 4.1.3.4 Components Robbed Serviceable from HeliService Aircraft

Components Removed Serviceable from a HeliService USA aircraft for onward supply outside the Company requiring the issue of an FAA Form 8130:

1. Make entries on the donor aircraft work pack and label the aircraft if applicable fully identifying the parts removed/received, by location, Part Number, Serial Number and TSO/TSN where applicable and as required.
2. Perform a visual inspection of the item prior to the issue an FAA Form 8130. The item may then be sent outside of the company.

**Under no circumstances** may any work be carried out on the component, other than visual inspection, while in the intervening stage between aircraft. Any such work is considered to be "Off Wing" and therefore subject to a declared component maintenance capability with the appropriate rating granted by the component authority.

#### 4.1.3.5 HeliService Components Removed Serviceable for transfer to Stores



Follow the process above, but attach a Yellow Status tag instead of an FAA Form 8130. This may be completed by staff without the authorization to issue an 8130, but requires a suitably authorized certifying staff member, i.e. authorized to issue a CRS for the aircraft type and trade for the component removed.

A function test may be necessary prior to removal of an item. Following a successful test, components that fall into this category are to be returned to stores with a correctly completed Yellow Status Tag. The component may then be transferred into stores and entered into Digital Airware. Parts transferred to Stores in this way shall only be used on HeliService USA fleet.

#### 4.1.3.6 Unapproved Parts/Bogus Parts

Any component suspected of being an unapproved or a bogus part is to be quarantined. All records received with the item including the Purchase Order are to be retained with the item.

Unapproved parts include, but are not limited to:

- Parts specified in the illustrated parts catalogues (IPC) of a type certificated aircraft, but which have been manufactured, reclaimed or reworked and then marked by an unauthorized source and provided with documents which indicate falsely that the part(s) are genuine and conform to the approved type design, or meet a particular industry standard and are offered for use as conforming with an aircraft manufacturer's authorized IPC.
- Parts shipped directly to users by manufacturers, suppliers, or distributors who do not themselves hold appropriate production approvals for the parts, and have not been authorized to make direct shipments to users by the Type Certificate holder,
- Parts which have not been maintained, overhauled or repaired in accordance with the requirements of approved airworthiness data and/or statutory requirements, or that have been maintained, overhauled or repaired by persons not authorized to perform and certify these functions.
- Parts without valid documentation and/or no traceability to its origin.

"QUARANTINE" procedure/treatment of a suspected unapproved part 'bogus part'

1. Identify part with RED Status Label clearly stating "**Suspected Bogus Part**"
2. Record within parts database (Digital Airware)
3. An Incoming Inspection Finding must be recorded.
4. Internal occurrence report

The Director of Maintenance or as delegated shall report the incident to the Competent Authority within 24 hrs after the discovery of the bogus part/component.

The Maintenance Manager will investigate the origin and manufacturer of the item in conjunction with the Competent Authority and/or the Type Certificate Holder.

Only if it is proved that the item is genuine by the Director of Maintenance or Maintenance Manager, the item will be returned to the Stores Department for use.



Unapproved/bogus parts will be disposed off under the advice of the Competent Authority and/or the Type Certificate Holder.

#### **4.1.4 Installation of components/standard parts/ materials**

Components, standard parts and materials shall only be fitted when specified in the applicable maintenance data. This could include parts catalogue (IPC), service bulletins (SB), aircraft maintenance manual (AMM), etc. So, the installation of a component, standard part and material can only be done after checking the applicable maintenance data.

This check shall ensure that the part number, modification status, limitations, etc., of the component, standard part or material are the ones specified in the applicable maintenance data of the particular aircraft or component (i.e. IPC, SB, AMM, CMM, etc.) where the component, standard part or material is going to be installed.

It is the responsibility of the involved C/S or S/S to ensure that following checks are performed before installation:

1. Verify the applicable maintenance data specifies the particular component, standard part or material.
2. Verification of the satisfactory condition and appropriate document for installation
3. The component is eligible to be fitted when different modification and/or airworthiness directive configuration may be applicable.
4. Verification prior to installation of standard parts on an aircraft or component (e.g. traceability, applicable standard as per maintenance data requirement)
5. Verification prior to use any raw or consumable material on an aircraft or component (e.g. due dates, applicable specification as per maintenance data requirement)

#### **4.1.5 Storage and release of aircraft components and material for maintenance**

##### **4.1.5.1 Storage Procedures**

Access to storage facilities is restricted to authorized personnel including Director of Maintenance, Maintenance Manager, Quality Manager, Accountable Manager, Engineers, Head of Stores, stores personnel or delegated C/S and S/S.

The Bonded Store should be kept dry and at a reasonably stable temperature preferably between 50 - 70 degrees Celsius. All aircraft components shall be retained on the shelving provided for that purpose except where this is precluded by its size or weight. Items to be stored at floor level must be protected from possible contamination or damage from all potential sources.

Sheet and bar metal should be retained in racks provided.

The Director of Maintenance or as delegated is responsible for ensuring procedures for maintaining satisfactory storage conditions according to manufacturer's recommendation for:



1. Aircraft components, consumables and raw material on suitable storage / racking facilities, adequate heating and lighting taking into account length of time in store and inhibiting requirements.
2. Special storage requirements e.g. for ESD sensitive devices and rubber.
3. Flammable fluids including oils and greases in metal/flameproof cabinet.
4. Engines and bulky assemblies
5. Segregation between serviceable, unserviceable and unsalvageable items and recording the position within the parts database.
6. Unserviceable components, which are identified as such and stored in a secured location (Quarantine Store) until a decision is made on the future status of such components.
7. Controlling and monitoring of the respective position in stores, shelf life / life limit and modification standard of components or materials which will be managed on the parts database (Digital Airware) and checked by the Head of Stores, assisted by a C/S or S/S if required.

A temperature monitoring system is located in the Bonded Store which alerts selected members of staff, by email. When an email is received items subject to temperature control will be re-located, to a more suitable environment, as soon as possible.

Company Procedures.

#### **4.1.5.2 Tagging**

All items that are received into stores from a supplier/repair organization must have an tag attached securely to the item.

The labels will have most of the information listed below, depending on the item:

1. Description
2. Part Number
3. Serial number
4. Quantity (on the Barcode)
5. Expire date
6. Batch number
7. Storage location (on the Barcode)

Unserviceable components shall be identified with a Green (Repairable) or Red (Condemned) Status Label, clearly stating status, and stored in Quarantine store until a decision is made on the future status of such components. The information on the label shall clearly state, as applicable, in-service times, maintenance status, preservation status, failures, defects or malfunctions reported or detected, exposure to adverse environmental conditions, and if the component was installed on an aircraft involved in an accident or incident.

When in agreement with the component owner, the component is disposed of, for legitimate non-flight uses, such as training and education aids, research and development, or for non-aviation applications,



mutilation may not be appropriate. In such case, the component must be marked indicating that it is unsalvageable, or the original part number or data plate information must be removed, or a record kept of the disposition of the component.

## 4.2 Acceptance of tools and equipment

This chapter describes the procedures for the acceptance of new, repaired, modified, calibrated tools and equipment received at HeliService USA, including borrowed/loaned tooling.

Selection, monitoring and granting of internal approvals of tool service providers (inspection /servicing/ calibration) will be complied with according to company and manufacturer requirements.

Tools and equipment acceptance procedure involves identifying and confirming:

1. The source as one from the approved tooling suppliers
2. Conformity with company requirements (e.g. certification, approvals)
3. Records accompanying the tooling and required documentation are present
4. Compliance with order / serviceable condition
5. Internal identification has been allocated in Digital Airware
6. The necessity of calibration and calibration dates are recorded
7. Control/issue for use is recorded in Digital Airware

The Maintenance Manager or as delegated is responsible for ensuring that all tooling (new, maintained, modified, calibrated tools/ equipment received and also the borrowed/loaned tooling) identified in the aircraft/component manufacturers maintenance manuals, tool listings and general aerospace support equipment is inspected and assessed upon receipt. An initial incoming inspection is performed satisfactorily.

The assessment shall include at least:

1. Accompanying documentation (e.g. calibration certificate, manual).
2. Required inspection intervals (for Ground Equipment in cooperation with the Maintenance/Ground Operations Manager)
3. Training or instruction required.
4. Calibration or scheduled maintenance.
5. Tools and equipment requiring calibration are given a calibration / validity label.

All tools are entered in the catalogue of tools and equipment in Digital Airware and given a unique tool ID and location number.

For calibrated tools the registration date, the date of calibration and the next calibration due date as well as the last calibration certificate must be added to Digital Airware.

For tools requiring periodic maintenance the last and next maintenance date, as per manufacturer's instructions, should be recorded.



Tooling which does not meet the above requirements shall be quarantined.

### 4.3 Calibration of tools and equipment

This chapter describes the procedures related to the controls, revisions, modifications, checking and calibrations of the tools/ equipment.

The Maintenance Manager or as delegated will ensure the registry, Inspection, servicing frequency and calibration program for tooling and equipment are maintained in Digital Airware.

The inspection, servicing and calibration time periods and frequencies are as advised by the manufacturers, and or as listed in accompanying documentation.

The calibration intervals for the applicable articles must be in conformity with:

- manufacturer's recommendations, or
- an officially recognized National or International standard, or
- a period prescribed by the Maintenance Manager based on monitoring, the previous calibration and the calibration reports or
- In the event of no other guidance, for pressure/mechanical items a maximum of 12 months and for electrical equipment a maximum of 24 months.

The due date for the next calibration is documented by means of a sticker on the tool, if the item is too small to bear a label, the label will be affixed to the item's container/case or attached to the item by a tag.

Items used infrequently may be calibrated on 'as required' basis and will be labelled "calibrate before use".

All items that require calibration/testing are to be removed from service prior to the expiry date and are identified with Green Status label attached detailing the item that requires calibration.

Calibrations forecasting and tracking is carried out by utilizing the Digital Airware tools module. A report is generated by Digital Airware, for tools and equipment, showing all which are expiring within the next 60 days.

The Maintenance Manager or as delegated shall ensure the item due calibration is quarantined in a timely manner and not available for use on the shop floor.

Upon return of the item from calibration, an updated calibration carried-out- date and due-date will be entered onto the database.

Calibration periods will be reviewed periodically or when any inaccuracies are discovered.

**Personal tools or tools designated "not for aircraft use" are not to be used on aircraft.**

In the event HeliService USA loans/rents a tool or equipment that requires calibration, the item shall be inspected for a valid calibration or test certificate prior to use. Items shall be identified on the relevant



aircraft paperwork along with the name of the loan/rental company.

The release of tools for loan/rental from/to other organizations will be authorized by the Maintenance Manager. The Maintenance Manager must ensure himself that the non-availability of such a tool does not affect HeliService USA's current maintenance operations.

Procedure for dealing with tools found out of tolerance during calibration will be removed from service and placed in Quarantine.

Expired calibrations or tools or measuring instruments which are outside their tolerance must be reported immediately to the Maintenance Manager by the discoverer. See also Remedial Measures below.

#### Articles in use

If a tool or measuring device which is in use is not calibrated, or is giving clearly false readings or measurements, it must be immediately removed from service, labelled as "Unserviceable", locked in the quarantine store, recorded as such in the register of tools (Digital Airware) and forwarded to the Maintenance Manager for appropriate actions (repair, recalibration, resale, scrapping). The Maintenance Manager must be informed.

#### Articles in calibration or test station

If an article is subject to calibration and / or test procedure and is reported as being out of calibration (temporarily) or outside the tolerance of the specified calibration range, then details of the extent of the calibration will be obtained and forwarded to the Maintenance Manager (e.g. extent of the tolerance deviation, place of use, frequency of use).

#### Remedial measures

In the case of an expired calibration or out of calibration the Director of Maintenance and Maintenance Manager assesses the use and the consequences of the incident (**when was the out-of-tolerance tool used? on which aircraft? after the expiry of the calibration? inform the customer/operator for further actions in case of safety concerns?, etc.**) and issues instructions for any necessary repetitions of aircraft or component maintenance measures, or replacement deliveries, taking into account:

- the quantity of items determined as being outside the tolerance and
- the tolerance that is allowed for the part, which may have been verified for this tool or measuring

After making an assessment, the Maintenance Manager generates a note which is kept with the calibration certificate of the article for future reference.

## 4.4 Use of tooling and equipment (including alternate tools)

This chapter describes all management procedures for tooling, distribution and return of the tooling after use.

Distribution of company tools is controlled from the tool store using a logbook entry system which records the user and the location/aircraft where it is being used. Each aircraft/work pack will also have a record of the tools used. Prior to an aircraft being released to service, the C/S or S/S will check that all



tools used have been returned back into the store.

Determining tool serviceability prior to issue of any tooling is the responsibility of the C/S or S/S. Training (if required) and control of personnel in the use of specialist tools and equipment including records of training for the same will be responsibility of the Maintenance Manager.

**Personal (own) instrument / tool control is not permitted.** All tools will be uniquely identified and either listed on an inventory or shadowed to clearly show any missing tools.

Use and control of alternative tools will involve:

1. Demonstration of equivalence between design/manufacturing data of alternative tools and the data/features of the tools recommended in the maintenance data of the manufacturers;
2. In-house identification of alternative tools (PN, SN);
3. Alternative tools will be validated by the Director of Maintenance and/or Maintenance Manager;
4. Register of alternative tools /tagging/relation between the references of origin tools and alternative tools on stores database;
5. Treatment of possible changes of maintenance data according to the new references of alternative tooling (modifications limited to the references of the tooling to be used and/or adaptation of maintenance data regarding alternative tooling);
6. Reference to the use/storage/maintenance manuals;
7. In-house approval of each alternative tooling before servicing;
8. Storage of the records of alternative tooling.

The Maintenance Manager identifies a potential tool manufacturer and, with the approval of the Director of Maintenance, selects the company and orders the tool that is to be manufactured based on a sample of the original tool.

Verification, inspection and validation are carried out by certifying personnel who are licensed for the type of aircraft on which the tool is to be used.

This alternative tool must be entered into the tool control system before the tool is used.

#### **Transitional regulation for the provision of tools between the Quonset base and various Line Stations**

To ensure the supply of type-specific special tools at the approved stations, a precise lay-time planning is carried out in advance in consultation with the station managers.

This prevents the occurrence of overlapping maintenance events of the same type. This rule applies until the capacity of the special tools at the respective locations is sufficient to carry out parallel maintenance events. All required standard tools are kept in sufficient quantities at the respective locations. If, under special circumstances, the special tools cannot be delivered from the Quonset site, they will be borrowed from the aircraft manufacturer (Leonardo) for the required period.

## **4.5 Control of man-hour planning vs. Scheduled maintenance work**

The Director of Maintenance or as delegated is responsible for producing a Maintenance Man-Hour Plan.



The plan should relate to the anticipated workload and include all necessary work such as, but not limited to, planning, maintenance record checks, production of work-sheets/cards in paper or electronic form, accomplishment of maintenance, inspection and the completion of maintenance records.

This plan is:

1. Reviewed at least every 3 months and updated when necessary;
2. Covering all staff (e.g. certifying staff, inspectors, mechanics, planners, quality auditors, etc.).

Particular attention is given to the situation when the same person is acting with different roles during a particular maintenance check (e.g. a person who is acting at the same time as an Hoist Operator during a particular base maintenance check). In such cases the man-hour plan for the particular maintenance check will take into account this aspect to ensure the person has allocated enough time to carry out the necessary activities required for each of the different roles he/she undertakes, and appropriate consideration is given to human performance limitations.

Hangar visit plan versus man-hour plan

1. Will be made available to demonstrate sufficiency of hangar space to carry out planned maintenance;
2. Relation is clear to include company management, non-commercial or other activities.

Man hours availability versus work load planning takes into account:

1. Human performance limitations;
2. Complexity of work;
3. Additional factors that affect personnel capabilities and work environment safety.

Organization of shifts will only be carried out when planning revision is required due to increased workloads.

Contracted personnel will compose never more than half the staff that perform maintenance in each workshop, hangar or flight line on any shift compared to employed personnel to ensure organizational stability. For the purpose of meeting a specific operational necessity, a temporary increase of the proportion of contracted staff is permitted to the organization by the Competent Authority.

The Maintenance Manager will notify the Director of Maintenance and Director of Operations of deviations exceeding 25% between the work load and the man hour availability.

## **4.5.1 Working at Height Policy**

### **4.5.1.1 Introduction**

#### **What is Working at Height?**

Working at height refers to any work where a person could potentially fall and injure themselves. A ladder, an opening on the floor, and even a loading dock can be considered working at height. In general industry, fall protection is required by OSHA for any height change of 4 feet or more. Here are safety tips



to help mitigate the risk of working at height.

**Purpose:**

This manual provides comprehensive guidelines for ensuring the safety of personnel working from heights in helicopter aviation. It aims to minimize the risks associated with height-related activities, emphasizing the safe operation of helicopters in such environments.

**Scope:**

This manual covers all activities involving personnel working at heights during helicopter operations, including maintenance work and cargo loading

**Overview:**

Helicopter aviation presents unique challenges when it comes to working at heights, especially due to the dynamic nature of operations, weather conditions, and the need for precise coordination between air and ground teams. This manual outlines the critical safety measures to mitigate risks associated with such work.

#### 4.5.1.2 Legal and Safety Regulations

Personnel working from heights in helicopter operations are required to comply with strict regulations to ensure their safety:

- **Occupational Health and Safety Standards :**  
All personnel involved in working at heights must adhere to national and international safety standards, including those outlined by organizations such as OSHA.
- **Aviation Safety Guidelines:**  
The Federal Aviation Administration (FAA) and other regulatory bodies provide specific guidelines for helicopter operations, including those involving personnel working at heights.
- **PPE Requirements:**  
All personnel must wear appropriate personal protective equipment (PPE), such as helmets, gloves, high-visibility clothing, and fall protection systems.
- **Duty of Care:**  
Employers have a legal obligation to ensure the safety of workers, which includes providing adequate training, ensuring proper equipment, and conducting safety briefings.

#### 4.5.1.3 Risk Assessment for Working at Heights

Risk assessments should be carried out before any work at height begins, to identify potential hazards and control risks:

- **Identifying Hazards:**  
Risks include falls, collision with rotating helicopter parts, wind-induced instability, poor weather, and insufficient safety equipment.
- **Risk Evaluation and Control :**  
Each identified hazard should be evaluated, and control measures should be implemented, such as



secure harnesses, clear communication protocols, and stable positioning.

- **Safety Briefing Requirements:**

All personnel involved in helicopter operations should attend a safety briefing covering the identified risks, emergency procedures, and expected actions during work at heights.

#### 4.5.1.4 Personal Protective Equipment (PPE)

##### Overview of PPE:

Helicopter aviation requires specialized PPE for personnel working at heights:

- **Helmets:**

Helmets must be worn for all jobs greater than 4 feet from the ground if no work stand with rails is available.

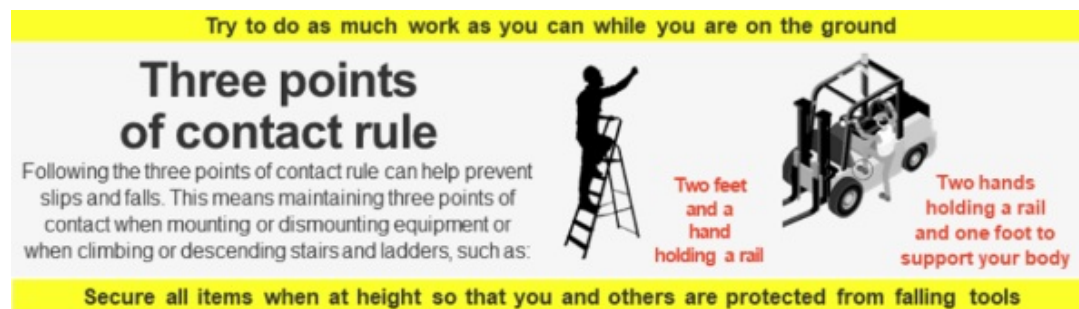
#### 4.5.1.5 Safe Working Practices

Personnel should follow strict guidelines when working at heights:

- **Securing Equipment:**

All tools and equipment should be securely fastened to prevent them from falling.

- Always have good footwear to reduce the risk of slip or trip
- Inspect the equipment to ensure handrails and other parts are safe and not faulty or damaged
- Check for any wet areas of slip hazards
- 3 Points of Contact



- Use appropriate work stands when possible
- Work stands with rails do not require the use of helmets



- When you can, use the railing and maintain 3 points of contact. Railing is a form of Passive protection, the easiest and most recommended way to keep you safe while achieving compliance.

#### 4.5.1.6 Proper Ladder Use

Ladders lie at the source of many industrial and workplace accidents simply because we take their use for granted. Before using a ladder, consider whether or not they are the best solution for the task.

1. Ensure you always have three points of contact with the ladder.
  - Two legs, one hand. Two hands, one leg. If you think about it, this means you do not carry something such as a bucket up or down a ladder. This would force you to break the three-point rule. Find another means like pulling it up by a rope after transitioning or holding smaller items in a belt or pocket.





#### 4.5.1.7 Emergency Procedures

In the event of a fall or other incident, immediate action is necessary:

- **Fall Prevention and Response:**  
If necessary call 911 for immediate help
- **First Aid:**  
First aid kits must be available at all work sites, and personnel must be aware of location.

#### 4.5.1.8 Documenting and Reporting

- **Safety Audits and Inspections:**  
Regular audits and inspections must be conducted to ensure compliance with safety protocols.
- **Incident Reporting:**  
All incidents, including near misses, must be documented and analyzed for future safety improvements.
- **Continuous Improvement:**  
Safety measures should be continuously reviewed and improved based on feedback and incident analysis.

By following these guidelines, helicopter operators and personnel can minimize the risks of working at heights, ensuring a safe environment for all involved in helicopter aviation.



## 4.6 Tool Control

### 4.6.1 Purpose

All tooling and equipment, including ground equipment must be used in accordance with manufacturer's instructions, relevant aircraft and/or component maintenance instructions, engineering standard maintenance practices, company procedures and local procedures.

Tools and equipment should be adequately identified, recorded, stored, and maintained in a serviceable condition and where appropriate calibrated.

The use of alternative tools and equipment from that specified in manufacturer's maintenance instructions will be controlled and authorized.

All tools and equipment used for aircraft maintenance must be accounted for and are therefore subject to tool control.

### 4.6.2 Responsibilities

It is in the responsibility of the Director of Maintenance or as delegated to ensure that sufficient tools, equipment and work stands are made available or provided to ensure maintenance can be properly carried out.

#### ALL Users

It is the responsibility of each individual to adhere to this procedure.

### 4.6.3 Standards

#### Serviceability

Prior to use, all tools and equipment must be assessed for serviceability. Tools assessed as unserviceable should be identified as such by means of an Unserviceable Status Label, and quarantined from use pending disposal/rectification action. When a tool becomes Unserviceable this must be reported immediately to the Maintenance Manager.

#### Identification

All tools and equipment should be clearly identified for their intended or designed purpose. The identification and segregation of items for specific maintenance tasks may be necessary to avoid contamination, miss-use, or degradation of calibrated qualities.

HeliService tools are assigned to a definitive area/storage location based on a code. The code is composed as follows:

| Organization | Place/Department | Numbering | Example |
|--------------|------------------|-----------|---------|
| HSI          | Quonset          | 1,2,....  | HSI-1   |
|              | Martha's Vinyard | 1,2,....  | HSI-2   |

## Storage

Tools and equipment should be allocated to an appropriate storage area, which is readily accessible and reasonably secured to maintain continuing serviceability and availability.

## Alternative Tooling

When the manufacturer specifies a particular tool or equipment, the use of an alternative tools or equipment, which will achieve the same result and/or inspection/ test criteria, will be authorized by the Maintenance Manager and Director of Maintenance.

The tool, where the size allows it, carries its unique bar code. If the article is stored on a tool board or shadow board, the place allocated to it also carries its unique bar code.

## Safety Precautions

Tools and equipment should be used in accordance with relevant safety precautions specified by the manufacturer and where appropriate industry practices and should never be used for other than their intended purpose. P.P.E. should be used where appropriate. Lifting equipment, component stands and working platforms should be used within their design and specification limitations.

## Tool Control

To further enhance flight safety, Heliservice USA has introduced a tool management system which minimizes the likelihood of any tooling causing a potential hazard to the safety or serviceability of aircraft. This system introduces Company tooling and tool control. The tool control system is based on toolboxes/cabinets and tool shadow boards. All tools must be individually identified. The tool control procedures detailed below has the Digital Airware and Snap-on tool asset management software in place. Furthermore, HeliService USA utilizes an state-of-the art Snap-on Level V tool box which performs many functions regarding control and safety.

## Tool Control System

The Tool Control System is based on toolboxes/cabinets fitted with engineered tool layouts in high visibility foam and asset management software. The system is designed to prevent the loss of tools and records, while accounting for the location of all tools held on site including aircraft specific special type tools. The system is also used to manage tool calibration/maintenance. The identification of personnel, aircraft registration and tooling are controlled by the allocation of proprietary software provided by Snap-on tools for all level V toolboxes.

Standard toolboxes have a Toolbox Inventory List with proper shadowing of all tooling.

All tools designated for aircraft / aircraft component use are stored in designated roll cabinets / toolboxes / tool boards and tools stores. The use of Tool Control is **mandatory** and personal tools or



tools designated "not for aircraft use" are not to be used on aircraft.

The Maintenance Manager or as delegated is responsible for the correct use of Tool Control.

### Tool Control Procedure

In HeliService USA's operations, all employees are provided with a unique identification card which includes a unique bar-code. Before a tool is issued from the toolkit to be used, the card is scanned, and the toolbox issues the tool to the associated user. It is the responsibility of everyone signing out the tool/toolbox to check it's serviceability and completeness.

In case Snap-on technology is not available, a Toolbox Checklist and Tool Issue and Return must be used.

After completing the work, the tool/toolbox/toolkit must be returned. This is done in the same way as checking the tool out. By scanning the tool back in the employee automatically signs for serviceability and completeness of the toolkit/toolbox/tool and that torque wrenches are set to "zero".

**NOTE: Before an aircraft is released for flight, the toolbox must be checked to ensue there are no tools signed out.**

### Special Tooling Control Procedure

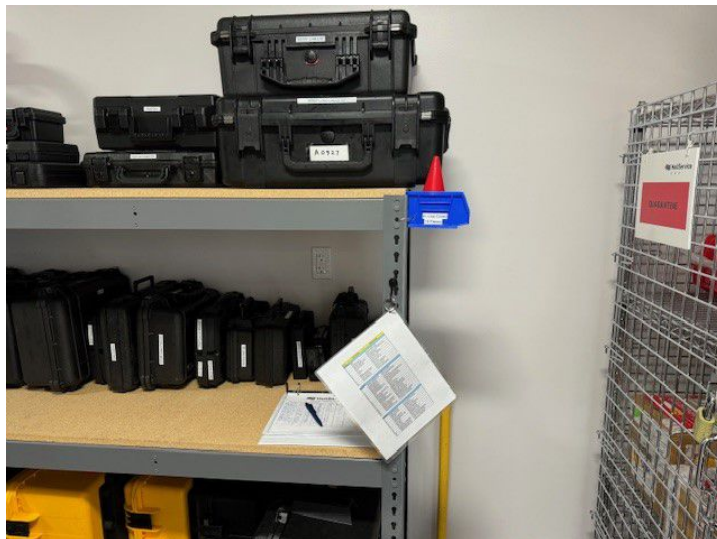
All special tooling is located in locked controlled room and access is restricted to authorized personnel only. Special tools are located on shelves or in vidmars designated for tooling only. All special tooling follows requirements listed above in 2.3. Special tooling checkout procedure is as follows:

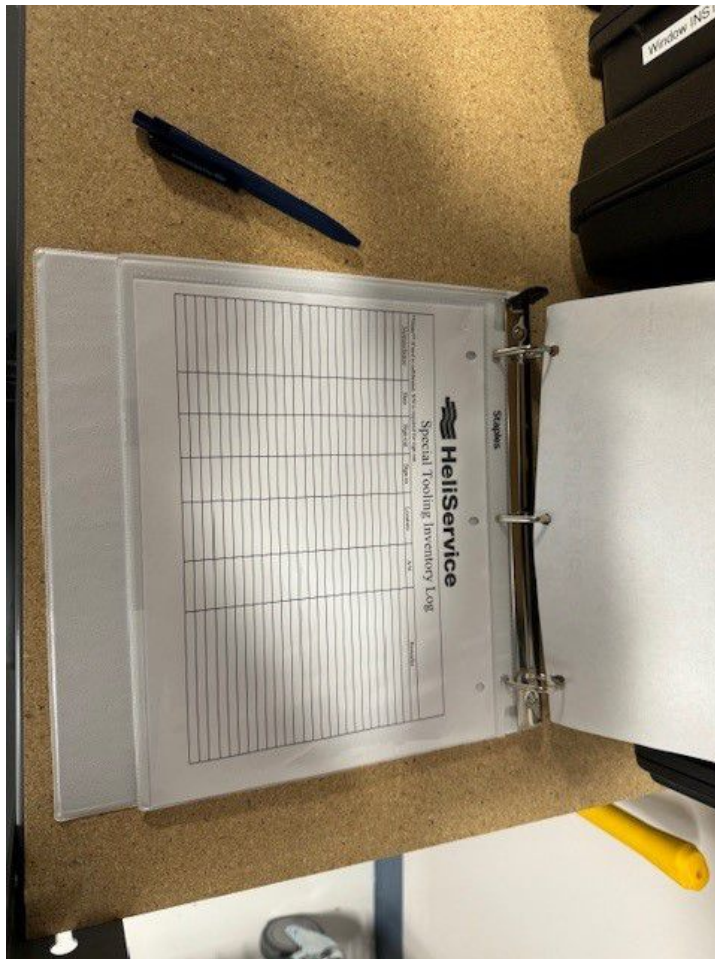
- Remove tool from shelf or drawer
- Sign tool out in sign in/out binder
- Place red cone if applicable in location tool was removed from

Special tooling check in procedure:

- Sign in tool in the binder
- Place tool on shelf
- Return cone designated bin







#### Lost or missing tool

In case of a lost tool the Director of Maintenance must be informed immediately;

- The Director of Maintenance identifies the missing tool and ask all personnel if the tool's last location is known;
- Carry out a loose article check on the affected aircraft and immediate surroundings;
- If the lost or missing tool cannot be located, the Director of Maintenance (or his/her deputy) is to be informed immediately. The Director of Maintenance (or his/her deputy) will assess if further searches are required;
- If the tool is still not found a log entry is to be raised against the affected aircraft stating, "Loose article check to be carried out" and the aircraft has to undergo a loose object inspection before the aircraft is released back into service;
- The Maintenance Manager must complete a Report in our SMS system in Fleetplan;
- Digital Airware management software must be updated to reflect that the tool is now Broken/Lost;
- The normal location of the lost tool in the tool cabinet (it's cut out) is to be clearly marked as "Missing Tool";
- The Director of Maintenance (or his/her deputy) will place an order to replace the tool that went missing;
- Once the replacement tool has been received, it is to be placed into the appropriate cut out in the



tool cabinet. Digital Airware and Snap-on is to be updated that the tool has been replaced. The newly procured tool must not have the same identification as the lost tool. Instead, it is marked with the additional letter "E" indicating that it is a replacement for the lost tool. This is done in case the lost tool is found, which prevents two tools with the same identification being in circulation.

### **Tool Serviceability**

Prior to use, all tools are to be assessed for serviceability. Tools assessed as unserviceable are to be quarantined and the normal location of the tool is to be blanked and marked as "U/S". The Maintenance manager (or his/her deputy) must inform the Director of Maintenance to purchase a replacement tool that conforms to the Tool Control standard.

The appropriate cut out is clearly annotated "Tool Broken" and Digital Airware and Snap-On is to be updated and the tool assigned to the location "Broken".

When the replacement tool is obtained, it is to be placed in the appropriate cut out in the tool cabinet and Digital Airware and Snap-On is to be updated.

### **Completion of Work Inspection**

It is the responsibility of every engineer on completion of their working day to ensure all tooling is complete and correct and all toolkits/toolboxes/tools are returned to Tools Store or dedicated location. This action is an integral part of Tool Control and Shift/Project Supervisors must ensure compliance with this instruction.

At the end of each shift the supervisor will check what tools have been used by checking Gannet. Any tools that have been used should either be returned or handed over.

### **Adding a Tool to the Tool Control System**

The permanent addition of tooling to the Tool Control System may only be approved by the Director of Maintenance, Maintenance manager (or his/her deputy), as particular needs is required.

### **Line Maintenance**

Under normal conditions when an aircraft has been released for a flight, tooling is not to be utilized on that aircraft. However, situations occur throughout the daily flying program that requires the use of tools on the aircraft flight line to conduct line maintenance.

Line maintenance is limited to low complexity maintenance.

Tools used for Line Maintenance must be under control of the appropriate tracking system. Tools for line maintenance, such as flashlights etc. may be kept out with the tool cabinet. They may for instance, be grouped together, ready for use, on wall mounted boards. Any tools stored like this must be treated as if they are part of a tool cabinet. Before signing the pre-flight inspection in the technical log, the engineer must ensure there are no accounted tools issued to that aircraft.

### **Training and monitoring of personnel for the use of tools and equipment**



If an alternative tool has been approved and deployed by the Director of Maintenance or Maintenance Manager, the entire staff will be trained in the function and use of this tool by their supervisor or a designated qualified person before the tool is used on an aircraft or a component.

## 4.7 Cleanliness standards of maintenance facilities and workshops

### 4.7.1 Purpose

The standards of housekeeping and cleanliness are often the first impression which is observed by visitors to the organization. Poor standards maintained in these areas may influence how the employees perceive the work in many other areas of their activities. Housekeeping and cleanliness is crucial to a **safe workplace and to emphasize HeliService USA's Culture.**

### 4.7.2 Responsibilities

It is the responsibility of **EVERY** employee, through the Maintenance Manager, to ensure that the facilities are kept to the highest standards of cleanliness.

### 4.7.3 Standards

**Exclusion program for foreign objects.**

The Director of Maintenance shall include foreign objects control as part of the duties of all supervisors. However it remains everyones responsibility to prevent FOD. The following guidelines are to be observed:

- The entire maintenance staff shall be trained in foreign object detection and preventive measures by the Supervisors. During the training, it must be stressed that all employees remain fully aware of the dangers of foreign objects and ensure that adequate precautions are taken to prevent objects falling into or being left in critical areas.
- The maintenance personnel of aircraft, and other people who have access to the flight and maintenance areas, must empty their pockets of foreign objects or securely close them before beginning work on aircraft.
- Items that are regarded as personal foreign objects include pens, pencils, labels and badges, shoes or boots with fittings or studs which come into contact with the floor, magnetic stripe cards, cash, keys etc.
- Objects such as cable ties, rivet sleeves, shavings or waste products must be disposed of in debris containers and may under no circumstances be placed on aircraft floors or structures.
- During the maintenance work, debris containers and tool boxes must be used by the maintenance staff for storing foreign objects. These must be cleaned at completion of the task.
- Fixing parts that have been removed and will be re-mounted for refits are to be kept in bags or containers that are specially marked for the purpose. The contents must be secured in order to prevent spillage.



- Ground support equipment must be thoroughly checked for foreign objects before use.

Hand tools are not foreign objects, if they are not mislaid or left in the vicinity of the aircraft. To prevent this, the following guidelines apply:

- The responsibility for a personally allocated tool lies with each individual employee. After every maintenance task has been completed, an examination of the area and the aircraft must be performed and all tools removed. This examination is assumed to have occurred when an employee of the certifying staff approves a task on the relevant work package.
- The shift supervisor will check all special tools and equipment in the tool store at the end of the shift and inform the incoming shift of all tools not yet returned.

Each employee is responsible for the cleanliness of the aircraft and their immediate work area. Components such as flight control surfaces, tail boom, upper deck, air intake, avionics bay, ... must be checked for cleanliness in stages. Any accumulation of dirt, salt stain and deposit, debris or other foreign material must be removed.

**NOTE: Before closing any access panels on the aircraft a loose object check must be performed.**

**Whenever a helicopter enters the hangar air intake covers must be installed. This to prevent any Foreign Object entering the air intake which might cause damage to the engine. This also applies for helicopters in base maintenance.**

### **Cleaning Program – Individual Responsibilities – Time-Scales**

Each employee is responsible for the cleanliness of aircraft and the immediate surrounding work area. Cleanliness shall be maintained at all times, as far as practicable, and not be left until completion of the task.

Components such as flight control surfaces, tail units and boxed in structures should be inspected progressively for cleanliness. Any accumulation of dirt, salt stain or deposit, foreign objects or other extraneous material must be removed.

**Note: If any part of the aircraft or engine control system is inside a closed-in compartment, consideration should be given to the requirements for independent inspections to be carried out.**

Situations may occur, when an aircraft is undergoing a major modification, that large amounts of swarf, and debris are generated. Where this is likely, areas of the aircraft not affected should be masked off prior to the commencement of work and debris cleaned away, on a continuous basis, in order to prevent migration of FOD. A thorough cleaning and internal inspection of the aircraft compartments affected must be carried out prior to closure of the areas.

### **Cleaning Methods**

Aircraft cleaning methods are detailed in manufacturer's Standard Practices manuals and/or Corrosion Control Publications. However, smaller structures such as horizontal stabilizers etc., should be rotated in



order to free any loose items inside. The use of industrial vacuum cleaners with suitable adaptors should be utilized where normal cleaning methods are inadequate or inappropriate.

The use of air jets should not be encouraged due to the possibility of FOD and loose dirt being blown into inaccessible areas. Compressed air may also contain high moisture content and use in certain areas may lead to corrosion.

#### **Cleanliness of Complete Aircraft**

Before completion of an inspection, a foreign objects check must ensure that all areas that have been opened have been checked for internal cleanliness. Special attention must be paid to all compartments where engine- and flight controls pass.

The cleanliness of warning notices, posters, date plates, transparent covers of information signs or displays and instruments must be checked.

#### **Disposal of waste materials**

Waste bins are provided in the hangar for general waste material, they must be emptied as needed.

Waste bins are provided in the hangar for oil contaminated material/hazardous waste, they are collected periodically by an approved waste disposal company.

Aircraft parts that are worn or damaged should be disposed of in accordance with this SOP CH-5.

## **4.8 Incoming Inspection and Storing**

### **4.8.1 Standards**

#### **Incoming goods Inspection**

The incoming goods inspections are carried out by Engineers, and generally include the following check points:

1. Check for external damage:
  - Damage to the packaging
  - Damage or loss of content
  - Damage to delivered unpacked material
2. Verification of delivery:
  - Compliance with the shipping documents in terms of type, number, part, and serial numbers
  - Compliance with the order (Modification status)
  - whether it is so-called "Bogus Parts"
3. Verification of the completeness of the delivery according to Purchase Order:
  - Correctness of the product description
  - Order quantity



- Delivered quantity
- 4. Check for the presence and completeness of the delivery documents
  - Delivery note
  - Quality Certificates
- 5. Verification of the quality certificates supplied
  - in accordance with the required evidence
  - complete, correct information
- 6. Verification of the marking of goods / products
  - in accordance with the information in the quality certificates
- 7. Check of condition and quality of goods / products
  - Damage
  - Deformation
  - Cracks
  - Quality of the surface
  - Corrosion
- 8. Complaint-free goods / products
  - Appropriate labeling
  - Transfer to storage
- 9. Information to the Quality- and Maintenance Manager
  - In case of a finding the Maintenance- and Quality Manager to be informed who will hold a MRB to further evaluate the issue
  - Finding to be entered in Incoming Inspection Finding List MX\_FO\_018
  - A Report in our SMS system in Fleetplan must be raised
  - According to the out-come of the MRB the goods will be returned to the supplier or stored in the quarantine area until further decision.

**Note: Special attention must be given to ESD sensitive parts! Refer to ESD procedure as per CP-4.**

**Remark:**

The scope of the inspection and the level of detail are defined individually, depending on the component's significance and the previous quality of the supplier.

**Storage regulations**

All products / materials will be stored I/A/W SOP Chapter 1.2

***General***

Before the storage of parts, it is necessary to check whether the manufacturer of the goods to be stored has specified specific storage conditions which must be adhered to. Spare parts are organized by part numbers, starting from A to Z, then numbers beginning with 0. Large, heavy or bulky parts may be exempted. Parts must be stored in a packaged and preserved state, as far as necessary to provide



protection against contamination, corrosion or other factors of influence to quality / risk. The product description and the test status must clearly be evident from the documents (identification tag, test certificate, etc.), which must be located on the parts.

**During storage, a clear segregation must always be made between "Certified" serviceable, airworthy components/parts and unserviceable/unsalvageable parts/components which must be stored in Quarantine.**

### *Basics*

Requirements for storage areas: If no special conditions are required, the following guidelines apply:

- Dry, dust-free, ventilation possible
- Temperature should be controlled i.a.w. the manufacturer's storage requirements
- Distance of the radiators at least 1.00 m from the stored items
- Relative humidity should be controlled i.a.w. the manufacturer's storage requirements
- Avoidance of direct sun exposure
- Artificial lighting according to the stored items
- Facilities free from harmful gases and vapors

The average preferred temperature range will be 50-70 degrees Fahrenheit.

Storage facilities for serviceable aircraft components should be clean, well-ventilated, and maintained at a constant dry temperature to minimize the effects of condensation. Manufacturer's storage recommendations should be followed for those aircraft components identified in such published recommendations.

The stores area is equipped with calibrated air temperature and air humidity measurement devices. Measurements are recorded online. In case of fluctuations of temperature- or humidity values an alert e-mail will be automatically generated to the Director of Maintenance and Maintenance Manager and actions will be taken to solve the issue in a timely manner.

### *Goods Receipt*

As proof of quality, an EASA Form 1, FAA Form 8130-3 or TCA Form 1 is accepted for new aircraft parts and C.o.C for standard parts/consumables.

Used parts purchased must either be delivered with an EASA Form 1, or if they come from the FAA or TCA area, be equipped with a so-called "dual release".

If the goods receipt inspection has been completed without complaints, the goods are collected in the storage card and stored in the material store as required and defined.

In the case of small parts, the parts are labeled in such a way as to ensure traceability of the parts with respect to the manufacturer, the batch number and the delivery date.

In general, it should be noted that First in - First out (FIFO) principle applies and where applicable First

expires - First Out(FEFO).

#### ***Parts with limited storage and usage time***

When storing products / goods with a limited storage or usage time, it must be proceeded in such a way that, on the one hand, the storage location is identical to the same products and, on the other hand, the expiry date of the stored goods is clearly and unambiguously marked and recognizable. The expiry dates for these products/goods are entered in Digital Airware and a weekly report is automatically produced and mailed to logistics, Purchase department, Maintenance- and Quality Manager and his/her deputy. This report includes all products/goods which expires within 30 days.

#### ***Raw material / semi-finished products***

Sheet metal, pipes, profiles, bars, etc. are to be secured and stored against bending in appropriate devices. In addition, high-quality material is to be stored in such a way that no damage such as scratches, cracks, scrapes and bends can occur.

#### ***Consumables***

Consumables, such as screws, nuts, rivets, brackets, etc:

In the case of complaints, the complete batch of the disputed batch number will be scrapped.

#### ***Products / goods made of elastomers***

Products / goods / products made of natural rubber (rubber) are subject to aging and require particular attention to achieve / usage of the maximum storage period:

- Protection against extreme temperature fluctuations,
- No direct sunlight,
- Special artificial lighting,
- Protection from ozone, moisture, and drafts,
- No storage together with oils, grease, fuels, acids, etc.
- Avoid the connection (on or on) with metals, especially copper, as stress - free storage of the products to avoid surface tension and thus crack formation, deformation, one - sided loading and kinking. The stacking of material is to be avoided because of the deformation of the lower layer. In the case of hose lines, hoses, tires, seals, as well as all elastomers at all, the expiry date of the storage period must be clearly visible on the part or the storage container.

#### ***Instruments / fuel / hydraulic system assemblies***

Instruments and other devices and systems with connections for pneumatic or hydraulic lines must be sealed with plugs at the connections. Connectors and connections must be fitted with protective caps. If necessary, moisture-absorbing substances or supplements (silicates) must be used in the containers of



these devices.

#### ***Flammable materials***

Flammable materials, such as paints, varnishes, adhesives, solvents, chemicals, etc., must be stored in a cool, dry place in rooms with ventilation. The containers are well sealed to hold, lobes or paper as a closure are not permitted.

#### ***Aggressive materials***

Aggressive materials such as acids, alkalis, various plastic materials and similar substances must be stored separately.

Acids and toxic substances must be labeled as such and kept locked.

#### ***Inspection of the stored goods***

The store personnel regularly checks, at least weekly, the stored products / goods to be kept in a serviceable condition; including:

- Monitoring of expiration dates of the storage time limited parts,
- Checking the components / parts for storage damage,
- Detection and removal of damage to the packaging,
- Determination and removal of moisture, corrosion, etc.
- Check that existing surface protection is still undamaged.

#### ***Pyrotechnic objects***

Pyrotechnic objects (e.g. explosive cartridges) may be received/ stored/ shipped. Storage takes place in specifically designated safe.

#### ***Tires***

Tires must be stored as per manufacturers documentation and must be rotated weekly. The Head of Stores is responsible to ensure this is performed.

#### ***Receiving goods outside of Base Station Quonset***

Orders to be shipped directly to customers/ subcontractors/ repair stations/ line stations shall be reported to the Director of Maintenance. The Engineer ensures the incoming goods inspection at delivery. The receiving inspector may be a Heliservice USA employee (if on-site) or personnel of the company receiving the delivery. After completion of the incoming goods inspection and confirmation via E-Mail to the Maintenance Manager, goods will be booked in Digital Airware at the respective location.



## 4.9 ESD Procedures

### 4.9.1 Standards

#### ESD Sensitive parts.

##### Background

The age of electronics brought with it new problems associated with static electricity and electrostatic discharge. Due to electronic devices becoming faster and smaller, their sensitivity to ESD increased. Today, ESD impacts productivity and product reliability in virtually every aspect of today's electronics environment, and in fact for all market segments, studies have estimated that up to 60% of device failures are ESD caused.

A major factor contributing to ESD damage is lack of awareness of the ease in which damage occurs. For example, the average person can feel a discharge on a doorknob if it is greater than 3500 volts; hear the discharge if it is greater than 5000 volts and see it if it is greater than 8000 volts. By comparison, damage to integrated circuits occurs at fractions of those voltages. This means that the person imparting the damage will not have felt, seen or heard the discharge, and because of this we must always assume we can potentially cause ESD damage unless those charges are properly grounded, as for example, in accordance with this Best Practice.

*ESD sensitive parts that have been damaged could manifest such damage in one of several ways upon being installed on the aircraft:*

- a) Immediate failure of the part when the component/system is tested.
- b) Premature failure of the part. E.g. the part fails much sooner than its designed Mean Time Between Failure (MTBF) figure.
- c) The part or its next higher assembly exhibits intermittent or random failures.

Depending on the part's criticality, ESD damage can affect safety, flight delays or cancellations, system reliability, create chronic or rogue parts, and increased maintenance costs. Parts which are subject to ESD damage can be positively identified by these typical markings".



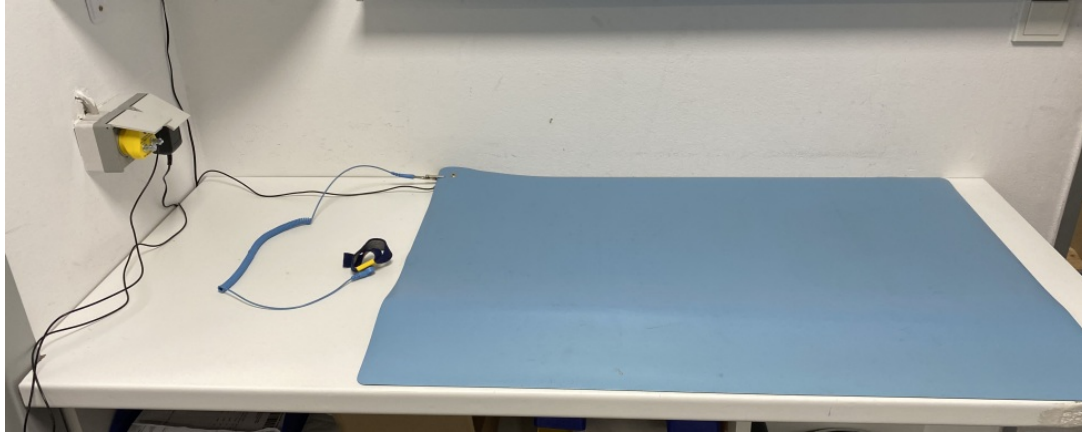
#### Equipment for an ESD Protected Area (EPA)

Consists of an ESD Wrist Strap and chord, Wrist Band Chords, grounding connection, and Table-top Conductive/Dissipative Work Surface Mat. All EPA components should be installed in accordance with the manufacturer's instructions.

Users may optionally choose to have additional Mats or Wrist Straps to support additional employees



working simultaneously.



### Packaging

When packaging ESD sensitive parts, consideration is given to one of three types of parts; Piece parts (such as IC Chips, Transistors), Circuit Boards, or LRU's (Line Replaceable Units, commonly referred to as Rotables or appliances). ATA Specification 300 offers simple additional guidance.

**CAUTION:** Staples should not be used to seal any ESD-protective packaging or used to attach documents as this may compromise the ESD protection.

**A. Piece Parts:** The Leads for these parts should be embedded in ESD protective foam, typically black. The parts are then placed in static shielding/dissipative bags (typically gray or silver, not pink which is not recommended due to its limited protective properties; they do not shield from Electrostatic Fields<sup>1</sup>). The parts may then be placed in standard protective shipping packaging.

**B. Circuit Cards:** Cards are placed in static shielding/dissipative bags (typically gray or silver, not pink which is not recommended due to its limited protective properties; they do not shield from Electrostatic Fields<sup>1</sup>). The cards may then be placed in standard protective shipping packaging.

**C. LRU's:** Assure ESD Protective Caps are placed over the connector pins. Once the LRU has these caps in place, the part can be handled outside of an EPA without fear of inducing ESD damage. With the ESD caps in place, placing the LRU in static shielding/dissipative bags is not required. It may then be placed in standard protective shipping packaging.

**D. LABELING:** ESD labels should be affixed to the outside of packages or containers to apprise handlers of the ESD sensitive nature of its contents.



### Handling, Warehousing

With the exception of being placed in the final standard shipping container/package, once ESD sensitive parts have been protected as described in the 'Packaging' paragraph, the parts may be safely handled outside of an EPA.

**ESD protective caps must be stored separately from any other protective caps.**



**All ESD Parts returned to stores and/or Vendor must protect with the proper ESD caps and ESD packaging!**

ESD parts are received without proper ESD protection must fail incoming inspection and must be reported.

## 4.10 Consumables

### 4.10.1 Standards

All incoming consumables shall be checked for the following:

- Expiration date
- Proper seal
- Leaks
- Damage

Any rejected item shall be discarded according to manufacturer and State regulation.

All Materials shall be stored in accordance with manufacture guidelines:

- Hazardous materials to be stored in an approved and labeled locker
- Items requiring refrigeration to be stored in a refrigerator labeled "Consumables Only"



All items shall be checked prior to utilization for the following:

- Expiration date
- Proper storage
- Leaks
- Deformities
- Serviceability

Any items found unserviceable or expired shall be removed from the locker and properly disposed of.

#### 4.10.2 Storage of opened consumables

All opened consumables shall be properly stored and labeled as appropriate.

**INCORRECT** way of storing consumables:





CORRECT way of storing consumables:





## 4.11 Maintenance Training

### 4.11.1 Introduction

The DOM is responsible for the overall administration of the maintenance/Dual Inspection/RII training programs.

The DOM or his designee conducts annual review of the program to ensure currency and applicability of the material presented.

This manual is divided into the following chapters:

- Chapter 1 – Maintenance Training Introduction
- Chapter 2 – Basic Indoctrination Training
- Chapter 3 – Initial, Transition, Differences, New Model and Competency Based Training
- Chapter 4 – Recurrent Training
- Chapter 5 – Ground Run Training
- Chapter 6 – Avionics and Avionics Equipment Training
- Chapter 7 – Non Certificated Structures Technician, Air Carrier Repairman and Crew Chief Training
- Chapter 8 – Contract and Contract Provided Maintenance
- Chapter 9 – Forms
- Chapter 10 – AMT Apprenticeship Program

Each chapter includes the type of training to be conducted or guidance associated with training and training issues.

If an FAA approved Repair Station is contracted to perform maintenance or inspections, the DOM will determine that it has appropriate ratings and the persons assigned to the work are properly trained,



using the HSUSA Maintenance Training Program or other acceptable means.

#### **4.11.1.1 14 CFR § 135.429, 135.433**

Maintenance training programs will ensure that each person, including inspection personnel, who determines the adequacy of maintenance or preventive maintenance work done for it is:

- Fully informed about procedures and techniques;
- Fully informed about new equipment in use; and
- Competent to perform his or her duties.

#### **4.11.1.2 Definitions**

For the purpose of this volume, "Major Maintenance and Inspection" means: "Any maintenance up to but not including Overhaul or Rebuilding as described in 14 CFR § 43.2.

For the purpose of this volume, "Routine Maintenance and Inspection" means: "Any maintenance up to but not including Major Alterations and Major Repairs as described in 14 CFR § 43 Appendix A.

For the purpose of this volume, "Appropriately Rated Person" means "Any person certificated under 14 CFR § 65.

For the purpose of this volume, "Apprentice" means any non-certificated individual working under an Appropriately Rated person.

For the purpose of this volume, "Instructor Technician" means "An Appropriately Rated Person that has demonstrated a knowledge of the aircraft or systems on which instructions are to be given."

Safety Management System (SMS) – Formal procedures to manage safety critical activities.

Aircraft Data Monitoring Systems (ADMS) – Aircraft drivetrain/rotor vibration monitoring system.

Audit Policy and Procedure Manual (APPM) – Policy Manual used by Quality Assurance.

#### **4.11.1.3 Manager, Maintenance Training and Standards Duties and Responsibilities**

The Manager, Maintenance Training and Standards reports to the DOM and develops procedures and policies for conduct of business with the Maintenance Training Office.

#### **4.11.1.4 Maintenance Instructors Duties and Responsibilities**

The Maintenance instructors report to the Manager, Maintenance Training and Standards.

##### **4.11.1.4.1 Qualifications for Maintenance Instructors:**

- Minimum two years maintenance experience on the HSUSA aircraft models or similar type.
- Manufacture's school on model or similar type preferred.



- Has good communication and written skills.
- Possess a valid Airframe and Powerplant (A&P) Certificate.
- Be knowledgeable regarding aviation safety standards and safe operating practices.
- Completion of Fundamentals of Instruction Course (FAA-H-8083-9 Aviation Instructors Handbook) or its equivalent.

#### **4.11.1.5 Needs Assessment**

HSUSA needs assessment is a two-part process that determines overall training requirements, as well as individual employee training requirements.

In determining the HSUSA overall training requirements, the DOM or his designee and the managers of each technical area must review the HSUSA Operation Specifications (OpSpecs), job position duties and responsibilities listed in the manual, technical job functions and tasks, customer requirements, past, current, and expected scope of work, employment procedures for hiring, and current employee experience levels.

This needs assessment will result in a description of the knowledge and skill standards for each defined job position or function as determined by the DOM or his designee.

HSUSA may identify additional training needs through:

- The needs assessment outlined in this manual;
- FAA or other external agency oversight findings;
- Investigations that lead to voluntary disclosures;
- Routine or special Q.S.C audits;
- Feedback from employees; and
- The results of audits related to maintenance human factors.

The DOM or his designee ensures the Training Department regularly reviews the results and reports for additional training needs through the Continuing Analysis Surveillance Program.

Aircraft Maintenance Technicians (AMT's) will be assessed and evaluated:

- The DOM or his designee will assess newly hired AMT's.
- Documents provided by the AMT will be used to ensure compliance with 14 CFR requirements for recency of experience.

HSUSA will establish skill levels and qualifications for each job position based upon technical functions and tasks.

In addition, HSUSA will use current Job Descriptions and Corporate Policies and Procedures to assist in



the assessment of each individual job requirements.

If the assessment and evaluation indicates additional training is required for new equipment, material, technical data, or processes, HSUSA training department will take steps to ensure appropriated training is provided.

#### **4.11.1.6 Maintenance Training Programs**

HSUSA training department shall be responsible for developing training course outlines, training guidelines and training record storage and security, for AMT's.

The Maintenance Instructors are responsible for implementing the programs outlined in this manual.

Maintenance Instructors shall generate documentation of the training as necessary to establish the records for certification authorization for qualified persons.

Maintenance training presentations are of a multimedia type, using Adobe Acrobat, Power Point and videos. Training aids, i.e., mock-ups and parts will be used to enhance the presentations.

The following must be completed for an individual to have maintenance qualification for HSUSA aircraft.

- The individual must be an Appropriately Rated Person.
- The individual must attend:
  - Basic Indoctrination Training.
  - Aircraft initial training on the model aircraft they will be maintaining.
  - Annual Recurrent training.

Process measurements are imbedded in the applicable training syllabi and curriculum.

#### **4.11.1.7 Helicopter Maintenance and Inspection Qualifications**

Qualifications are identified as follows:

- Code A – Qualified/Authorized for Major Maintenance and Inspections. This means there are no special restrictions on the work or inspection procedures, which the individual may perform. Acquired through airframe specific formal classroom training and 85 percent OJT completion.
- Code B – Qualified/Authorized for Routine Maintenance and Inspections. This means the individual is restricted to those common procedures, which are done repeatedly. Acquired through airframe specific formal classroom training or 85 percent OJT completion.
- Code C – Qualified/Authorized for Daily Inspections only. This designation is for a specific model and may be authorization by their manager. Acquired through OJT.
- Code D – Qualified/Authorized for engines only. Must have Powerplant certificate. Acquired through one-year engine maintenance OJT with HSUSA and recommendation by the manager.



- Code E – Qualified/Authorized for airframe only. Must have Airframe certificate. Acquired through one-year airframe maintenance OJT with HSUSA and recommendation by the manager.
- Code G – (Supplemental Qualified/Authorized) A “G” code indicates an authorization to ground run that model aircraft. Acquired through ground run training as stated in Chapter 5.
- Code S – Structures Technician – Authorized to perform sheetmetal and composite structural repairs. Acquired through one-year maintenance OJT with HSUSA or specific formal classroom training.
- Code R – Air Carrier Repairmen – Qualified/Authorized to perform work on aircraft within the Repairmen Certificate Limitation. Must have Repairmen Certificate. Acquired through one-year OJT with the certificated holder or specific formal classroom training.
- Code RII – Required Inspection Item – Authorized to perform Required Inspection Item inspections on ten or more passenger aircraft and applicable 14 CFR’s. The RII designation is granted and issued under the authority of the Chief Inspector.
- Code DI – Dual Inspection – Authorized to perform Dual Inspections for nine or less passenger aircraft as defined in OM-A, Chapter 11. The DI designation is granted and issued under the authority of the Chief Inspector.

#### **4.11.1.8 Maintenance Training Records**

The training department retains and distributes training records as applicable to AMT’s that perform maintenance, preventive maintenance and alterations on HSUSA aircraft.

Maintenance training record keeping includes the following:

- Training dates;
- Who performed the training;
- The number of hours of training performed; and
- The content of the training performed.

HSUSA training department maintain a database that shows the qualifications and authorizations of HSUSA AMT’s.

- AMT training records and qualifications are reviewed annually for any discrepancies and is performed by either the Manager, Maintenance Training and Standards, Training Coordinator or a Maintenance Instructor.

Upon completion, the form is placed in the AMT’s training record until the next review is accomplished.

- When a new qualification or authorization has been added to an AMT’s “Helicopter Maintenance and



Inspection Qualification” record, the training department notifies the DOM of the new qualification.

#### **4.11.1.8.1 Primary sources for maintenance qualifications**

##### **Test Results**

- On the Job Training (OJT)
- Training provided by an outside source
- Contract Maintenance Provider training

Test results are filed in the individuals training records as proof of training and maintained by HSUSA training department.

- Testing – passing grade of 75 percent is required.
- Initial or transition training completion, the individual is given a “B” authorization to perform maintenance and inspections on HSUSA aircraft.
- The HSUSA training department may issue a Certificate of Achievement.

OEM’s, vendors, previous employer, or individuals who are in specialized fields of maintenance are considered outside sources of training.

- The training from one of the above can be documented in the form of a certificate issued by the training facility or records of training from a previous employer or some other document, which is acceptable to the DOM or his designee.
- A waiver will be placed in the record indicating the prior training is acceptable.
- The documented training is added to the individuals training records.

When maintenance is to be conducted at a contract facility, outside of HSUSA certificate, training will be conducted and recorded in a manner acceptable to the DOM.

- Completion of the training will be document and a copy of the documentation shall be included in the individuals training records.
- A waiver will be placed in the record indicating that prior training is acceptable.

Selected individuals at the contracting facilities will attend the “Basic Indoctrination” course as specified in this manual. The training may be adjusted in order to accommodate the requirements base on an assessment of training needs to fulfill the contractual obligations of the contracting facility.

Long tenured AMT’s which have received HSUSA provided training that pre-dates formal requirements to document specific training may not have the records and/or forms called out within this manual in their training records.

- In the event an AMT has undocumented and/or unrecorded training based on the above paragraph, a



training documentation waiver may be used to satisfy the documentation requirements.

- The AMT's current training record, work history, performance evaluation and training needs assessments will be reviewed by the AMT's immediate supervisor, Manager, Maintenance Training and Standards, and the Chief Inspector and a decision made as to the individual being properly trained and qualified.

#### **4.11.1.9 Auditing**

##### **4.11.1.9.1 Surveillance**

The Manager, Quality is responsible for executing maintenance-training audits.

- Documentation, Reporting, Corrective Action and Management Evaluation, conducted as per the Audit Policy and Procedure Manual (APPM) and Safety Management Manual (SMS) Manual.

#### **4.11.1.10 Electronic Record Keeping – Maintenance Training Records**

Electronic maintenance training record keeping is based on guidance contained in FAA Advisory Circular AC 120-78 (as revised). Subject: Acceptance and Use of Electronic Signatures, Electronic Record Keeping Systems, and Electronic Manuals, and meets the applicable record keeping requirements of 14 CFR Parts, 43, 91 and 135.

#### **4.11.2 Basic Indoctrination Training**

##### **4.11.2.1 General**

Basic Indoctrination is given to newly hired maintenance employees within 180 days after being hired.

Basic indoctrination is given to individuals working for a contracted Maintenance Provider as determined by the DOM.

Basic indoctrination consists of the following elements:

- Safety Management System (SMS)
- Operations Manual (OM-A)
- Required Inspection Item/Dual Inspection (RII/DI)
- Human Factors/Maintenance Resources Management (MRM)
- Digital Airware Training
- Materials Receiving Inspector Training
- ADMS Training
- Aircraft Data Monitoring Systems (ADMS) Training



Re-hires that have been gone more than 36 months will attend all elements of training.

Re-hires, at the discretion of the DOM or his designee, may not need all elements listed above.

Individuals that are employed by HSUSA, but are inactive for an extended period shall be treated as a re-hire unless otherwise specified by the DOM or his designee.

Appropriately rated individuals will attend classes that are listed on Form Basic IndocTraining. The form is completed as the individual completes the classes specified on the form.

The classes listed as optional on are not required for completion.

The DOM or his designee may add additional training to the form.

When the training is completed, the form is added to the individual's training record.

The indoctrination training will include instructions on the use of operator's manuals, policies, procedures and forms. During all training, there will be emphasis to the importance of following documented procedures without deviation.

#### **4.11.2.2 Scope**

Basic Indoctrination is an overview of the policies and procedures used by HSUSA to familiarize AMT's for the purpose of conducting, maintenance, preventative maintenance, and alteration/repairs on the HSUSA aircraft.

Elements in paragraph 2.1 may be rearranged in order to accommodate instructor, classroom and material available to conduct the training. The specific class determines total hours.

#### **4.11.2.3 Process Measurement**

Successful completion of the training is determined by the requirements of each element of the training.

#### **4.11.2.4 Competency**

HSUSA assesses competency by providing tests at the end of Basic Indoc and Recurrent training with a required passing rate of 80% or better.

#### **4.11.2.5 Feedback**

At the completion of the course, the employee can provide feedback via the "14 CFR 135 Employee Training Course Evaluation" provided by the instructor.

#### **4.11.2.6 Safety Management System – Initial**

HSUSA's objective is to ensure that appropriate personnel are trained and competent to perform their SMS duties. The scope of training is provided be commensurate with the individual's involvement in the SMS to include AMT's assigned to ground run an aircraft.



This process is designed to provide assurances that SMS training necessary to meet competency requirements are appropriately developed, documented and regularly evaluated.

Training process inputs are obtained through the outputs of the SMS and the personnel competency expectations of the SMS.

Basic course duration is approximately four hours.

Instruction given will familiarize the AMT with the four components of the SMS, Safety Policy, Safety Promotion, Safety Risk Management and Safety Assurance.

Instruction given will familiarize the AMT with the ten Life Saving Behaviors.

Instructions given will familiarize the AMT with HSUSA "Just Culture" process.

#### **4.11.2.7 Maintenance Manuals**

The objective of this training is to provide the newly hired or contracted AMT's, an increased awareness of the requirement for each individual to comply with the policies and procedures contained in the HSUSA Maintenance Manuals (OM-A, SOP).

Instructions given will familiarize the AMT's with OM-A and SOP impact maintenance on HSUSA aircraft.

Basic course duration is approximately eight hours. Subject matter contains General Sections, Record Keeping, Forms, Tool Management, General and Aircraft Memos.

#### **4.11.2.8 Required Inspection Item**

RESERVED FOR FUTURE UTILIZATION - Correspondence below is subject to change once HSUSA has issued and approved GMM.

The objective of this training is to give HSUSA AMT's or other contracted appropriately rated person, training on the Required Inspection Item inspector process as required by 14 CFR § 135.429, 135.433.

Maintenance Managers select AMT's and submit names to the Chief Inspector. After review of the individual's records, the Chief Inspector approves and forwards the form to the training department.

The new inspector is issued a Required Inspection Item / Dual Inspection Authorization document indicating the individual's limitations from the Chief Inspector.

Course duration is approximately four hours and consist of the following subjects:

- Special Tools
- Inspection Fundamentals
- Applicable sections of 14 CFR Parts 43, 91 and 135 and their impact upon the RII inspector.
- HSUSA GMM section for RII personnel duties and responsibilities.



- Logbook Entries
- A test is given at the completion of the course, a passing grade of 75 percent is required.

After completion of the test, the training instructor signs the certificate of training block on the test form. This serves as notification to the RII candidate of the successful completion of the course. The RII candidate signs the acknowledgement section of the training form.

Acknowledgement section, when signed by the RII candidate, serves as confirmation from the RII candidate, that the RII candidate is appropriately certificated, qualified, trained, authorized and current as airframe and/or powerplant mechanic.

The document is placed in the RII candidates training records.

Contract Maintenance personnel will receive the same training as HSUSA RII inspectors.

#### **4.11.2.9 Human Factors/Maintenance Resource Management**

The objective of this course is to familiarize the AMT with Human Factors and Maintenance Resource Management to the level of understanding to manage all available resources and to develop skills in recognizing and responding to the conditions that lead to an error.

Course duration is approximately four hours and consists of the following subjects:

- Human factors in aviation.
- Operational Integrity
- Situational Awareness
- Error Chain Recognition
- Communication Skills
- Briefings
- Synergy/Team Concepts
- Leadership
- Conflict Resolution
- Decision Making
- Stress Management

#### **4.11.2.10 Digital Airware**

The objective of this course is to familiarize the AMT with the Digital Airware software used by HSUSA for electronic record keeping aircraft logbook.

Course duration is approximately 4 hours and consists of the following subject matter:



- Terminology/Definitions
- Screen Layouts
- Procedures for;
- Materials
- Material Request
- Maintenance Entries
- Non-standard Tasks
- Standard Procedures
- Work Units – Airframe and Component
- Component Replacements

#### **4.11.2.11 Materials Receiving Inspection Training**

The objective of this course is to familiarize AMT's that will be accepting parts and materials for use on HSUSA aircraft by inspecting documents and material for acceptance as airworthy to place in storage.

Course duration is approximately four hours and consists of the following subject matter:

- Materials receiving documents
- Detecting and Reporting of Suspected Unapproved Parts
- Materials Receiving, Handling, Storage and Shipping Processes

#### **4.11.2.12 Aircraft Data Monitoring Systems (ADMS) Training**

The objective of this course is to familiarize AMT's with different aircraft ADMS systems, interpretation of vibration data and HSUSA processes.

Course duration is approximately one hour.

### **4.11.3 Initial/Transition/Differences/New Model/Competency Based Training**

#### **4.11.3.1 Initial Training**

Initial training and transition training attendee's must have completed the "Basic Indoctrination" course.

Initial training is given to familiarize AMT's with one model of HSUSA aircraft to the level of performing routine maintenance, inspections and trouble shooting.

Initial training course duration is approximately 80 hours.

Initial training course subject matter consists of the following elements:



- Model type general information
- Ground Handling and Servicing
- Weight and Balance
- Hydraulic Systems
- Flight Controls
- Fuel Systems
- Main Rotor
- Main Rotor Drive
- Tail Rotor
- Tail Rotor Drive
- Powerplant
- Electrical Systems
- Landing Gear and Float Systems
- Anti-Ice and Rain Protection
- Heating and Air Conditioning
- Indicating and Record Systems
- Track and Balance
- Automatic Flight Control Systems
- Technical Directives
- Airworthiness Directives
- Unique systems relevant to the model type

#### **4.11.3.2 Waiver of Training**

A waiver of training may be granted if the employee has attended a formal factory school, and that the employee is current on the aircraft model being waived. The DOM or designee may grant this waiver. Waiver is documented and retained in the employee's training file.

The Standards Manager, through interview and assessment testing, will determine currency on aircraft.

#### **4.11.3.3 Transition Training**

Transition training is given to AMT's and inspectors who already are authorized to maintain or inspect



one model aircraft.

- To add another model aircraft to a person's authorizations and receive transition training, that person must hold a "B" authorization on at least one other aircraft operated by HSUSA.

Transition training course is approximately 40 hours.

Upon completion of the training course, the individual will receive a Certificate of Achievement.

#### **4.11.3.4 Differences Training**

In the event that HSUSA lists on its Operation Specifications, aircraft that are similar in type, but vary in specific airframe and/or powerplant areas, a differences training course will be given to train AMT's on those differences. HSUSA maintenance training department determines this type training and duration.

Upon completion of the differences training course, the individual will be given a "B" qualification and a Certificate of Achievement.

#### **4.11.3.5 New Model Aircraft Training**

New model aircraft purchased by HSUSA, require training of AMT's and inspectors for the purpose of maintenance and inspections, prior to placing the new model aircraft in service.

Upon completion of new model aircraft training, the individual will receive a "B" qualification.

- Completion of training can be by formal classroom through Initial Training or actual hands on working on the new model aircraft through Competency Based Training (OJT).

#### **4.11.3.6 Competency Based Training (OJT)**

Competency Based training is used to familiarize AMT's with HSUSA aircraft to the level of performing maintenance and troubleshooting.

AMT's who have no experience or who have experience on other type aircraft that HSUSA does not operate, these AMT's are allowed hands on practical experience, under the guidance and instructions of an instructor technician in order to gain experience and eventually work unsupervised.

After initial introduction, the AMT is assigned to field maintenance location and an instructor technician is assigned.

- The AMT is issued a OJT progress card. OJT Instruction Progress cards are available for each model aircraft.
- The AMT is responsible for upkeep and recording of tasks that are completed. The OJT Progress card is kept in the possession of the AMT until it is forward to Maintenance Training upon completion.

The instructor technician will have at least a "B" qualification for the aircraft on which instructions are given.



The instructor technician will give guidance in the use of applicable technical data, special tools and company policy and procedures.

The instructor technician assess the AMT for competence by "on the job" evaluation. Upon completion of the "Daily" section of the OJT progress card, a "C" qualification will be issued to the AMT.

Upon 85 percent completion of the OJT progress card, a "B" qualification will be issued to the AMT.

Upon 85 percent completion of the OJT progress card along with completion of airframe specific classroom training, and "A" qualification will be issued to the AMT.

## **4.11.4 Recurrent Training**

### **4.11.4.1 Maintenance Recurrent**

Qualified and appropriately rated persons shall attend recurrent training at 24-month intervals.

AMT's will receive training for each aircraft that they hold an "A" or "B" qualification on. Air Carrier Repairmen will receive training for each aircraft that they hold an "R" qualification on.

Recurrent training curriculum consists of the following elements;

- Safety/SMS
- Just Culture
- Avionics
- Human Factors/MRM
- RII/DI
- ADMS
- GMM
- Each of the different aircraft models that the certificate holder operates.
- OM-A
- Technical Directives
- Airworthiness Directives

Recurrent training course duration is approximately 12 hours.

In the event formal classroom training for recurrent training cannot be conducted, correspondence training may be used in place of classroom training.

## **Requirements for Recurrent Training**



– A person possessing an "A", "B" or "R" authorization shall attend annual recurrent training, or a person that previously held an "A", "B" or "R" authorization and lost that authorization within the past 17 months, may attend recurrent training for the purpose of being re-qualified on a specific aircraft.

– If immediate reauthorization is required, a correspondence test may be issued.

Following successful completion of the correspondence test and as soon as practicable, attendance in classroom recurrent training is required.

– A person previously possessing a 14 CFR 135 Helicopter Maintenance and Inspection Qualification, having lost the authorization between 18 and 35 months earlier, shall attend classroom recurrent training.

– Any person having lost the authorization 36 months or more, at the discretion of the DOM, attend Initial or Transition training on the respective aircraft.

– Training that is less than forty hours in duration, may be considered recurrent training, and may justify a Certificate of Attendance.

#### **4.11.4.2 Maintenance Ground Run Recurrent**

AMT's that are ground run authorized must receive recurrent training for each aircraft type at intervals of 24 months.

– Failure to complete the training within the allotted time would cause loss of the ground run authorization.

Ground run recurrent training consists of classroom training followed by aircraft practical training.

– A ground run instructor using the applicable aircraft ground run training program will conduct the training.

– The instructor will conduct the training and document the results of the training using.

– Comply with the guidance on the form for proper disposition of the form.

– Two starts supervised by a ground run instructor is required.

**NOTE:** The ground run instructor will be a training department Maintenance Instructor. The Maintenance Instructor must hold a "Ground Run Instructor" qualification on the aircraft model in which instruction is to be given.

In the absence of the above noted contact the training department. A qualified pilot may perform this training if approved by the DOM and Chief Pilot.

Maintenance ground run recurrent includes but is not limited to the following subjects:

– Main rotor blades untied and rotor brake off. Verify the blades are free to rotate and airframe is tied down and secured before attempting start. Perform uninterrupted 360-degree walk around.

– Starter duty limitations as per the Rotor Flight Manual (RFM). Possible damage to starter/generator,



electrical systems. Battery reserves.

- Aircraft starting system. If the aircraft has automatic latching, understand when it will engage and how to disengage as required. Emergency start abort.
- The use of ground power if available. Apply power to the aircraft as per the RFM. How a Ground Power Unit (GPU) can prevent issues during start. Proper use of a GPU and indications that the GPU is not powering the aircraft.
- Communicate your intent to begin the starting operation. Announce "Clear" and wait, assuring that other personnel are clear and in a safe condition.
- Completion of two ground run starts to shut down. Proper post run monitoring of aircraft and engine. Security of aircraft after each ground run.

Course duration is approximately one hour and 30 minutes.

## **4.11.5 Ground Run Training**

### **4.11.5.1 General**

When the Maintenance Manager determines a need to have personnel conduct ground runs, the Maintenance Manager will notify the training department.

- AMT's selected by the Maintenance Manager must have a "B" qualification on the model aircraft selected for, and must have one or more years' experience on the model aircraft selected.

The training department reviews and determines the applicant's eligibility.

- The training department may waive the requirements in paragraph 5.1. Upon approval, the training event is scheduled.

A training department instructor pilot will conduct the training program using the appropriate forms.

Results of the training using the applicable forms are forwarded to the training department and a notice of authorization is sent to the AMT and the AMT's Maintenance Manager.

Course duration is approximately two classroom/simulator hours.

If a simulator for particular aircraft model is not available, training will be accomplished at a physical aircraft.

Classroom material is provided to the AMT by the training department.

### **4.11.5.2 AMT Ground Run Training**

Pre start classroom discussion.

- Engine operation and engine cockpit displays



- Checklist
- Procedures
- Aircraft tied down
- Wheeled aircraft, chocked and parking brake on
- Crew restraint on and adjusted
- No one under the rotor during start or shutdown
- Environment
- RFM Limitations (Wind, temperature, etc.)
- Wind velocity no greater than 25 knots
- Allowed ground runs
- Engine wash
- Leak checks
- System checks
- Ground rotor track and balance
- Prohibited checks
- Manual starts
- Crew brief for all involved personnel
- Hazards
- Weak aircraft battery or weak ground power unit
- Hot starts
- JSA's
- Duties
- Check procedures
- RFM preflight
- Uninterrupted 360 degree walk around prior to start
- Electrical Power
- Battery
- Ground Power Unit



- Lighting
- Controls
- No collective application
- Collective down and frictioned, lock applied if applicable
- All other controls frictioned as required
- Trims at neutral/centered as required
- Normal operations
- Start
- Run
- Shutdown
- Malfunctions
- Premature light off
- Power failure during start
- Hung start
- Hot start
- After fire
- Unusual vibrations/noises
- Control motoring
- Power interruption during shutdown

#### **4.11.5.3 Instructor Guided Training**

##### Classroom Period

- Review training

##### Simulator Period

- Perform the following from both positions:
- Normal Start
- Hung Start
- Hot Start
- After Fire



– Normal Shutdown

#### **4.11.5.4 Maintenance Ground Run Instructor**

Maintenance Instructors will be selected by the Maintenance Standards Manager to be trained and qualified as an airframe model specific "Ground Run Instructor" in order to provide Maintenance Ground Run Recurrent to AMT's.

The Ground Run Instructor qualification gives Maintenance Instructors the authority to perform aircraft ground runs and Maintenance Ground Run Recurrent Training on the aircraft models in which they are qualified.

In order to maintain the "Ground Run Instructor" qualification, Maintenance Instructors must receive Ground Run Instructor Recurrent Training at intervals of 24 months, not to exceed 28 months; on each aircraft model they hold the instructor qualification.

A training department Instructor Pilot will conduct the Initial and Recurrent Ground Run Instructor Training using the appropriate Ground Run Instructor Training using the appropriate Ground Run Instructor Qualification Progress Report form.

The completed forms for Initial training are placed in the Maintenance Instructors training record and the qualification is issued.

The completed forms for Recurrent Training are placed in the Maintenance Instructors training record and maintained for three years. The Hands-On section of the form is not required for completion of Recurrent Training.

#### **4.11.5.5 Maintenance Ground Run Instructor Training**

The Maintenance Ground Run Instructor Initial and Recurrent Training will be accomplished following the content of the Ground Run Instructor Qualification form listed in 5.4.

Training will consist of classroom time, a minimum of one SIM/FTD session and a minimum of one ground run session in a physical aircraft.

– If a SIM/FTD is not available for a particular aircraft model, the SIM/FTD session will be substituted utilizing an aircraft.

Training must be accomplished until the Maintenance Instructor is proficient in all categories noted on the Ground Run Instructor Qualification form.

Ground Run Instructor Qualification form categories and breakdown:

- Preparation (SIM Session) – Prestart discussion and knowledge of aircraft systems and limitations.
- General (SIM Session) – General Evaluation of Instructor throughout training.
- Run-Up (SIM Session) – General Starting and Shutdown of aircraft.



- Unexpected (SIM Session) Malfunctions – Discussion of possible malfunctions and appropriate action to take during ground run. Requires normal aircraft shutdown.
- Emergency Procedures (SIM Session) – Discussion and review of emergency procedures and appropriate action to take during ground run. Requires immediate aircraft shutdown. The following emergency procedures shall be demonstrated and evaluated during the SIM session.
- Hung Start
- Hot Start
- After Fire
- Hands On (Aircraft Session) – Evaluation at physical aircraft to include items not possible in SIM.

If SIM session is supplemented with an aircraft, the Emergency Procedures will only include discussion and review.

Once training is completed to a level of proficiency for performing maintenance ground run instruction, the Maintenance Instructor and Instructor Pilot will sign the form acknowledging as such.

#### **4.11.6 Avionics and Avionics Test Equipment Training**

##### **4.11.6.1 Avionics Training**

Basic avionics training is intended to familiarize Repairmen or AMT with the repair and installation of avionics systems. In addition, training includes where to find information for testing and troubleshooting, following the troubleshooting, repair using manufacturer's electrical standards manuals, inspections and checks required to return the system to service.

The Basic Avionics Training course will be designated classroom training, duration is approximately eight hours and the course subject matter will consist of the following elements, but not limited to:

- Title 14, Code of Federal Regulations
- General Maintenance Manual – OM-A
- Use of Manufacturer Electrical Stand Practice
- AC 43.13
- Repair of aircraft wiring
- Installation and Repair of Avionics Systems
- Testing, Checks and Inspection of Avionics Systems

Aircraft Specific Avionics training consists of the basic avionics training noted above, and model specific avionics systems training, to include theory, testing and troubleshooting that allows the Repairmen or AMT to certify equipment for return to service after maintenance and inspection.



The Aircraft Specific Avionics Training course will be designated classroom training, duration is approximately 40 hours and the subject matter will consist of avionics systems and wiring in the following elements, but not limited to:

- Electrical Systems, components, and wiring related to all ATA chapters
- Aircraft Avionics Package
- Autopilot System
- Optional Equipment
- HUMS System
- Related Technical Directives and Airworthiness Directives
- Unique Avionics Systems relevant to model type

#### **4.11.6.2 Avionics Test Equipment Training**

The objective of this course is to train selected AMT's in the proper use of specific avionics test equipment to perform maintenance, testing and troubleshooting on the certificate holder's aircraft transponder and Air Data Computer (ADC) systems.

The AMT is instructed on how to connect the test equipment to the aircraft, test equipment switch locations, and switch functions.

AMT's are trained on one or more of the following test equipment:

- Laversab 6300 ADC Test Set
- TR-220 Transponder Test Set
- IFR 6000 Test Set

Course duration for training on one of the above test equipment is approximately four hours.

Course material consists of applicable sections of GMM and pertinent parts of 14 CFR's.

At the conclusion of training, a memo of the training is placed in the individuals training records.

#### **4.11.6.3 ATC Transponder and Air Data Computer (ADC) Training**

The objective of this course is to provide training to the AMT and contract personnel to perform on-wing test and inspection of transponders and ADC's used on the certificate holder's aircraft.

Personnel that are authorized to perform Transponder Test and Inspections or Air Data Computer Test and Inspection, attend the applicable course(s) that cover operation, testing and test equipment procedures used by the certificate holder.

Course material consists of GMM policies, procedures, and applicable 14 CFR's.



At the conclusion of training, a memo of the training is placed in the individuals training records.

#### **4.11.7 Non Certificated Technician, Air Carrier Repairman Training**

##### **4.11.7.1 Non Certificated Technician**

The objective of this course is to train and give guidance about applicable 14 CFR's, GMM, and OM-A policies and procedures.

Trainee's must meet the minimum standards of experience as required by 14 CFR § 65.101.

All Non-Certificated Technicians shall attend Basic Indoc as outlined in the "Basic Indoctrination Training" chapter of this manual, to include the following elements.

- Regulations
- Operation Specifications
- General Maintenance Manual/OM-A/SOP
- Logbook Entries

Additional elements will be documented.

##### **4.11.7.2 Air Carrier Repairman Training**

RESERVED FOR FUTURE USE: The content below is subject to change once HSUSA officially implements the procedures below.

The objective of this course is to train and give guidance about applicable 14 CFR's, GMM, OM-A, and SOP policies and procedures.

Trainee's must meet the minimum standards of experience as required by 14 CFR § 65.101.

The Air Carrier Repairman are limited to the following specific job functions:

- Avionics
- Sheet Metal and Composite Repair

All Air Carrier Repairman shall attend Basic Indoc as outline in the "Basic Indoctrination Training" section of this manual.

Air Carrier Repairmen with the job function of Sheet Metal and Composite Repair shall receive training in the following areas:

- Title 14, Code of Federal Regulations
- Supplemental Books



- AC 43.13
- Manufacturer Structural Repair Manual
- Aircraft Logbook and Digital Airware Entries

Avionics Air Carrier Repairman shall receive training in the following areas as outlined in the “Avionics and Avionics Equipment Training” section of this manual:

- Basics Avionics Training
- Avionics Test Equipment Training
- ATC Transponder and Air Data Computer (ADC) Training

Additional training elements for Air Carrier Repairman will be documented.

#### **4.11.8 Contract and Contract Provided Maintenance Training**

##### **4.11.8.1 Contract Maintenance Training**

Contract maintenance training is training which is acquired from sources outside of HSUSA training staff and/or facilities, i.e., OEM’s, vendors or training from sources involved in specialized fields of training.

The training may involve, on the job training (OJT) with instructors or representatives, vendors, or specialized fields of maintenance.

- A “specialized field of maintenance” is defined as a specific method, process or procedure used during the performance of maintenance on an aircraft, systems or components, using special test equipment or equipment specifically designed for unique tasks.

Completion of the training will be documented and a copy of that documentation shall be included in the AMT’s training records.

DOM or his representative may review the facilities, course material and instructors qualifications to ensure that they are adequate for maintenance training.

##### **4.11.8.2 Contract Provided Maintenance**

When maintenance is to be conducted at a contracted facility, training will be conducted, documented and recorded in a manner acceptable to the DOM of his designee.

DOM or his designee will ensure the contracted facility has appropriately rated and qualified staff.

Selected individuals at the contract facilities will attend HSUSA “Contract Maintenance Provider – Basic Indoctrination” course.

##### **4.11.8.3 Contract Maintenance Provider – Basic Indoctrination**



Contract Maintenance Provider – Basic Indoctrination is given to individuals working for a contracted Maintenance Provider as determined by the DOM.

Contract Maintenance Provider may require training in only certain elements of the Basic Indoctrination course as determined by the DOM or his designee.

During all training there will be emphasis applied to the importance of following documented procedures.

Contract Maintenance Provider – Basic Indoctrination training will include instructions in the use of HSUSA manuals, policies, procedures and forms.

#### **4.11.9 Training Forms**

Reserved for Training Forms

#### **4.11.10 AMT Apprenticeship Program**

Reserved for AMT Apprenticeship Program

### **4.12 Continuing Analysis and Surveillance System (CASS)**

#### **4.12.1 Regulatory Compliance**

Title 14 of the Code of Federal Regulations (14 CFR) Part 135, § 135.431 require an air carrier operating under Part 135 to establish and maintain a Continuing Analysis and Surveillance System (CASS).

These Federal Regulation sections also allow the Federal Aviation Administration (FAA) to require revisions to an operator's maintenance program based on deficiencies or irregularities revealed by the CASS.

HSUSA has established and maintains a system for the continuing analysis and surveillance of the *performance* and *effectiveness* of its maintenance program (consisting of an inspection program and the program covering other maintenance, preventive maintenance, and alterations), for the correction of any deficiency in those programs.

HSUSA maintenance program consists of ten required elements. Detailed information regarding these elements can be found in the current edition of AC 120-16; the elements are:

- i. Airworthiness Responsibility
- ii. Maintenance Manuals
- iii. Maintenance Organization
- iv. Maintenance Schedule
- v. Required Inspection Items (RII)



- vi. Contracted Maintenance
- vii. Personnel Training
- viii. Accomplishment and Approval of Maintenance
- ix. Maintenance Recordkeeping System
- x. CASS

## **4.12.2 Implementation**

The CASS is one of the maintenance program's ten elements. It engages the entire program functioning as an air carrier maintenance program management tool.

This is done through a continuous system safety based, closed-loop cycle, consisting of the following four major activities:

### **4.12.2.1 Surveillance**

- i. Measuring performance through transaction audits
- ii. Measuring effectiveness through collecting operational data
- iii. See chapter two in this manual for surveillance procedures

### **4.12.2.2 Data Analysis**

- i. Analyzing surveillance data to identify deficiencies, hazards and/or risks
- ii. See chapter three in this manual for data analysis procedures

### **4.12.2.3 Corrective Action**

- i. Correcting maintenance or inspection program deficiencies, hazards and/or risks
- ii. See chapter four in this manual for procedures

### **4.12.2.4 Follow-Up**

- i. Monitoring corrective action measures taken
- ii. Verification of surveillance planning
- iii. See chapter five in this manual for procedures

These four major activities (depicted in figure 1-1 at the end of this chapter) continually monitor and evaluate the performance and effectiveness of HSUSA maintenance program:

- i. Performance monitoring ensures that everyone, including all of the air carrier's maintenance providers, comply with the air carrier's manual, program and all applicable regulations.



ii. Effectiveness monitoring ensures the maintenance program is producing airworthy aircraft.

### 4.12.3 Organizational Structure

The CASS involves the functions of HSUSA Maintenance Department and Safety Department (including the Quality Systems Compliance group), but is operated and maintained by

Maintenance Department and adheres to the organizational structure of the company.

Figure 1-2 (at the end of this chapter) shows the CASS organizational flow chart for HSUSA highlighting key personnel for its execution.

### 4.12.4 Duties and Responsibilities

In relation to the CASS, the duties and responsibilities of key personnel involved are as follows:

The DOM

1. The overall responsibility for the CASS lies with the DOM.
2. The responsibility for the operation of the CASS lies with the DOM.

The CASS Administrator

1. The responsibility and authority of the CASS resides with the CASS Administrator.
2. The CASS Administrator is responsible for accomplishing CASS Performance and Effectiveness monitoring.
3. The CASS Administrator carries out data collection, analysis, reporting and monitoring of data as defined by this documentation relative to the ongoing 14 CFR Part 135 operations, and vendors supporting those operations.
4. Daily responsibilities include:
  - Data integrity assurance
  - Identification of deficiencies, potential problems, alerts, trends, exceedances, etc.
  - Development and execution of reporting schemes to submit program findings based on available data.

The Manager, Quality

1. The Manager, Quality is responsible for executing CASS Audits.

The QSC group carries out surveillance for performance monitoring of the ten elements of an air carrier maintenance program (seen in Chapter 2 of this manual), specifically auditing at:

1. Field bases
2. Vendor facilities
3. Training

#### **4.12.5 Reporting Process**

For the purposes of the certificate holder, the CASS involves a reporting function to provide management with oversight of and input into the CASS findings, trending, analyses, corrective actions and results.

The details of the CASS reporting function are found in the following section:

##### **4.12.5.1 Monthly Report**

The CASS Monthly Report is a compilation of descriptive statistics from the data analysis of day-to-day monitoring.

- The CASS Administrator is responsible for the completion of the Monthly Report.
- The report is accomplished through computer programs such as Microsoft® Excel and/or Word.
- The report is accomplished on a monthly basis (stored on HSUSA Teams Network) and sent to the DOM via electronic means.

##### **4.12.5.2 Quarterly Report**

The CASS Quarterly Report is a compilation of descriptive statistics from the data analysis of long-term monitoring and serves primarily for information sharing and to engage members of the Maintenance Review Board for input.

- The CASS Administrator is responsible for the completion of the Quarterly Report.
- The report is accomplished through computer programs such as Microsoft® Excel and/or Word.
- The report is accomplished on a quarterly basis (stored on HSUSA Teams Network) and sent to the Maintenance Review Board via electronic means.

##### **4.12.5.3 Maintenance Review Board**

The CASS Maintenance Review Board (MRB) serves to provide information and an overview of CASS progress as well as provide a platform for input into ongoing findings and resulting CAPs.

- The CASS Administrator is responsible for the MRB meetings, agendas, responses to inquiries and meeting's minutes.
- The MRB meeting is accomplished by way of presentation (stored on HSUSA Teamsnetwork).
- The MRB meets Bi-Annually (minimum) for review.

#### **Members**

The sustaining members of the MRB include:

1. DOM
2. CASS Administrator



3. Manager, Quality
4. Chief Inspector
5. A Maintenance Training Instructor

– Additional personnel may be requested to attend in order to provide administrative assistance or some other technical guidance concerning specific agenda items.

#### **Attendance**

- MRB attendance is required by the sustaining members, individual attendance may be delegated to an appropriate party to represent that individual due to schedule conflicts.
- In the event that additional scheduling conflicts arise, a reasonable delay of the meeting is accepted.

#### **Minutes**

- Meeting minutes are recorded and distributed within one week (preferably) following the convening of the MRB.

#### **4.12.5.4 Management Review**

The Management Review is an operational leadership team meeting to review trends, lessons learned and/or improvement opportunities for data collected from various inputs within the company.

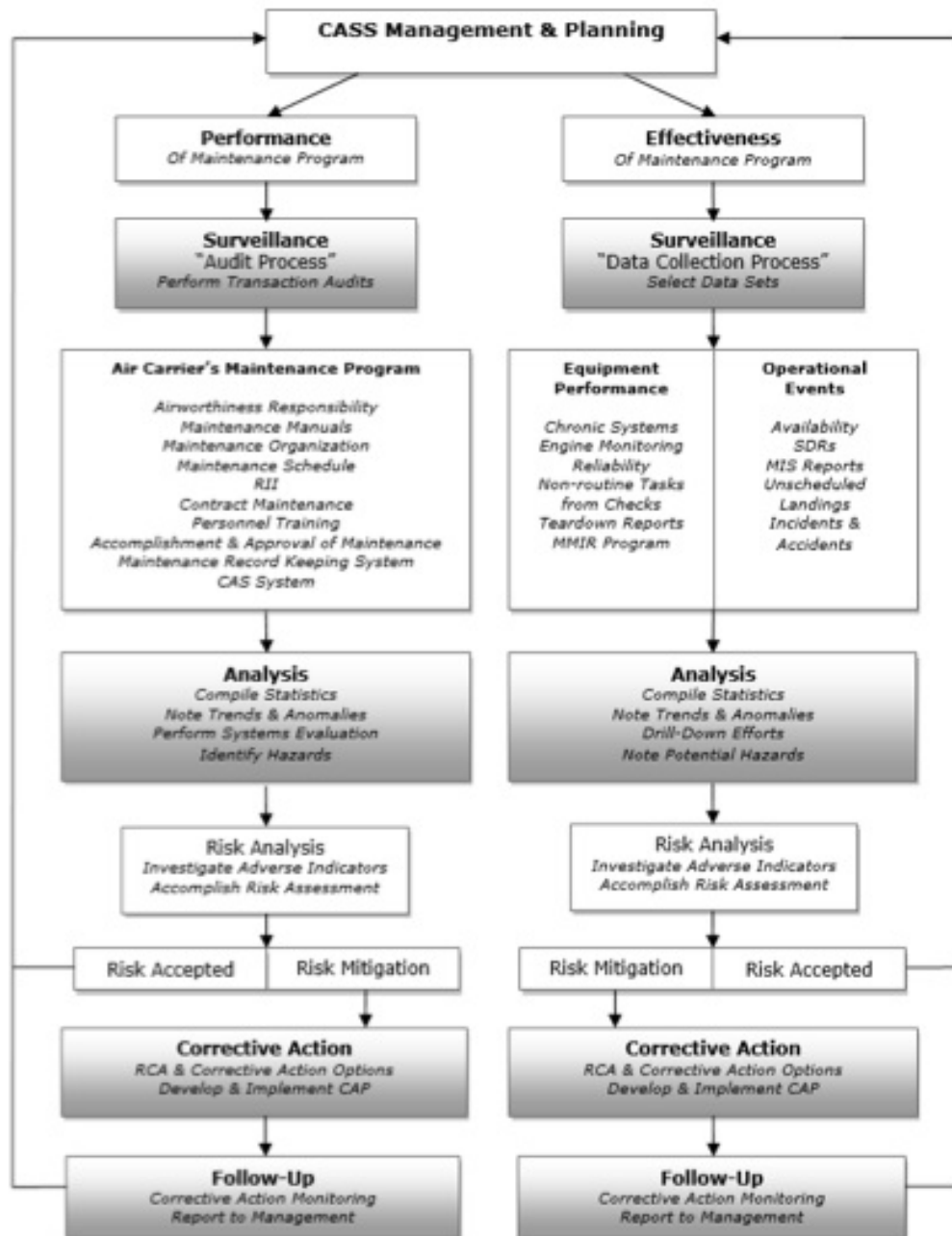
CASS data is a requirement of the Safety Management System (SMS) and Quality Management System (QMS) and as such, is included in the management review.

- The Director of Health, Safety, Environmental and Quality is responsible for the management review.
- The Management review is accomplished by way of presentation (stored on HSUSA Teams network).
- The review is conducted on a quarterly basis.

#### **4.12.5.5 Reporting and Flow Charts**



**Figure 1-1 CASS Activities and Process Flow Chart**



Legend: Shaded boxes the four major activities of the CASS



**Figure 1-2 CASS Organizational Flow Chart**

\*The Manager, Quality executes the audit function of the CASS but not in part of the Maintenance Department.

Legend: Shaded boxes denote CASS personnel.



## 4.12.6 Surveillance

### 4.12.6.1 Overview

HSUSA conducts surveillance to collect data for use in evaluating its maintenance program.

The CASS accomplishes surveillance from two regulatory perspectives: performance and effectiveness.

This chapter outlines the means by which surveillance is carried out so that the information and data collected can be analyzed (see chapter three of this manual for Analysis procedures).

### 4.12.6.2 Performance Surveillance

Surveillance to verify performance is accomplished using audits to ensure the maintenance program, manuals and regulations are being followed.

– Performance surveillance is accomplished by conducting CASS audits on all ten elements of HSUSA maintenance program.

1. The Quality Systems Compliance (QSC) group carries out the CASS Audits of elements one to ten (except for element six).
2. The CASS Administrator carries out the CASS Audit of element six.

– QSC auditors perform work-in-progress audits using checklists to ensure the requirements of each element are in compliance with the program's manual and regulations. These checklists are extracted directly from the maintenance program.

Findings of CASS Audits are sent to the CASS Administrator via electronic means on a monthly basis when conducted.

– CASS Audits are carried out on six-month and annual based intervals.

1. These distinctions are noted by a "C" or an "L" in the numbering system of program element audit questions.

Specifically, the ten maintenance program elements and the corresponding CASS Audit questions are as follows:

#### 4.12.6.2.1      **Airworthiness Responsibility**

- Is there a current Aircraft Status Report for each aircraft?
- Does each aircraft have an up-to-date Aircraft Due List?
- Are the Aircraft Logbook entries complete prior to flight?
- Do the Aircraft Logbooks have proper entries and referenced data?
- Do the Aircraft Logbook entries have proper RII or Dual inspection entries?
- Do the Aircraft Logbooks have proper Airworthiness Release entries?
- Do the Aircraft Logbooks have proper post maintenance checks entries?

#### 4.12.6.2.2      **Maintenance Manuals**

- Are the correct maintenance manuals / technical publications / forms available for all model types?
- Are maintenance manuals / technical publications / forms available to the maintenance staff?
- Are maintenance manuals / technical publications / forms current?
- Is there a method to verify the currency of maintenance manuals / technical publications / forms?

#### 4.12.6.2.3      **Maintenance Organization**

- Is the maintenance organizational chart up-to-date?
- Is the description of the duties and responsibilities of each position as listed on the organization chart up-to-date?
- Is the environment within which individuals conduct their tasks adequate; hangars, offices, flight line etc.?
- Are the SMS policies and procedures of the maintenance organization followed and respected?
- Are there adequate resources in order to achieve safety and production goals; staffing, equipment etc.?
- Does the maintenance organization investigate system failures and take all needed remedial action to avoid a repetition; occurrence reporting, safety meetings, safety reports?
- Does the maintenance organization investigate system failures and take all needed remedial action to avoid a repetition; occurrence reporting, safety meetings, safety reports?
- Does the maintenance organization ensure that personnel with RII and Dual Inspection authority do not self-inspect?

#### 4.12.6.2.4      **Maintenance Schedule**

- Does the maintenance schedule set item, task and interval of scheduled maintenance efforts?

- Is the maintenance schedule for each aircraft maintenance followed?
- Is the maintenance schedule for each component maintenance followed?
- Is the maintenance schedule updated in a timely manner?
- Does CASS data collection use analysis findings to modify the maintenance schedule?

#### **4.12.6.2.5 RII / Dual Inspection**

- Are training and qualification records available for all inspectors?
- Is there a process to ensure RII / Dual Inspection may not self-inspect?
- Do maintenance records indicate RII's and Dual Inspections are not self-inspecting their work?
- Does the manual contain procedures that limit countermand decisions of RII / Dual Inspection inspector to a supervisory level?
- Are RII and Dual Inspection procedures as referenced in OM-A, Chapter 11 properly followed?
- Is the list of RII persons up-to-date?

#### **4.12.6.2.6 Maintenance Providers**

- Is there a documented Maintenance Provider Surveillance Program?
- Does the surveillance program provide guidance to evaluate a maintenance vendor in order to determine audit requirements (onsite, mail out) prior to adding to Approved Vendor Listing?
- Does the surveillance program provide guidance to evaluate current/approved maintenance vendors in order to determine audit requirements (onsite, mail out)?
- Does the surveillance program's audit checklists contain details to ensure that maintenance providers and suppliers are qualified and provide services IAW the certificate holder's GMM / OM-A?
- Does the surveillance program provide guidance to document audit discrepancies, corrective action response timelines, and closure requirements?
- Does the surveillance program provide guidance for audit follow up for verification and validation of corrective actions?
- Are records of audits, and applicable certificates, maintained for all approved maintenance providers and suppliers?

#### **4.12.6.2.7 Personnel Training**

- Does AMT have a valid A&P?
- Is AMT current relative to their 135 maintenance qualifications?



- Is AMT current relative to their 135 RII / DI authorizations?
- Has AMT been instructed on proper RII / DI procedures?
- Are AMT's following the OJT program of the company?
- Are AMT's evaluated during their probation period?
- Has AMT successfully attended recurrent training?
- Has AMT successfully attended model type training?
- Has AMT participated in Process Audit?
- Has AMT attended new hire indoctrination class?

#### **4.12.6.2.8 Accomplishment and Approval of Maintenance**

- Are the maintenance facilities and equipment adequate to perform maintenance?
- Are parts and components properly, labeled, stored, handled and dispensed?
- Do special tools and equipment meet calibration requirements?
- Are required special tools provided with instructions for use?
- Are maintenance manuals, OM-A and GMM used to perform maintenance and maintenance references appropriate to manuals utilized?
- Are turnover/pass-down notes being used if applicable?
- Does random sampling of maintenance records properly document work stoppages?
- Are major repairs or alterations properly classified where the guidance is defined?
- Are logbook entries completed by a certified AMT's who are authorized by the air carrier?
- Are airworthiness releases properly executed?

#### **4.12.6.2.9 Maintenance Record Keeping System**

- Are maintenance records and current status properly created and stored?
- Are Airworthiness Directives properly evaluated, documented, tracked and accomplished?
- Are major repairs and alterations properly documented and accomplished?

#### **4.12.6.2.10 The CASS itself**

- Is there a review of audits and associated data of the maintenance program elements one – ten?
- Is there a review of CASS Data analysis by senior management?



#### 4.12.6.3 Effectiveness Surveillance

Surveillance to verify effectiveness is accomplished by collecting operational data and aircraft systems failure data, and measuring availability through routine and non-routine maintenance program activities.

Effectiveness surveillance is conducted through three main categories: Emergency Responding, Day-to-Day Monitoring and Long-Term Monitoring. These categories are a part of systematic method to not only review operational data but to provide collected information into the CASS for analysis and trending as outlined in chapter three of this manual.

Emergency Responding covers all data related to critical and/or emergency safety-related situations such as Incidents and Accidents.

- The Operational Leadership Team (OLT), consisting of the Director of Operations, DOM, Director of Health Safety Environmental and Quality and the Chief Pilot, is responsible for the immediate review and course of action regarding such events.
- Occurrence Reporting is used to capture and identify those events that meet critical or emergency conditions.
- Surveillance is done on a daily basis (event-driven) and data is compiled quarterly (provided to CASS via quarterly SMS presentations) for analysis.

Day-to-Day Monitoring covers all data related to maintenance discrepancies and/or delays.

- The CASS Administrator is responsible for the review and analysis of day-to-day events.
- Occurrence Reporting and AOG Activity Tracker are used to identify events causing mechanical or operational interruptions.
- Surveillance is done on a weekly basis and data is compiled monthly for analysis. Long-Term Monitoring covers all aspects of operational data analysis in that it reviews overall performance and the need for corrective action.
- The CASS Administrator is responsible for performing long-term surveillance.
- Digital Airware and AOG Activity Tracker are used to identify data from scheduled and unscheduled events.
- Surveillance is done on a monthly basis and data is compiled on a quarterly basis for analysis.

#### 4.12.7 Operational Data Collection

Operational data is collected through a variety of inputs but all stem from routine and non-routine activities (i.e. scheduled and unscheduled events).

- Scheduled data is captured during routine activities that include, but are not limited to:



i. Inspections (findings and repairs)

ii. Scheduled removals and overhauls

iii. Engine monitoring programs

iv. QSC audit findings

– Unscheduled data is captured during non-routine activities that include, but are not limited to:

i. Aircraft on Ground (AOG) events

ii. Discrepancy write-ups

iii. Minimum Equipment List (MEL)

iv. Unplanned replacement of parts

v. Vendor Teardown Reports

vi. Incidents and/or accidents

The primary operational data source queried for routine and non-routine activity data is the company-wide, web-based solution called Digital Airware Application.

Specifically, data sources that hold operational-related data include:

– Technical Logs

– Occurrence Reports

– Repair Orders

– Material Requests and Issues

– Hangar Work Orders

Other data sources from which the CASS collects operational data include:

– HSUSA MS TEAMS AOG Activity Tracker application accessible via HSUSA network

– HSUSA MS TEAMS MEL Activity Tracker application accessible via HSUSA network

– Aircraft Data Monitoring Systems (ADMS) department data Operational data, in general, is categorized by Air Transport Association (ATA) codes, as provided by the FAA Joint Aircraft System/Component Code (JASC).

Once data is collected by the CASS it progresses into the Analysis phase (see chapter 3 for procedures of analysis).

#### 4.12.8 Data Retention

Data residing in the Digital Airware application, the AOG and MEL Activity Tracker application and the



QSC database is stored on HSUSA network.

Operational data collected by CASS is stored on HSUSA network.

The Information Technology (IT) department handles the backing-up of HSUSA network on which all collected CASS data is stored.

#### **4.12.9 Performance Analysis**

Analysis to verify performance is done by analyzing the results of conducted audits.

- The CASS Administrator is responsible for the analysis of audit findings.
- Analysis is accomplished through computer programs, such as Microsoft © Excel and/or Word.
- i. Descriptive statistics are used to group data by finding category (or some other meaningful metric) to identify the top detractors of a given data set.
- ii. Trends are calculated for the identified detractors and used to mark improvement or decline of previous deficiencies and to identify the possible onset of new deficiencies.
- iii. Drill down is then performed to identify causal factors of the noted trends

Audit findings are centered around the ten maintenance elements (as noted in chapter two of this manual) and analyzed by descriptive statistics such as counts, rates and trends, in order to:

- Verify that the certificate holder is in compliance with the manual, regulations and requirements and to:
- Identify system deficiencies, assign causal factors to facilitate:
  - i. Risk Assessment
  - ii. Corrective Action Planning

#### **4.12.10 Effectiveness Analysis**

Analysis to verify effectiveness of the certificate holder's maintenance program is done by analyzing the collected operational data and aircraft systems failure data.

- The CASS Administrator is responsible for performing the analysis of operational and aircraft systems failure data.
- Analysis is accomplished through computer programs, such as Microsoft © Excel.
- i. Descriptive statistics are used to group data by ATA Chapter (or some other meaningful metric) to identify the top five detractors of a given data set.
- ii. Trends are calculated for the identified detractors and used to mark improvement or decline of previous deficiencies and to identify the possible onset of new deficiencies.
- iii. Drill down is then performed to identify casual factors of the noted trends.



Operational and aircraft systems failure data are centered around routine and non-routine activity, in order to:

- Ensure that HSUSA maintenance program is releasing airworthy aircraft and to provide the maximum level of availability for operations.
- Identify system deficiencies, to facilitate:
  - i. Risk Assessment
  - ii. Corrective Action Planning

#### **4.12.11 Data Analysis Techniques**

Analysis involves both quantitative and qualitative techniques to arrive at an unbiased, objective and meaningful conclusion.

The most common kinds of analyses previously mentioned in this chapter are briefly explained in the following section:

##### **4.12.11.1 Descriptive Statistics**

- Descriptive statistics are techniques used to summarize sets of data so that observations can be made. Such summaries may be either quantitative or visual (“Descriptive statistics”, <[http://en.wikipedia.org/wiki/Descriptive\\_statistics](http://en.wikipedia.org/wiki/Descriptive_statistics)>).
- Relative to the CASS, descriptive statistics are mostly tabular (e.g. tables with numeric values) and visual (graphs) with supporting summaries, including but not limited to:
  - i. Range of Data
  - ii. Counts, Means & Percentages
  - iii. Rates
  - iv. Standard Deviations

##### **4.12.11.2 Trend Analysis**

- Trend analysis is a technique, which attempts to spot a pattern that would indicate an underlying problem within information over some given period of time (“Trend analysis”, <[http://en.wikipedia.org/wiki/Trend\\_analysis](http://en.wikipedia.org/wiki/Trend_analysis)>).
- Relative to the CASS, trend analysis uses descriptive statistics to generate metrics such as:
  - i. Availability
    - a. Availability is the proportion of time a system is in a functioning condition during an interval of time.



b. Availability is quantified by calculating a percentage, namely one minus unavailability for a given data set.

c. Specifically, unscheduled and scheduled availability of aircraft fleets are useful metrics of availability.

ii. Reliability

a. Reliability describes the probability of a system completing its expected function during an interval of time

b. Reliability is quantified by calculating the failure rate of a system or item during an interval of time.

c. Specifically, AOG rates of aircraft fleets or systems and mean time between failures (MTBF) for unscheduled removals of components are useful metrics of reliability.

#### 4.12.11.3 Drill-Down

– Drill-Down is a technique, which depends on data granularity. Specifically, it is the process of exploring multidimensional data from one level of detail to the next (“Drilldown”, <[http://en.wikipedia.org/wiki/Trend\\_analysis](http://en.wikipedia.org/wiki/Trend_analysis)>).

– Relative to the CASS, drill down includes but is not limited to:

i. Using pivot and/or tabular query to search, sort, filter underlying data in multiple ways to draw meaningful conclusions.

ii. Engaging in technical assistance (i.e. certificate holder’s repair shops, OEM technical representatives and/or engineering)

iii. Causal factoring (i.e. seeing cause and effect between events)

These data analysis techniques can be used on any data having individually identifiable events such as AOG, discrepancy, MEL, unscheduled removal data, audit findings etc.

#### 4.12.12 Prioritization of Analysis

Analysis is done on collected data with priority given to those data that best represent the overall “health” of the maintenance program; regarding performance, Audit data is the primary indicator of program compliance; regarding effectiveness, AOG data is seen as the leading indicator as it quantifies availability and reliability.

In the event that analysis reveals deficiencies that affect or could affect airworthiness, passenger or crew safety:

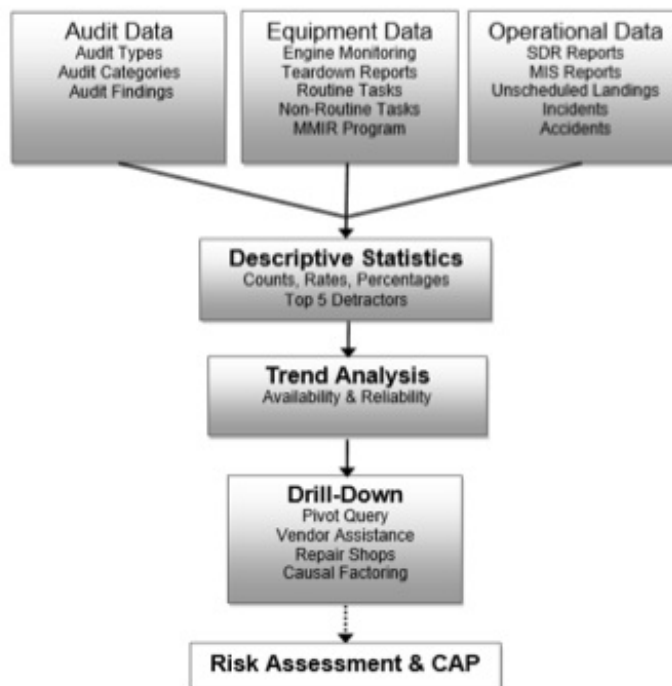
– They will be reported to the DOM or some other appropriate manager, via voice, paper or electronic means as soon as possible.

While CASS analysis can be used day-to-day or month-to-month, it is most meaningful when applied over the long term (e.g. six months or more).



Program deficiencies that are identified as non-safety related are prioritized according to scope, importance, financial impact and/or company convenience.

**Figure 3-1 CASS Flow of Analysis**



### 4.12.13 Corrective Action Development

Several processes are engaged to facilitate the possible development of a Corrective Action Plan (CAP). These processes aid in corrective action options and/or decision-making. They include, but are not limited to:

#### 4.12.13.1 Risk Management

Risk Management (RM) is the process, by which safety related risk (hazards) or deficiencies are identified, analyzed, assessed, and controlled (Risk management", <[http://en.wikipedia.org/wiki/Risk\\_management](http://en.wikipedia.org/wiki/Risk_management)>).

– Safety related risk has clear association with critical and/or emergency operational data (i.e. incidents and accidents) as well as audit data, which may be classified as such.

i. The Director of Safety and Quality is responsible for executing RM.

ii. RM procedures are found in the SMS Program

iii. RM is accomplished continuously and is event-driven.

– Any additional RM considerations that deal with program deficiencies are non-compliance risk (audit data) and low availability risk (operational data). These are addressed during the Analysis Assessment.

#### 4.12.13.2 Analysis Assessment

- Analysis Assessment is the process by which results of CASS analysis are used to determine what data are at an acceptable or unacceptable level relative to the program to determine if an RCA is needed.
  - i. The DOM is responsible for reviewing the results of analysis to determine acceptability.
  - ii. Analysis Assessments are done through use of the CASS Analysis Form, See Section 4.12.17 for an example of this form.
  - iii. For both day-to-day and long-term monitoring Analysis Assessments are accomplished continuously as required.
- Rather than an relying solely on risk management principles, other factors are considered when performing the assessment of analysis results; these include, but are not limited to:
  - i. Safety considerations (severity)
  - ii. AOG considerations (likelihood)
  - iii. Added value (impact of change)
  - iv. Industry vs operator (ability to fix)
  - v. Cost benefit considerations
- When analysis review reveals that identified areas of improvement are deemed acceptable, then no action is taken. Conversely, when the review of analysis deems deficiencies as unacceptable, a root cause analysis is performed to see what corrective action (if any) can be applied to correct the deficiency.

#### **4.12.13.3 Root Cause Analysis**

- RCA helps identify what, how and why something happened; assessing an incident at the most fundamental level, thus revealing the source of a problem ("Root cause analysis", <[http://en.wikipedia.org/wiki/Root\\_cause\\_analysis](http://en.wikipedia.org/wiki/Root_cause_analysis)>).
- Relative to the CASS, an RCA can engage either program performance (audit findings) or effectiveness (operational data and trends).
  - i. The DOM is responsible for root cause analysis for the purpose of corrective action
  - ii. RCAs are done through the CASS Analysis Form. See Section 4.12.17 for an example of this form.
  - iii. For both day-to-day and long-term monitoring RCAs are accomplished continuously as required.
- A CASS RCA is meant to determine a reasonably identifiable root cause (e.g. part, procedure, staffing etc.) by incorporating methods that may include but are not limited to:
  - i. Additional Drill-Down analysis
  - ii. Why-Why diagramming
  - iii. Investigation techniques



- Ultimately, the RCA is used to facilitate the possible development of a CAP; if the RCA warrants a change to the program, then a CAP is developed and implemented.

The above efforts are incorporated in the development of CAP, to both clearly identify deficiencies and to aid in developing the best possible remedy for those deficiencies.

#### **4.12.14 Corrective Action Plan**

The CAP is the actual process to implement a change to the maintenance program that corrects a fundamental problem and to prevent reoccurrence.

- The DOM is responsible for selecting and implementing a CAP.
- This is done through the CASS Analysis Form. See Section 4.12.17 for an example of this form.
- CAPs are developed and implemented on a continuous basis as required.

CAPs are implemented through but not limited to:

- i. Maintenance Alert Notices (MAN)
- ii. Inspection program revisions
- iii. Revisions to the training syllabus
- iv. Revisions to the General Maintenance Manuals (GMM), OM-A, and/or SOP

#### **4.12.15 CASS Analysis Form**

The CASS Analysis Form serves as a single source to record and track the process of analysis and actions taken, once the CASS identifies a maintenance program deficiency ("finding").

- The CASS Administrator is responsible for completing and maintaining the CASS Analysis Forms.
- Data is keyed electronically via computer into the form and when completed, stored on HSUSA's SharePoint.
- The form is completed when a maintenance program deficiency is identified through the CASS.

The form is referenced also found in Section 4.12.17 of this manual for reference; outlining of the basic flow of processes, (previously outlined in this chapter) that are engaged to develop, implement and track a CAP.

#### **4.12.16 Follow-up Process**

##### **OVERVIEW**

The CASS performs follow-up once a corrective action plan is implemented to quantify improvement in the maintenance program resulting from the CAP.



Follow-up for both the performance and effectiveness functions of the CASS occur as a result of CAPs but also includes reporting the results and progress of the CASS to the company's management and/or operational leadership.

The rest of this chapter outlines follow-up procedures carried out by the CASS, as well as the reporting that is accomplished for the performance and effectiveness of the maintenance program.

#### **FOLLOW-UP PROCESS**

Follow-up for the performance and effectiveness portions of the CASS are done by re-analyzing data (as per chapter three of this manual) after:

- A CAP is fully implemented.
- An assigned period of time has passed (as noted on the CASS Analysis Form Follow-Up section).

The results obtained from re-analyzing the data are then used to quantify improvement or no improvement from the implemented CAP.

- The CASS Administrator manages and maintains a working list of all findings with implemented CAPs that require a follow-up.

- This is accomplished by using the CASS Analysis Form. This form is exclusive to the CASS Administrator.

Intranet. (See section 4.12.17 on the following pages for an example of this form)

- It is in this form that a follow-up is assigned and scheduled if warranted.

### **4.12.17          RESERVED FOR FORMS**

### **4.12.18          Corrosion Control Prevention**

#### **Aircraft Corrosion Control Prevention Program details**

Environment Classification: Severe

Effective Date: 6/30/2025

Review Cycle: Annually or upon environmental/mission change

#### **1. PURPOSE**

To define detailed procedures for detecting, preventing, and treating corrosion on aircraft operating in severe environments, preserving safety and asset longevity.

#### **2. SCOPE**

Applies to all HeliService-USA aircraft and maintenance personnel working on aircraft exposed to hot,



humid, coastal, and chemically aggressive conditions.

### 3. RESPONSIBILITIES

- Maintenance Management Personnel: Oversees program implementation and ensures compliance.
- Maintenance Technicians: Execute inspections, cleanings, CPC applications, and record findings.
- QA Department: Audits inspection reports and recommends schedule adjustments.

### 4. DEFINITIONS

- CPC: Corrosion Preventive Compound
- Severe Environment: High humidity (>75%), salt exposure, high temperature (>28°C), or industrial pollution zones.

### 5. GENERAL POLICY

Aircraft shall be maintained under "severe" corrosion environment standards. Inspections and cleanings must be conducted on a structured timeline, and corrosion issues remediated immediately. Documentation of findings is mandatory for each interval.

### 6. WEEKLY TASKS (Every 7 Days)

- - External Cleaning: Wash and rinse all external surfaces using approved detergents. Focus on rotor hubs, fuselage seams, fasteners, and hinges.
- - Nose & Main Landing Gear Bays – Inspect & Clean: Remove visible contaminants. Check bay walls, joints, and cabling for corrosion.
- - Apply CPC to Nose/Main Landing Gear Bays: Apply a thin film of CPC. Focus on fasteners, seams, brackets, and coating degradation.
- - Landing Gear Assemblies: Inspect struts, braces, and linkages for rust, pitting, and hydraulic line CPC breakdown.
- - Pitot Tubes & Supports: Check for CPC loss, salt crusts, and clear drain holes.
- - Windshields and Transparencies: Examine for paint flaking, discoloration, or frame corrosion.
- - Rescue Hoist & Brackets: Inspect housing, cable exit, and structure interface.
- - Main & Tail Rotors (Visual Check): Check head, blades, and hubs for salt, paint bubbling, and discoloration.
- - Drain Lines & Scuppers: Flush and inspect for blockages and salt accumulation.
- - Searchlight, Flotation, Life Raft & Cargo Hook Installations: Inspect for cracks, water accumulation, or CPC wear.
- - Forward and Aft Upper Cowling: Visually inspect for corrosion, seal condition, and panel edges.



## 7. MONTHLY TASKS (Every 30 Days)

- - Inspect all access doors, floors, compartments, gearboxes, and control panels for corrosion signs and CPC degradation.

## 8. THREE-MONTH TASKS (Every 90 Days)

- - Perform detailed internal inspections of rotor heads, landing gear bays, tail boom, and avionic areas for corrosion and water ingress.

## 9. SIX-MONTH TASKS (Every 180 Days)

- - Conduct comprehensive checks on cabin/cockpit lower structures, hoist and flotation systems, avionic bays, and exposed airframe sections.

## 10. FOUR-YEAR TASKS (Every 48 Months)

- - Perform internal fuel tank inspection, checking for corrosion, salt, and structural integrity. Tank removal required.

## 11. DOCUMENTATION & RECORDKEEPING

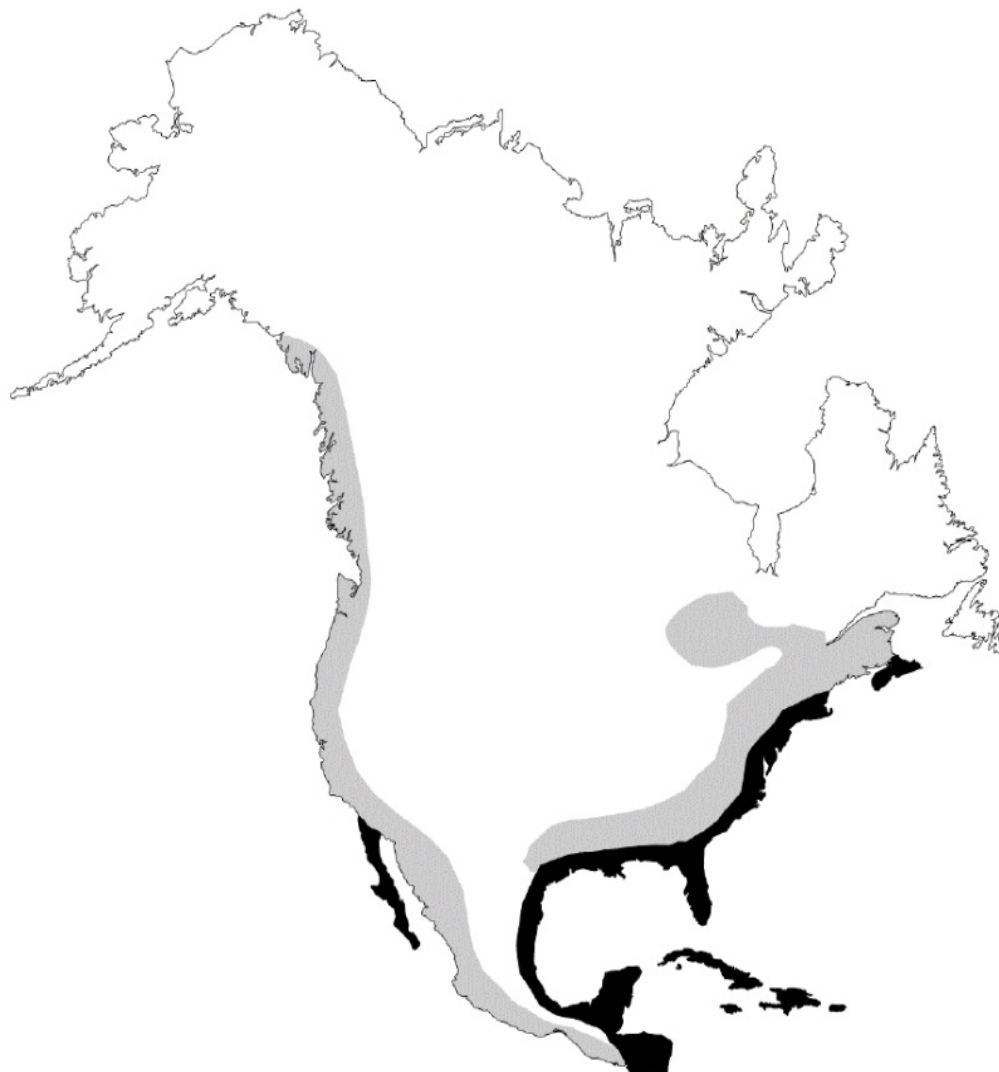
- Record all inspections, CPC applications, and corrosion findings in aircraft Digital Airware logs.
- Photos and detailed notes are required for all discrepancies.

## 12. REFERENCES

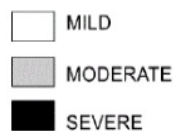
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- - Maintenance Planning Information Publication (69-A-AMPI-00-P)

## Corrosion Severity Zone Map





CORROSION SEVERITY ZONE



#### 4.12.19 List of References

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<[http://en.wikipedia.org/wiki/Risk\\_management](http://en.wikipedia.org/wiki/Risk_management)>



## 5 Hoist Procedures

### 5.1 Cargo Transfer Procedures

#### 1. Communication and Approach

- **Cargo Manifest:** A manifest detailing the actual weight of each cargo bag must be provided to the flight crew before departure. The maximum weight for cargo transfer flights may not be more than 200 kg
- **Establish Radio Communication:** If able, Establish two-way radio communication with the on-site coordinator 10-15 minutes before arrival.
- **Hoist Area Check:** The ground crew or first crew on the ground should check the hoist area for any loose objects or debris.
- **Initial Hover and Assessment:** Before commencing the hoist, fly to the rest position to visually check the platform for safety and perform a power check.

#### 2. Hoisting Procedures

- **Commands:** The hoist operator will provide verbal conning commands to guide the helicopter into position.
- **Hooking Up:** The ground party connects the load to the hoist hook. The hoist operator should confirm that the static line is free while raising the hoist hook.
- **Lifting:** Once cargo is attached, the hoist operator will provide advisory commands while lifting the cargo. The helicopter shall stay over the hoist site until cargo is just below the helicopter foot rail. If corrections should be made, an "easy" command should be made to gently maneuver the helicopter. For example, easy left or easy forward. If oscillations are noticed, lower the cargo back down to the deck, center the helicopter over the load, and re-raise cargo.
- **Securing the Cargo:** Prior to forward flight, the cargo should be raised until, if able, at the foot rail, or just slightly below the foot rail if the cargo is too large. The safety line should then be attached to the forward handle to prevent the cargo from moving or oscillating during transit.
- **Lowering the Cargo:** Once the helicopter is in the rest position at the hoisting location, the hoist operator will disconnect the safety line and provide standard conning commands while also lowering the cargo until it is fully on deck.

#### 3. In-Flight Monitoring

##### NOTE

**Handling Cargo Oscillations:** If the cargo begins to oscillate at any point, immediately notify the pilots to reduce airspeed. The helicopter should slow down until the oscillations stop and then maintain that airspeed until reaching the destination.

#### Progress Check's



During each airspeed progress check, the hoist operator will report on the cargo's stability and overall flight characteristics.

- **Progress check - Forward flight 0-20 knts:** Once cargo is secured at or just below the rail and safety line is attached, notify pilots ready for forward flight. Monitor cargo, give advisory commands. If cargo begins to oscillate, notify pilots and slow air speed and slightly raise or lower cargo.
- **Progress check - Forward flight 21-40 knts:** Monitor cargo and give advisory commands. If cargo begins to osculate, slow airspeed until cargo stops osculating and remain at that airspeed until at hoisting site.
- **Progress check - Forward flight 41-60 knts:** Monitor cargo and give advisory commands. If cargo begins to osculate, slow airspeed until cargo stops osculating and remain at that airspeed until at hoisting site.
- **Progress check - Forward flight 61-80 knts:** Monitor cargo and give advisory commands. If cargo begins to osculate, slow airspeed until cargo stops osculating and remain at that airspeed until at hoisting site.

#### 4. Post-Mission Documentation Procedures

- **Logging:** After the mission, log the number of hoist cycles, and the maximum cable length used. An entry must be created in DAW for any cargo weighing over 150kg to document that a hoist conditioning check needs to be completed.
- **Hoist Conditioning Check:** For any cargo over 150 kg, a conditioning check shall be performed on the hoist to ensure it is in good working order after the lift and before hoisting any HEC.

## 5.2 Remote Hook Operation

### 1. Pre-Mission Planning

- **Risk Assessment:** Conduct a full risk/hazard analysis (weather, obstacles, number of bags, and weight of bags).
- **Crew Briefing:** Review mission objectives, roles, communication plan, emergency procedures, and go/no-go criteria.
- **Weight & Balance:** Verify load weights are within aircraft and hook limits; account for bag configuration.
- **Equipment Inspection:**
  - Check hoist system, remote hook, tactical tethers/ securing devices, and release mechanisms.
  - Ensure bags are properly rigged with rated slings or nets.
- **IF HOISTING FROM OR TO A MANNED PLATFORM - Platform Coordination:** Confirm platform personnel are briefed, clear on weight limitations, and hand signals/communication.



#### 1. Pre-Takeoff / Departure

- Pre-Flight hoist bag- Verify that all required hoisting equipment (Remote hook, batteries and arm sleeve with remote) is present and safe for takeoff.
- Insert batteries into Remote Hook to power it on, then press the "open" button on the remote control to verify the hook opens fully and displays green light. Then press the "close" button on the remote control and verify the hook closes fully and displays blue light.
- Bags secured inside or near aircraft for hook-up.
- Perform standard helicopter hoisting checklists.

#### 1. Battery Installation

- Insert the battery pack into the remote hook system, ensuring it is fully seated and securely locked into position.
- Verify power activation by confirming that the BLUE LED indicators illuminate, signifying the unit is powered and operational.
- Apply a secondary retention measure by placing tape over the plastic battery cover to prevent accidental dislodgement during operation.

#### 1. 4. Approach to Pick up Point

- Pilot establishes a stable hover over designated pick-up or drop-off zone.
- Hoist Operator visually clears the area below.
- Communicate drift corrections: "Easy Forward, easy back, easy left, easy right, hold."
- Hook is lowered for bag hook-up / attachment.

#### **CAUTION – Static Discharge Cable Use with Remote Hook Operations**

Do not attach the static discharge cable when conducting operations with the remote hook system. During load handling, the cable may become trapped beneath the load, potentially requiring the aircraft to forcibly extract it. This action could result in cable damage or breakage, posing a risk to equipment and operational safety.

#### 5.A- Load Hook-Up- IF HOISTING FROM AIRCRAFT TO PLATFORM

- Hoist Operator will ensure bags are securely attached to remote hook via approved hoisting methods/ hoisting carabiners.
- Hoist Operator confirms positive hook engagement. The LED light should be BLUE.



- Report "Load Ready."
- Commence hoisting and follow standard hoisting procedures.
- Once the load is safely on deck, the hoist operator shall manually release the hook locking mechanism by pressing the **"Open"** button on the mini remote hook controller, which is secured to their forearm. This action disengages the locking mechanism, allowing for safe and controlled release of the load. THE LED light will turn RED, indicating the action is in the process of disengaging the locking mechanism. Once the mechanism is fully unlocked and ready for hoisting, the LED light will turn GREEN.

#### **5.B- Load Hook-Up- IF HOISTING FROM PLATFORM TO AIRCRAFT**

- Hoist Operator will lower the remote hook over the desired cargo to be hoisted. Ensure the cargo hook is OPEN. The LED light should be GREEN; this signals that the system is securely opened.
- The Hoist Operator will aim to lower the opened remote hook, on top or near the lifting cargo carabiner.
- The remote hook has a magnet which generates a magnetic field that attracts, centers and orients the sling, or rigging gear. Upon contact between the hoisting carabiner and the magnetic interface of the remote hook, the system will automatically activate the locking mechanism, securing the load. Alternatively, the operator may manually engage the locking mechanism using the remote control, ensuring positive lock confirmation prior to lift.
- Hoist Operator confirms positive hook engagement. The LED light should turn BLUE.
- Report "Load Ready."
- Commence hoisting and follow standard hoisting procedures.

#### **1. Transit between Bases or Platforms with Cargo**

- Refer to Cargo Transfer and Hoisting Procedures.

#### **1. Post-Flight**

- Conduct equipment inspection for wear/damage.

### **Remote Hook System – Technical Specifications**





- **Manufacturer:** Elebia
- **Model:** EVO2
- **Maximum Load Capacity:** 2,500 kg (5,511 lbs)
- **Service Life:** Rated for up to 100,000 operational cycles
- **Color Codes:** The Elebia EVO2 remote hook uses a 4-color LED system to indicate its operational status. Here's what each color typically represents:
  - **Blue LED – Power On / Hook Locked Closed**
  - Indicates the unit is powered and operational.
  - Confirms successful battery installation and system readiness.
  - Indicates that the hook is locked closed.





- **Green LED – Hook Open**
- Confirms that the hook is fully opened.



- **Red LED – Hook not Secured/ Hook In-Transit / Error**
- Indicates the hook is not fully secured or fully open, the load is not engaged or there is a fault/error condition.
- Activates whenever hook is transiting between fully open and fully locked.
- May also signal low battery or operational issues.



- **Yellow/Amber LED – Battery Low / Warning**
- Serves as a cautionary indicator, often related to battery status or transitional states.

*These color codes help operators quickly assess the hook's status during operations.*

- **Magnet Operation:** The EVO2 remote hook is equipped with a proprietary magnetic system consisting of 32 **neodymium permanent magnets**. These magnets are arranged in a precise configuration that generates an oriented magnetic field with two primary functions:
- **Attraction:** Draws the hoisting element (e.g., carabiner) into position.



- **Centering:** Aligns the load for optimal engagement.

*This magnetic system is the core component enabling remote load engagement, enhancing both safety and operational efficiency during aerial hoisting procedures.*

### Remote Hook Control- eMINI



The eMINI is a small and compact remote control. With the eMINI remote control, the operator can open and close the evo lifting hook and control its battery level. The eMINI also displays sensor information and maintenance messages.

- **Display Information:**
  - 0 to 9: Battery level of the evo lifting hook.
  - A: Low battery level of the eMINI remote control.
  - B: The eMINI remote control does not detect the evo lifting hook.
  - C: Maintenance review of the evo lifting hook.

## 5.2.1 Emergency Procedure

### Emergency Procedure: Remote Hook Connectivity Reset

Use this procedure if the eMINI remote displays "B" (No Detection) or the hook fails to open.

1. **Retrieve Equipment:** Hoist the cargo and remote hook up to a position directly in front of the cabin door.
2. **Access Battery:** Remove the secondary retention from the battery cover.



3. **Hard Reset:** Extract the battery pack, then re-insert it ensuring it is fully seated and the **BLUE LED** illuminates.
4. **Secure:** Re-apply secondary retention over the battery cover.
5. **Test & Release:** Lower the cargo and remote hook back down to the deck.
  - Verify the eMINI no longer displays "**B**".
  - Press the "**Open**" button.
  - Confirm the **GREEN LED** (Hook Open).



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## 6 Offshore Crew Transfer Operations

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### 6.1 Vessel Plates

#### 6.1.1 Aeolus



|                                                                                     |              |                                                                                                                                                                                                                                                                                                       |                       |                                            |  |
|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------|--|
|    |              | <h2 style="text-align: center;">HELIDECK INFORMATION PLATE</h2>                                                                                                                                                                                                                                       |                       |                                            |  |
| HELIDECK<br>Elev <b>Check</b>                                                       | VAR<br>Check | POSITION<br>Mobile                                                                                                                                                                                                                                                                                    | XAEL<br><b>Aeolus</b> |                                            |  |
| HEIGHT OF INSTALLATION: <b>268</b>                                                  |              | VHF<br><b>Check</b>                                                                                                                                                                                                                                                                                   | NDB<br>N/A            | Issue Date<br><b>30 May 2023</b>           |  |
| HIGHEST OBSTACLE WITHIN 5NM: <b>Check</b>                                           |              |                                                                                                                                                                                                                                                                                                       |                       |                                            |  |
| FUELLING INSTALLATION: <b>No</b>                                                    |              | Operating Company                                                                                                                                                                                                                                                                                     |                       | Issued By                                  |  |
| STARTING EQUIPMENT: <b>No</b>                                                       |              | <b>Van Oord</b><br>Shipmanagement b.v.                                                                                                                                                                                                                                                                |                       | <b>Helideck</b><br>Certification<br>Agency |  |
| Norway 1.25D - 'H' <b>Dh= 17.76 / SLA = 22.2</b>                                    |              |                                                                                                                                                                                                                                                                                                       |                       |                                            |  |
| HELIDECK D value: <b>22.2m</b>                                                      |              |                                                                                                                                                                                                                                                                                                       |                       |                                            |  |
| P/R/H Category: <b>1</b>                                                            |              |                                                                                                                                                                                                                                                                                                       |                       |                                            |  |
| Max Weight: <b>12.8t</b>                                                            |              |                                                                                                                                                                                                                                                                                                       |                       |                                            |  |
| Circle & H Lights: <b>No</b>                                                        |              |                                                                                                                                                                                                                                                                                                       |                       |                                            |  |
|  |              |                                                                                                                                                                                                                                                                                                       |                       |                                            |  |
| Wind (T°)                                                                           | Kts          | Limitation /Comment                                                                                                                                                                                                                                                                                   |                       |                                            |  |
|                                                                                     |              | Jack Up<br>• Daylight only operations - Aiming circle & "H" lights not fitted<br>• No helicopters when vessel on transit - Main Crane infringes 210 sector when on rest<br>• Crane boom may infringe 210 sector outside of planned helicopter operations<br><b>*CERTIFICATE EXPIRY – 15 Mar 2025*</b> |                       |                                            |  |
|                                                                                     |              | <b>Non Compliance</b>                                                                                                                                                                                                                                                                                 |                       |                                            |  |
|                                                                                     | 210°         | Crane boom when stowed in its rest.                                                                                                                                                                                                                                                                   |                       |                                            |  |
|                                                                                     | Misc         | TD/PM+H lights not fitted                                                                                                                                                                                                                                                                             |                       |                                            |  |



## 6.1.2      Bokalift 1

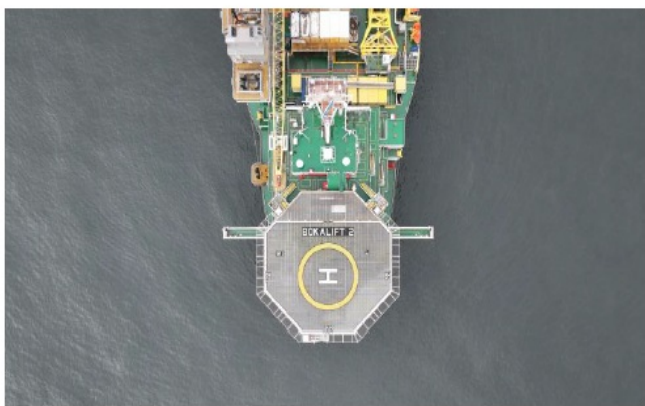


|                                                                                                                                                                                                                                                                                                                                                 |              |                                                                                                                                                                |                                   |            |                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------|------------------------------------------------------|
|                                                                                                                                                                                                                                                                |              | <b>HELIDECK INFORMATION PLATE</b><br><b>Certificate Expiry Date: 14<sup>th</sup> November 2025</b>                                                             |                                   |            |                                                      |
| HELIDECK<br>Elev <b>130 ft</b>                                                                                                                                                                                                                                                                                                                  | VAR<br>Check | POSITION<br>Mobile                                                                                                                                             | XBL1<br><b>Bokalift 1</b>         |            |                                                      |
| HEIGHT OF INSTALLATION:<br>HIGHEST OBSTACLE WITHIN 5NM:                                                                                                                                                                                                                                                                                         |              | 423 ft<br>Check                                                                                                                                                | VHF<br>Check                      | NDB<br>N/A | Issue Date<br>25 Jan 2024                            |
| FUELLING INSTALLATION:<br>STARTING EQUIPMENT:                                                                                                                                                                                                                                                                                                   |              | No<br>No                                                                                                                                                       | Operating Company<br><br>Boskalis |            | Issued By<br><br>HeliDeck<br>Certification<br>Agency |
| Norway 1.25D - 'Dh' =<br>HELIDECK D value:<br>P/R/H Category:<br>Max Weight:<br>Circle & H Lights:                                                                                                                                                                                                                                              |              | 17.92 / SLA=22.4<br>22.4<br>1<br>12.8t<br>Yes                                                                                                                  |                                   |            |                                                      |
|     |              |                                                                                                                                                                |                                   |            |                                                      |
| Wind (T°)                                                                                                                                                                                                                                                                                                                                       | Kts          | Limitation /Comment                                                                                                                                            |                                   |            |                                                      |
|                                                                                                                                                                                                                                                                                                                                                 |              | Heavy lift vessel<br>• Possible turbulence from turbine exhausts (port & stbd)<br>• Turbulence reports requested<br>• rev.9b HMS and repeater lights installed |                                   |            |                                                      |
|                                                                                                                                                                                                                                                                                                                                                 |              | <b>Non-Compliance</b>                                                                                                                                          |                                   |            |                                                      |
|                                                                                                                                                                                                                                                                                                                                                 | 150°         | Windsock 3.15m adl at 5.3m from SLA                                                                                                                            |                                   |            |                                                      |
|                                                                                                                                                                                                                                                                                                                                                 | Misc         | Approved friction surface - no net<br>Dangerous Goods not to be uplifted from this location                                                                    |                                   |            |                                                      |



### 6.1.3      Bokalift 2



|                                                                                                                                                                             |              |                                                                                                                                                                  |                           |                                              |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------|--|
|                                                                                            |              | <h2 style="text-align: center;">HELIDECK INFORMATION PLATE</h2> <p style="text-align: center; color: red;"><b>Certificate Expiry Date: 09 September 2023</b></p> |                           |                                              |  |
| HELIDECK<br>Elev <b>137 ft</b>                                                                                                                                              | VAR<br>Check | POSITION<br>Check                                                                                                                                                | XBL2<br><b>Bokalift 2</b> |                                              |  |
| HEIGHT OF INSTALLATION: <b>499</b>                                                                                                                                          |              | VHF<br>Check                                                                                                                                                     | NDB<br>Nil                | Issue Date<br><b>09 Jun 2023</b>             |  |
| HIGHEST OBSTACLE WITHIN 5NM: <b>Check</b>                                                                                                                                   |              |                                                                                                                                                                  |                           |                                              |  |
| FUELLING INSTALLATION: <b>No</b>                                                                                                                                            |              | Operating Company                                                                                                                                                |                           | Issued By                                    |  |
| STARTING EQUIPMENT: <b>No</b>                                                                                                                                               |              | <b>Bokalis</b>                                                                                                                                                   |                           | <b>Helideck<br/>Certification<br/>Agency</b> |  |
| Norway 1.25D - 'Dh' = <b>22.2 / Keel laid 1999</b>                                                                                                                          |              |                                                                                                                                                                  |                           |                                              |  |
| HELIDECK D value: <b>22.2</b>                                                                                                                                               |              |                                                                                                                                                                  |                           |                                              |  |
| P/R/H Category: <b>3</b>                                                                                                                                                    |              |                                                                                                                                                                  |                           |                                              |  |
| Max Weight: <b>12t</b>                                                                                                                                                      |              |                                                                                                                                                                  |                           |                                              |  |
| Circle & H Lights: <b>No</b>                                                                                                                                                |              |                                                                                                                                                                  |                           |                                              |  |
| <br> |              |                                                                                                                                                                  |                           |                                              |  |
| <b>Wind (T°)</b>                                                                                                                                                            | <b>Kts</b>   | <b>Limitation /Comment</b>                                                                                                                                       |                           |                                              |  |
|                                                                                                                                                                             |              | Heavy Lift Vessel - Bow deck<br>• Daylight operations only                                                                                                       |                           |                                              |  |
|                                                                                                                                                                             |              | <b>Non-Compliance</b>                                                                                                                                            |                           |                                              |  |
|                                                                                                                                                                             |              | Circle H lights not fitted                                                                                                                                       |                           |                                              |  |



6.1.4 Charybdis



# HELIDECK INFORMATION PLATE

Page 1 of 2

| Jack Up                                                                                                                                                          | Charybdis                                          | XCYB                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------|
| <b>GENERAL INFORMATION:</b>                                                                                                                                      |                                                    |                                        |
| <b>OPERATOR</b><br>Blue Ocean Energy Marine                                                                                                                      | <b>HIP ISSUE DATE</b><br>06 Oct 2025               | <b>HLAC EXPIRY DATE</b><br>22 Dec 2025 |
| <b>HELIDECK ELEVATION</b><br>Check                                                                                                                               | <b>INSTALLATION HEIGHT</b><br>Check                | <b>POSITION</b><br>Mobile              |
| <b>MAX WEIGHT</b><br>12.6 t                                                                                                                                      | <b>D VALUE</b><br>22.2 m                           | <b>VAR</b><br>Check                    |
| <b>TD/PM &amp; H LIGHTING</b><br>Yes                                                                                                                             | <b>PITCH / ROLL / HEAVE CATEGORY</b><br>1          | <b>HIGHEST OBSTACLE</b><br>Check       |
| <b>NORWAY SPECIFIC INFORMATION:</b>                                                                                                                              |                                                    |                                        |
| <b>YEAR KEEL LAID</b><br>2020                                                                                                                                    | <b>SAFE LANDING AREA</b><br>22.2 m                 | <b>Dh</b><br>17.76 m                   |
| <b>COMMUNICATIONS:</b>                                                                                                                                           |                                                    | <b>TURNAROUND:</b>                     |
| <b>VHF FREQUENCIES</b><br>Traf: Check    Log: Check                                                                                                              | <b>NDB</b><br>Freq: Check    Ident: 14655-21-00367 | <b>FUELING</b><br>No                   |
|                                                                                                                                                                  |                                                    | <b>STARTING</b><br>No                  |
| <b>LIMITATIONS / COMMENTS:</b>                                                                                                                                   |                                                    |                                        |
| <b>TEMPORARY LIMITATION NOTICES:</b><br>Nil                                                                                                                      |                                                    |                                        |
| <b>LIMITATIONS:</b>                                                                                                                                              | <b>WIND DIRECTION (T°)</b>                         | <b>WIND SPEED (Kts)</b>                |
| Table 1(T) if overflight of 5:1 items is unavoidable                                                                                                             |                                                    |                                        |
| Vessel in Transit - No helicopter operations                                                                                                                     |                                                    |                                        |
| Vessel Jacked up - Main crane boom MUST be in the raised position clear of OFS (210°) for all helicopter operations                                              |                                                    |                                        |
| Possible turbulence 130 - 250 deg relative to helideck heading - crane rest & when cargo deck loaded with stacked turbine columns - Turbulence reports requested |                                                    |                                        |
| Turbine exhausts 240 deg relative to the helideck heading                                                                                                        |                                                    |                                        |
| Maximum single pax weight 180kg due to inbound pax scale weight range                                                                                            |                                                    |                                        |
| No aeronautical radio operators - No helicopter operations in UKCS                                                                                               |                                                    |                                        |
| *To operate this vessel in either the UKCS or the NCS please contact the HCA*                                                                                    |                                                    |                                        |
| <b>COMMENTS:</b><br>Nil                                                                                                                                          |                                                    |                                        |

HCA-FRM-04 Rev 02

This Helideck Information Plate is non-transferable and may not be replicated or used by any 3rd party without express permission of HCA.

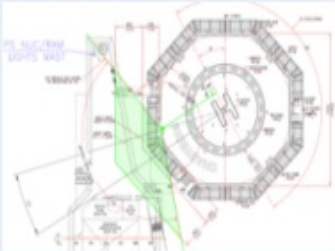
August 2024



NON-COMPLIANCES:

|                                                                  |                                                                                                                                 |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>210°</b><br>Nil                                               | <b>150°</b><br>Nil                                                                                                              |
| <b>5:1</b><br>Floodlight at fwd access extends 2.8m from the SLA | <b>Misc</b><br>HMS system non compliant<br>Helideck landing net to be fitted - NCS<br>NCS non - compliances for Dual Agent unit |

IMAGE LIBRARY:

|                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>PLAN DRAWING</b>                                                                 | <b>ADDITIONAL (IF AVAILABLE)</b>                                                     |
|   |   |
| <b>PORT SIDE</b>                                                                    | <b>STARBOARD SIDE</b>                                                                |
|  |  |



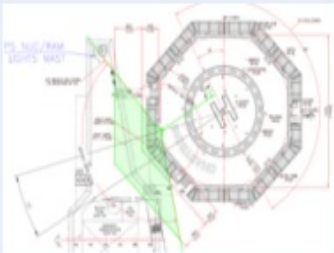


NON-COMPLIANCES:

|                                                    |                                           |
|----------------------------------------------------|-------------------------------------------|
| 210°                                               | 150°                                      |
| Nil                                                | Nil                                       |
| 5:1                                                | Misc                                      |
| Floodlight at fwd access extends 2.8m from the SLA | HMS system non compliant                  |
|                                                    | Helideck landing net to be fitted - NCS   |
|                                                    | NCS non - compliances for Dual Agent unit |

IMAGE LIBRARY:

PLAN DRAWING



ADDITIONAL (IF AVAILABLE)



PORT SIDE



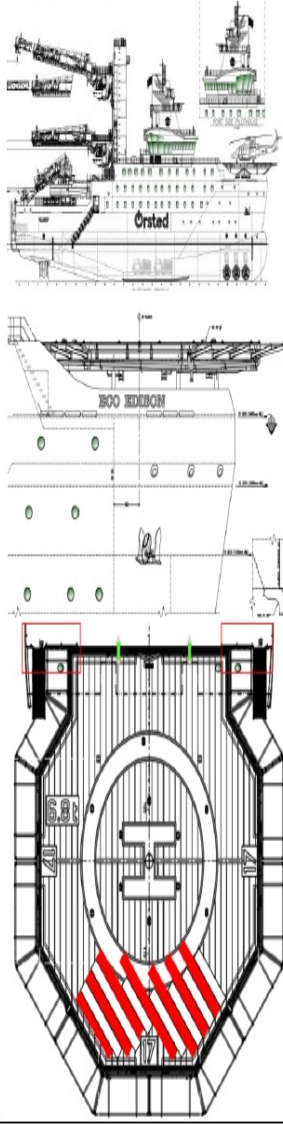
STARBOARD SIDE





## 6.1.5 Eco Edison

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                              |                                                                                                                                           |                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 1. Operating Company Logo<br>                                                                                                                                                                                                                                                                                                                                                                                                            | 2. Position<br>Call +1 985 213 6816          | 3. Location Name<br><br><b>M/V ECO EDISON</b>                                                                                             |                                             |
| 5. Operating Company<br><b>Orsted A/S</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6. Helideck D-Value<br>17 meters<br>55.77 ft | 7. Helideck Elevation<br>14.3 m / 46.92 ft<br>@ 19 ft draft                                                                               | 8. Lease Block Number<br><br>N/A            |
| 9. VHF Radio Frequency<br>122.85                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10. Helideck Dimensions<br>17 m x 17 m       | 11. Max Allowable Weight<br>7 tons<br>14,000 lbs.                                                                                         | 12. Region<br>Northeastern<br>U.S. Atlantic |
| 13. Limitations<br>MedEvac in the region is supported by the U.S. Coast Guard Air Station Cape Cod using the MH-60 helicopter with a D-Value of 64ft/19.51m and MTOW of 21,884 lbs./ 10.94 tons which exceeds the capacity of this helideck.<br><u>Hoist/Winch Only operations shall be conducted for all Medical Evacuations using this aircraft.</u><br><br>The aircraft nose should not be positioned within the Prohibited Landing Sector during landing, positioning to land or departure. See figure in section 27. |                                              | 14. TDPM/Aiming Circle<br>Centered<br>Lighted TD/PM & H                                                                                   | 15. Weather/ATIS/AWOS<br>FREQ: N/A          |
| 17. Remarks<br>Orsted and ECO procedures in place.<br>-20-minute call<br>-5-minute call<br>-Deck Status (Green Deck) updated upon gear check (visual)<br>-Largest approved helicopter type is AW-169                                                                                                                                                                                                                                                                                                                      |                                              | 18. Services<br>Helicopter Refueling : N<br>Heli Starter Equip : N<br>Heli Towing Equip : N<br>Helideck Status Light : Y (see section 28) |                                             |
| 19. Installation Picture<br>                                                                                                                                                                                                                                                                                                                                                                                                           |                                              |                                                                                                                                           |                                             |
| 20. Non-Compliances:<br>The non-folding handrails and firefighting hose reel are 210-degree OFS infringements. The Handrails have been painted in contrasting yellow/black striping for high contrasting. Additionally, a Prohibited Landing Sector / No Nose Zone has been painted on the adjacent side of the helideck in accordance with CAP 437 Ed. 9. Refer to Section 27.                                                                                                                                           |                                              | Parking Lot : N/A<br>:<br>:<br>:<br>:<br>:                                                                                                |                                             |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 26. Helideck Picture<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 27. Vessel overview schematic<br> |
| 28. Additional information:<br>- Helideck has lighted TDPM & H<br>- Helideck equipped with HMS repeater Light System that communicates to the pilots by showing one of three possible colors.<br><b>Steady Blue</b> = Safe to land; (Pitch roll and inclination, (SHR and MSI/WSI all within limits)<br><b>Steady Amber</b> = Land with caution (MSI/WSI only out of limits).<br><b>Steady Red</b> = DO NOT LAND (Pitch, Roll, inclination or SHR out of limits)<br><br>All Limit perimeters should be reviewed on and agreed to by the aircraft operator, vessel crew and the client. |                                                                                                                      |
| 29. Link to NOTAM site<br>No current NOTAM system available for region.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 30. Helideck Contact Telephone Number<br>+1 985 213 6816                                                             |



## 6.1.6 Global Symphony

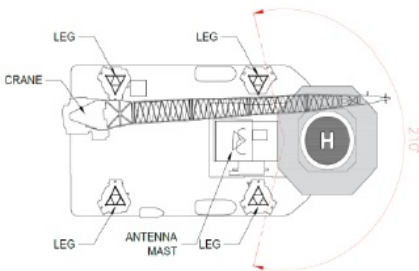
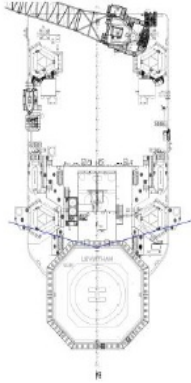


|                                                                                                                                  |                            |                                                                                                                                                            |                                          |                                              |  |
|----------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------|--|
|                                                 |                            | <h2 style="text-align: center;">HELIDECK INFORMATION PLATE</h2>                                                                                            |                                          |                                              |  |
| <b>HELIDECK</b><br>Elev <b>46 ft</b>                                                                                             | <b>VAR</b><br><b>Check</b> | <b>POSITION</b><br><b>Check</b>                                                                                                                            | <b>Global Symphony</b>                   |                                              |  |
| <b>HEIGHT OF INSTALLATION:</b> <b>125</b><br><b>HIGHEST OBSTACLE WITHIN 5NM:</b> <b>Check</b>                                    |                            | <b>VHF</b><br><b>Check</b>                                                                                                                                 | <b>NDB</b><br><b>410</b><br><b>C6YY3</b> | <b>Issue Date</b><br><b>27 Feb 2020</b>      |  |
| <b>FUELLING INSTALLATION:</b> <b>No</b><br><b>STARTING EQUIPMENT:</b> <b>No</b>                                                  |                            | <b>Operating Company</b>                                                                                                                                   |                                          | <b>Issued By</b>                             |  |
| Norway 1.25D - 'D'<br><b>HELIDECK D value:</b> <b>20.88</b><br><b>P/R/H Category:</b> <b>3</b><br><b>Max Weight:</b> <b>12.8</b> |                            | <b>Global Marine Systems</b>                                                                                                                               |                                          | <b>Helideck<br/>Certification<br/>Agency</b> |  |
|                                                                                                                                  |                            |                                                                                                                                                            |                                          |                                              |  |
|                                               |                            |                                                                                                                                                            |                                          |                                              |  |
| <b>Wind (T°)</b>                                                                                                                 | <b>Kts</b>                 | <b>Limitation /Comment</b>                                                                                                                                 |                                          |                                              |  |
|                                                                                                                                  |                            | Offshore Construction Vessel<br>• Table 1(T) if overflight of Nav mast unavoidable<br>• According to Norwegian regulations – 1,25 D<br>• Daylight ops only |                                          |                                              |  |
|                                                                                                                                  |                            | <b>Non Compliance</b>                                                                                                                                      |                                          |                                              |  |
|                                                                                                                                  | 150°                       | Marginal infringement of vent housing painted in obstruction markings                                                                                      |                                          |                                              |  |
|                                                                                                                                  | 5:1                        | Nav mast when lowered approx. 2.8m from edge of SLA                                                                                                        |                                          |                                              |  |
|                                                                                                                                  | Misc                       | No circle and h lights fitted                                                                                                                              |                                          |                                              |  |



## 6.1.7 Leviathan

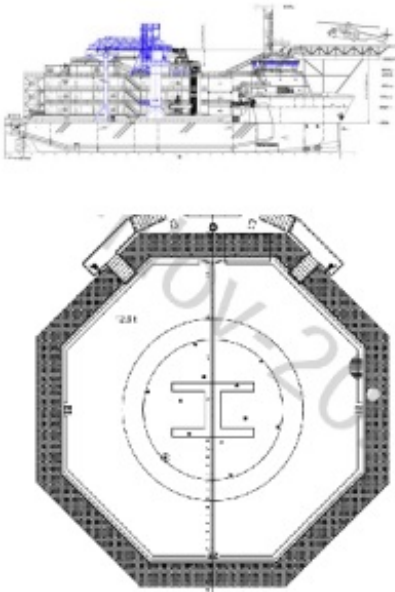


|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |                                                                                                                                 |                                         |            |                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------|------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              | <b>HELIDECK INFORMATION PLATE</b><br><b>Certificate Expiry Date: 4<sup>th</sup> April 2026</b>                                  |                                         |            |                                                |
| HELIDECK<br>Elev. Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | VAR<br>Check | POSITION<br>Check                                                                                                               | <b>XSLE</b><br><b>HEA Leviathan</b>     |            |                                                |
| HEIGHT OF INSTALLATION: Check<br>HIGHEST OBSTACLE WITHIN 5NM: Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |                                                                                                                                 | VHF<br>Check                            | NDB<br>Nil | Issue Date<br><b>10<sup>th</sup> June 2024</b> |
| FUELLING INSTALLATION: Yes<br>STARTING EQUIPMENT: Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |                                                                                                                                 | Operating Company                       |            | Issued By                                      |
| Norway 1.25D - 'Dh Norway'<br>HELIDECK D value:<br>P/R/H Category:<br>Max Weight:<br>Circle & H Lights:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              | 17.76m/SLA 22.2m<br>22.2m<br>1<br>12.6t<br>Yes                                                                                  | <b>Zakher Marine International Inc.</b> |            | <b>Helideck Certification Agency</b>           |
| <div style="display: flex; justify-content: space-around; align-items: flex-start;"> <div style="text-align: center;">  </div> <div style="text-align: center;">  </div> <div style="text-align: center;">  </div> </div> <div style="display: flex; justify-content: space-around; align-items: flex-start; margin-top: 20px;"> <div style="text-align: center;">  </div> <div style="text-align: center;">  </div> </div> |              |                                                                                                                                 |                                         |            |                                                |
| <b>Wind (T°)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Kts</b>   | <b>Limitation /Comment</b>                                                                                                      |                                         |            |                                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              | <b>4-Legged Jack-up Ship</b><br>• Landing prohibited unless main crane boom & jib is in the raised position clear of OFS (210°) |                                         |            |                                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              | <b>Non-Compliances</b>                                                                                                          |                                         |            |                                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Misc.        | Crane boom/jib may be at rest on port side of deck outside of notified helicopter flight times.                                 |                                         |            |                                                |

## 6.1.8 MV HOS Riverbend

|                                                                                                                                                                                                                                             |     |                                                                                                                                                                                                                                                                  |                                                                          |                                              |                                                           |                       |     |                    |     |                   |     |                       |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------|-----------------------|-----|--------------------|-----|-------------------|-----|-----------------------|-----|
| 1. Operating Company Logo<br>                                                                                                                              |     | 2. Position<br><b>Call – Bridge Cell</b><br><b>+1-985-276-0431</b>                                                                                                                                                                                               |                                                                          | 3. Location Name<br><b>M/V HOS Riverbend</b> |                                                           |                       |     |                    |     |                   |     |                       |     |
| 5. Operating Company<br><b>Hornbeck Offshore Services</b>                                                                                                                                                                                   |     | 6. Helideck D-Value<br><b>21 meters</b><br><b>68.9 ft</b>                                                                                                                                                                                                        | 7. Helideck Elevation<br><b>20.42 m / 67 ft</b><br><b>Standard draft</b> |                                              | 8. Lease Block Number<br><b>N/A</b>                       |                       |     |                    |     |                   |     |                       |     |
| 9. VHF Radio Frequency<br><b>122.85</b>                                                                                                                                                                                                     |     | 10. Helideck Dimensions<br><b>21 m x 21 m</b>                                                                                                                                                                                                                    | 11. Max Allowable Weight<br><b>12.6 tons</b><br><b>25,200 lbs.</b>       |                                              | 12. Region<br><b>Northeastern</b><br><b>U.S. Atlantic</b> |                       |     |                    |     |                   |     |                       |     |
| 13. Limitations<br><b>None</b>                                                                                                                                                                                                              |     | 14. TDPM/Aiming Circle<br><b>Centered</b><br><b>TD/PM &amp; H</b>                                                                                                                                                                                                |                                                                          | 15. Weather/ATIS/AWOS<br><b>FREQ: N/A</b>    |                                                           |                       |     |                    |     |                   |     |                       |     |
|                                                                                                                                                                                                                                             |     | 16. Marking Standard<br><b>CAP-437 Ed. 9</b>                                                                                                                                                                                                                     |                                                                          |                                              |                                                           |                       |     |                    |     |                   |     |                       |     |
| 17. Remarks<br><b>HOS procedures in place.</b><br><b>-20-minute call</b><br><b>-5-minute call</b><br><b>-Deck Status (Green Deck) updated upon gear check (visual)</b>                                                                      |     | 18. Services<br><table border="0"> <tr> <td>Helicopter Refueling:</td> <td>: N</td> </tr> <tr> <td>Heli Starter Equip</td> <td>: N</td> </tr> <tr> <td>Heli Towing Equip</td> <td>: N</td> </tr> <tr> <td>Helideck Status Light</td> <td>: Y</td> </tr> </table> |                                                                          |                                              |                                                           | Helicopter Refueling: | : N | Heli Starter Equip | : N | Heli Towing Equip | : N | Helideck Status Light | : Y |
| Helicopter Refueling:                                                                                                                                                                                                                       | : N |                                                                                                                                                                                                                                                                  |                                                                          |                                              |                                                           |                       |     |                    |     |                   |     |                       |     |
| Heli Starter Equip                                                                                                                                                                                                                          | : N |                                                                                                                                                                                                                                                                  |                                                                          |                                              |                                                           |                       |     |                    |     |                   |     |                       |     |
| Heli Towing Equip                                                                                                                                                                                                                           | : N |                                                                                                                                                                                                                                                                  |                                                                          |                                              |                                                           |                       |     |                    |     |                   |     |                       |     |
| Helideck Status Light                                                                                                                                                                                                                       | : Y |                                                                                                                                                                                                                                                                  |                                                                          |                                              |                                                           |                       |     |                    |     |                   |     |                       |     |
| 19. Installation Picture<br>                                                                                                                            |     |                                                                                                                                                                                                                                                                  |                                                                          |                                              |                                                           |                       |     |                    |     |                   |     |                       |     |
| 20. Non-Compliances:<br><b>The 180 degree 5:1 Falling gradient does have an infringement.</b><br>Foam monitor platforms port and starboard sides approx. 3.3m from edge of SLA Forward emergency access point approx. 2.6m from edge of SLA |     |                                                                                                                                                                                                                                                                  |                                                                          | Parking Lot N/A                              |                                                           |                       |     |                    |     |                   |     |                       |     |

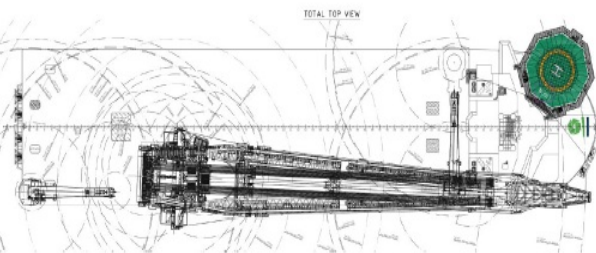


|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <div>26. Helideck Picture</div> <div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <div>27. Vessel overview schematic</div> <div></div> |
| <div>28. Additional information:</div> <div>LIT TD/PM</div> <div>- Helideck has lighted TDPM &amp; H</div> <div>Helideck equipped with HMS repeater Light System that communicates to the pilots by showing one of three possible colors.</div> <div>Steady Blue= Safe to land; (Pitch roll and inclination, (SHR and MSI/WSI all within limits.)</div> <div>Steady Amber = Land with caution (MSI/WSI only out of limits).</div> <div>Steady Red = DO NOT LAND (Pitch, Roll, inclination or SHR out of limits)</div> <div>All Limit perimeters should be reviewed on and agreed to by the aircraft operator, vessel crew and the client.</div> |                                                                                                                                        |
| <div>29. Link to NOTAM site</div> <div>No current NOTAM system is available for NE wind farm region.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div>30. Helideck Contact Telephone Number</div> <div>+1-985-276-0431</div>                                                            |

6.1.9

Orion



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |                                                                                                                                                                                                                                                                                                                        |                       |                                              |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------------|--|--|--|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |              | <h2 style="text-align: center;">HELIDECK INFORMATION PLATE</h2>                                                                                                                                                                                                                                                        |                       |                                              |  |  |  |
| HELIDECK<br>Elev. 136 ft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | VAR<br>Check | POSITION<br>Check                                                                                                                                                                                                                                                                                                      | <b>XORI<br/>Orion</b> |                                              |  |  |  |
| HEIGHT OF INSTALLATION: 324                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              | VHF<br>Check                                                                                                                                                                                                                                                                                                           | NDB<br>Check          | Issue Date<br>09 Aug 2022                    |  |  |  |
| HIGHEST OBSTACLE WITHIN 5NM: Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                                                                                                                                                                                                                                                                                                                        |                       |                                              |  |  |  |
| FUELLING INSTALLATION: No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              | Operating Company                                                                                                                                                                                                                                                                                                      |                       | Issued By                                    |  |  |  |
| STARTING EQUIPMENT: Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              | <b>DEME Offshore</b>                                                                                                                                                                                                                                                                                                   |                       | <b>Helideck<br/>Certification<br/>Agency</b> |  |  |  |
| (Norway 1.25D - 'DH'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |                                                                                                                                                                                                                                                                                                                        |                       |                                              |  |  |  |
| HELIDECK D value: 22.2m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |                                                                                                                                                                                                                                                                                                                        |                       |                                              |  |  |  |
| P/R/H Category: 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |                                                                                                                                                                                                                                                                                                                        |                       |                                              |  |  |  |
| Max Weight: 12.8t                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |                                                                                                                                                                                                                                                                                                                        |                       |                                              |  |  |  |
| Circle & H Lights: Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |                                                                                                                                                                                                                                                                                                                        |                       |                                              |  |  |  |
| <div style="display: flex; justify-content: space-around;">     </div> <div style="display: flex; justify-content: space-around; margin-top: 20px;">   </div> |              |                                                                                                                                                                                                                                                                                                                        |                       |                                              |  |  |  |
| <b>Wind (T°)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Kts</b>   | <b>Limitation /Comment</b>                                                                                                                                                                                                                                                                                             |                       |                                              |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              | <b>Self-propelled Heavy Lift Vessel</b> <ul style="list-style-type: none"> <li>• Table 1(T) if overflight of Fwd &amp; Stbd access platforms is unavoidable</li> <li>• HMS Rev.9B &amp; lights fitted</li> <li>• Approved high friction surface - no net</li> <li>• No DG's to be uplifted from this vessel</li> </ul> |                       |                                              |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              | <b>Non Compliance</b>                                                                                                                                                                                                                                                                                                  |                       |                                              |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5:1          | Forward and starboard foam monitor / access platforms infringe sector                                                                                                                                                                                                                                                  |                       |                                              |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Misc.        | No trained DGBA shipper on board.                                                                                                                                                                                                                                                                                      |                       |                                              |  |  |  |



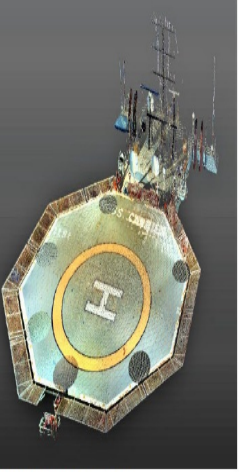
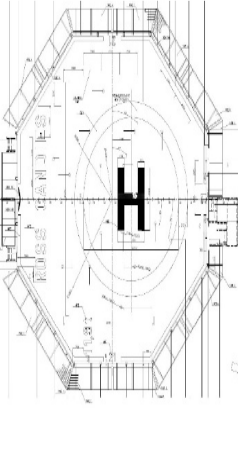
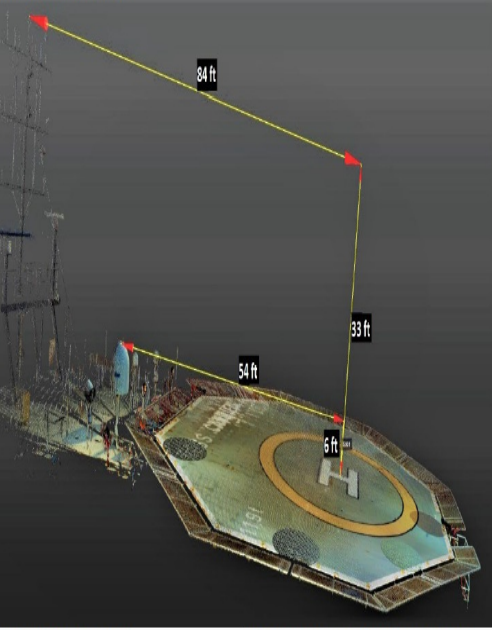
6.1.10 Paul Candies



6.1.11 Ross Candies

| HELIDECK INFORMATION PLATE                                                                                        |                                                                                   |                                                                                                                                                                       |                                              |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 1 Operating Company Logo<br>     | 2 Position<br>Contact Vessel<br>4 Helideck Identification Marking<br>Ross Candies | 3 Location Name<br><b>Ross Candies</b>                                                                                                                                |                                              |
| 5 Operating Company<br><b>Otto Candies LLC</b>                                                                    | 6 Helideck D-Value<br><b>69 ft / 21m</b>                                          | 7 Helideck Elevation<br><b>115 ft. 35m AMSL</b>                                                                                                                       | 8 Lease Block Number                         |
| 9 VHF Radio Frequency<br><b>123.075 MHz</b>                                                                       | 10 Helideck Dimensions<br><b>69x69ft. / 21x21m</b>                                | 11 Max Allowable Mass<br><b>26k / 11.9t</b>                                                                                                                           | 12 Region<br><b>Gulf of Mexico</b>           |
| 13 Limitations<br><b>Helideck rated for S-92</b>                                                                  |                                                                                   | 14 TDPM/Aiming Circle<br><input checked="" type="checkbox"/> Yes<br><input type="checkbox"/> No<br><input type="checkbox"/> Displaced 0.1D                            | 15 Weather/ATIS/AWOS<br>ID:<br>TEL:<br>FREQ: |
|                                                                                                                   |                                                                                   | 16 Marking Standard<br><input checked="" type="checkbox"/> CAP 437<br><input type="checkbox"/> HSAC RP<br><input type="checkbox"/> Other                              |                                              |
| 17 Remarks<br><b>Octagon, Steel Helideck, 18 tiedowns</b>                                                         |                                                                                   | 18 Services<br>Helicopter Refueling <b>No</b><br>Helicopter Starter Equipment <b>No</b><br>Helicopter Towing Equipment <b>No</b><br>Helideck Status Lights <b>Yes</b> |                                              |
| 19 Installation Picture(s)<br> |                                                                                   |                                                                                                                                                                       |                                              |
|                                                                                                                   |                                                                                   | 21 Parking Area Information<br>Type <b>NA</b><br>D-Value <b>NA</b><br>Dimensions <b>NA</b><br>Max. Allowable Mass <b>NA</b><br>Procedures <b>NA</b>                   |                                              |
| 22 NDB Frequency<br><b>NA</b>                                                                                     | 23 NDB Call Sign<br><b>NA</b>                                                     | 24 Issue Date<br><b>6/26/2023</b>                                                                                                                                     | 25 Revision Number<br><b>REV-001</b>         |

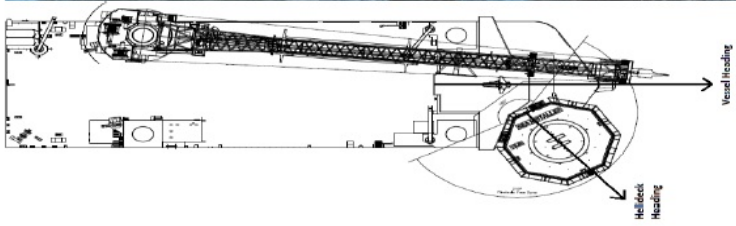
  

| HELIDECK INFORMATION PLATE                                                                                       |                                                                                                              |
|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 26 Helideck<br>                | 27 Platform Schematic<br> |
| 28 Additional Information<br> |                                                                                                              |
| 29 Link to NOTAM website                                                                                         | 30 Helideck Contact Telephone Number<br><b>Bridge Phone # 504-323-3556</b>                                   |

## 6.1.12

## Sea Installer



|                                                                                                                               |            |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                               |                  |                                                       |
|-------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------|-------------------------------------------------------|
|                                              |            | <h2 style="text-align: center;">HELIDECK INFORMATION PLATE</h2> <p style="text-align: center; color: red;"><b>Certificate Expiry Date: 22 May 2025</b></p>                                                                                                                                                                                                                                                                                   |                                               |                  |                                                       |
| HELIDECK Elev ft                                                                                                              | VAR Check  | POSITION Check                                                                                                                                                                                                                                                                                                                                                                                                                               | XSEI<br><b>Sea Installer</b>                  |                  |                                                       |
| HEIGHT OF INSTALLATION:<br>HIGHEST OBSTACLE WITHIN 5NM:                                                                       |            | Check<br>Check                                                                                                                                                                                                                                                                                                                                                                                                                               | VHF Check                                     | NDB<br>410 OYPY2 | Issue Date<br>29 Jan 2024                             |
| FUELLING INSTALLATION:<br>STARTING EQUIPMENT:                                                                                 |            | No<br>No                                                                                                                                                                                                                                                                                                                                                                                                                                     | Operating Company<br><br><b>Deme Offshore</b> |                  | Issued By<br><br><b>Helideck Certification Agency</b> |
| (Norway 1.25D - 'H' = D-22.2 - Dh=17.76/SLA=22.2<br>HELIDECK D value:<br>P/R/H Category:<br>Max Weight:<br>Circle & H Lights: |            | 22.2<br>1<br>12.8<br>Yes                                                                                                                                                                                                                                                                                                                                                                                                                     |                                               |                  |                                                       |
|                                            |            |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                               |                  |                                                       |
|                                            |            |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                               |                  |                                                       |
| <b>Wind (T°)</b>                                                                                                              | <b>Kts</b> | <b>Limitation /Comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |                                               |                  |                                                       |
|                                                                                                                               |            | Jack Up<br>• Helicopter operations allowed only when jacked-up<br><i>Jack Up</i><br>• <i>Helicopter operations allowed only when jacked-up</i><br><b>NOTE:</b><br>• <i>Landing prohibited unless crane boom/jib is in the raised position clear of 210 degree OFS</i><br>• <i>Possible turbulence when the WTG's &amp; Blades are loaded and wind speeds are in excess of 25Kts: 120-230 Rel to vessel HDG. Turbulence reports requested</i> |                                               |                  |                                                       |
|                                                                                                                               |            | <b>Non-Compliance</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                               |                  |                                                       |
|                                                                                                                               | 210°       | OFS reduced to 188 deg when WTG blades are loaded in cradles.                                                                                                                                                                                                                                                                                                                                                                                |                                               |                  |                                                       |
|                                                                                                                               | 5:1        | Vessel structure port and starboard aft infringe sector WTG Blades when loaded in the cradles                                                                                                                                                                                                                                                                                                                                                |                                               |                  |                                                       |
|                                                                                                                               | Misc       | Crane boom/jib may be at rest on port side of deck outside of notified helicopter flight times<br>Shut-down at pilots discretion - no starter unit                                                                                                                                                                                                                                                                                           |                                               |                  |                                                       |



|                                                                                   |                                                  |
|-----------------------------------------------------------------------------------|--------------------------------------------------|
|  | <h2>HELICOPTER LANDING AREA<br/>CERTIFICATE</h2> |
|-----------------------------------------------------------------------------------|--------------------------------------------------|

### Sea Installer

The above-named helideck has been inspected in accordance with CAP 437, BSL D 5-1, and HCA requirements for Offshore Helidecks.

The helideck has been found suitable for helicopter operations subject to:

1. Such non-compliances and restrictions as may be listed below; and,
2. Authorization by the helicopter operator.

| Wind (T°)                                                                               | Kts  | Limitation /Comment                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                         |      | Jack Up<br>• Helicopter operations allowed only when jacked-up<br><b>NOTE:</b><br>• <i>Landing prohibited unless crane boom/jib is in the raised position clear of 210 degree OFS</i><br>• <i>Possible turbulence when the WTG's &amp; Blades are loaded and wind speeds are in excess of 25Kts: 120-230 Rel to vessel HDG. Turbulence reports requested</i> |
|                                                                                         |      | <b>Non-Compliance</b>                                                                                                                                                                                                                                                                                                                                        |
|                                                                                         | 210° | OFS reduced to 188 deg when WTG blades are loaded in cradles.                                                                                                                                                                                                                                                                                                |
|                                                                                         | 5:1  | Vessel structure port and starboard aft infringe sector WTG Blades when loaded in the cradles                                                                                                                                                                                                                                                                |
|                                                                                         | Misc | Crane boom/jib may be at rest on port side of deck outside of notified helicopter flight times<br>Shut-down at pilots discretion - no starter unit                                                                                                                                                                                                           |
| Valid for helicopters with:<br>Maximum 'D' value:                                       |      | 'D' = 22.2 (Norway 1.25D - 'Dh' = 17.76 )                                                                                                                                                                                                                                                                                                                    |
| Maximum take-off weight:                                                                |      | 't' = 12.8                                                                                                                                                                                                                                                                                                                                                   |
| This certification shall remain in force until (unless previously revoked or suspended) |      | 22-May-25                                                                                                                                                                                                                                                                                                                                                    |

**Notes:**

1. This certificate is non-transferable.
2. The certificate holder is responsible for ensuring that the helideck, its environs and related equipment are at all times fit for purpose and that the helideck crew are suitably qualified, equipped and trained in the exercise of their duties.
3. This certificate shall cease to be valid if:
  - ♦ Changes of ownership or name of installation/vessel are made without notification to the HCA.
  - ♦ Changes to the helideck, its environs and/or related equipment are made without prior agreement of the HCA.
  - ♦ Levels of Helideck crew qualifications/competency are not maintained to the levels described in the OEUK Guidelines for Management of Offshore Helidecks or suitable alternative standards.
4. Any proposed changes are to be accompanied by drawings in plan and elevation with photographs where possible, particularly when such changes concern:
  - ♦ Modification to installation/vessel physical characteristics within the 150°, 210° and 180° falling gradient obstacle protected surfaces; and/or structural modifications to other areas of the installation/vessel that may affect or alter the airflow or turbulence experienced over the helideck
5. The Norwegian 1.25D requirement is only relevant to vessels constructed (keel laid) after 1 January 2008.



David Rae  
Helideck Certification Agency

Date: 29-Jan-24



### 6.1.13 Wind Scylla

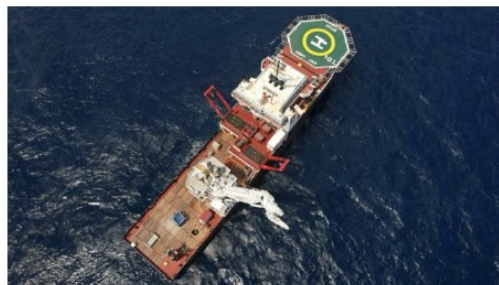


|                                                                                                                                  |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     |                    |                                                      |
|----------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------|------------------------------------------------------|
|                                                 |              | <b>HELIDECK INFORMATION PLATE</b><br><b>Certificate Expiry Date: 09 Mar 2025</b>                                                                                                                                                                                                                                                                                                                                                                                                     |                                     |                    |                                                      |
| HELIDECK<br>Elev 0 ft                                                                                                            | VAR<br>Check | POSITION<br>Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | XSSY<br><b>Wind Scylla</b>          |                    |                                                      |
| HEIGHT OF INSTALLATION:<br>HIGHEST OBSTACLE WITHIN 5NM:                                                                          |              | Variable<br>Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | VHF<br>Check                        | NDB<br>SYL - 579.5 | Issue Date<br>16 Apr 2024                            |
| FUELLING INSTALLATION:<br>STARTING EQUIPMENT:                                                                                    |              | No<br>Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Operating Company<br><br>Cadeler AS |                    | Issued By<br><br>Helideck<br>Certification<br>Agency |
| (Norway 1.25D - 'H' = D=22.2 - 17.76m / SLA = 22.2)<br>HELIDECK D value:<br>P/R/H Category:<br>Max Weight:<br>Circle & H Lights: |              | 22.2m<br>1<br>12.6t<br>Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                    |                                                      |
|                                               |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     |                    |                                                      |
| <b>Wind (T°)</b>                                                                                                                 | <b>Kts</b>   | <b>Limitation /Comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                    |                                                      |
| 150 - 240                                                                                                                        |              | <b>Jack-Up</b><br>• Table 1(T) if overflight of 5:1 items is unavoidable<br>• Possible turbulence when cargo deck loaded with stacked turbine columns. Turbulence reports requested.<br>• Large crane-boom rest abeam starboard side of helideck<br>• Check height of helideck and top of installation for each location<br>• No helicopter operations unless jacked up<br>• Main crane boom <b>MUST</b> be in the raised position clear of OFS (210°) for all helicopter operations |                                     |                    |                                                      |
|                                                                                                                                  |              | <b>Non-Compliance</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                     |                    |                                                      |
| 5:1                                                                                                                              |              | Port access platform<br>Access platform and main deck starboard                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     |                    |                                                      |
| Misc                                                                                                                             |              | NCS non - compliances for Dual Agent and C02 extinguishers                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                    |                                                      |



## 6.1.14 Wyatt Candies

| HELIDECK INFORMATION PLATE    |                   |                           |          |  |
|-------------------------------|-------------------|---------------------------|----------|-------------------------------------------------------------------------------------|
| Vessel                        | HELIDECK          | VAR                       | POSITION |                                                                                     |
| <b>Wyatt Candies</b>          | Elev. @50 ft. ASL | Check                     | Mobile   |                                                                                     |
| HEIGHT OF INSTALLATION:       | @50 FT            | VHF 122.7 MHZ             | NDB      | Issue Date                                                                          |
| HIGHTEST OBSTACLE WITHIN 5 NM | Verify            |                           | None     | 18 DEC 2023                                                                         |
| FUELING INSTALLATION:         | No                | OPERATING COMPANY         |          | Issue By                                                                            |
| STARTING EQUIPMENT:           | No                |                           |          |                                                                                     |
| HELIDECK D Value:             | 17 D              | <b>Otto Candies, LLC.</b> |          | <b>Aviation Maritime Services, a DEDAWA Inc. Company</b>                            |
| P/R/H/ Category:              | 3                 |                           |          |                                                                                     |
| Max Weight:                   | 7.0t              |                           |          |                                                                                     |
| Circle & H Lights:            | Not Fitted        |                           |          |                                                                                     |



| Wind (T°) | Kts | Limitation / Comment                                                                                                                                                                                                                                                                                                                                                   |
|-----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |     | Offshore Supply Vessel, Cable installation & multipurpose vessel <ul style="list-style-type: none"> <li>• Cat 3 Bow Deck</li> <li>• Approved Friction Surface</li> <li>• Aiming Circle and H not lighted, Night Operations – Emergency Only</li> <li>• Bow Mounted Status Light visible from Bridge, operated ONLY during flight operations, presently INOP</li> </ul> |
|           |     | <b>Non-Compliance</b>                                                                                                                                                                                                                                                                                                                                                  |
|           |     |                                                                                                                                                                                                                                                                                                                                                                        |
| Misc.     |     | Lighted Landing Circle and Lighted H not fitted, emergency night operations only.                                                                                                                                                                                                                                                                                      |



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## 7 Ops Memos

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### 7.1 Ops Memo 25-01

OPERATIONS MEMO NO. 25-01

18 March 2025

TO: ALL PILOTS

FROM: DO

Please ensure that baggage is NOT blocking the smoke detector.

For Offshore:

- This shall be checked before the baggage door is secured, by the Pilot outside monitoring the crew change.
- If there is too much baggage, then notify the HLO and have them reload or take some baggage out.
- If you must remove baggage you can strap it to a vacant seat if there is room. If not, it will have to be left behind.
- Notify Dispatch when you return.

For Onshore:

- Check the baggage compartment prior to departing to ensure the smoke detector is not blocked.
- If it is, notify Dispatch.

### 7.2 Ops Memo 25-02

OPERATIONS MEMO NO. 25-02

19 March 2025

TO: ALL PILOTS and FLIGHT CREW

FROM: DO

SUBJECT: **Standby status when there are no scheduled flights for MVY .**

With HEMS service being 24/7 it is critical that we always maintain our readiness posture. Prior to HEMS our MVY duty crews were not required to be standing by if there were no scheduled flights. The requirement to always have an aircraft readily available for HEMS has changed that.



Here is the new policy:

1. Duty aircraft at Quonset is operational and there is a spare aircraft at Quonset:
  - a. The crew may remain at home unless otherwise tasked by Operations or Maintenance.
2. Duty aircraft at Quonset is operational but there is no spare at Quonset:
  - a. The crew will maintain a standby status in Falmouth in case there are any issues with the duty aircraft at Quonset.
3. Duty aircraft at Quonset is AOG and there is no spare at Quonset:
  - a. The crew will maintain a standby status at MVY.
  - b. The aircraft will need to be run up with all daily checks completed (to include the hoist check), sufficient fuel to reposition to KOQU, and will remain on the ramp if weather permits.
4. Standby hours will be for the 12-hour period that the crew would normally be at MVY conducting HHO flights and adjusted for shorter periods of daylight during the winter

## 7.3 Ops Memo 25-03





## OPERATIONS MEMO

NO. 25-03

26 April 2025

TO: ALL PILOTS

FROM: DO

SUBJECT: **Premiere aircraft retractable landing gear**

If you have the opportunity to fly N942CF or the other Premiere aircraft that will be joining our hangar soon, keep in mind that it has retractable landing gear, which is a new tool that many of you are not accustomed to.

**IF YOU DO NOT HAVE PRIOR EXPERIENCE WITH RETRACTABLE GEAR THEN YOU ARE NOT AUTHORIZED TO GO "GEAR UP" WITHOUT DO/CP APPROVAL.**

If you do have prior experience and decide to cycle the landing gear while operating, don't forget the following:

- VLO is 130 KIAS
- We do not have "Landing Gear" in our checklist, anywhere.
- If the gear is up on approach, at approximately 200' RA, the aural warning will bark at you "LANDING GEAR", and a CAUTION will illuminate on the CAS
- Landing gear emergencies are in the QRH

Placard that is in the cockpit:

**$V_{LE} = V_{NE}$  POWER ON LESS 8 KIAS**  
 **$V_{LO} = 130$  KIAS**

ICN-69-A-151000-G-A0126-01273-A-01-1



## 7.4 Ops Memo 25-04





## OPERATIONS MEMO

NO. 25-04

29 April 2025

TO: ALL PILOTS

FROM: DO

SUBJECT: **Night HEMS shift**

Night HEMS shift will do the following at the beginning of their shift:

1. Conduct a thorough "Shift Change" brief with the entire crew to include Communications Specialist.
2. Start a new manifest and complete the HAA/SAR Risk Assessment and **SAVE** it.
  - a. Choose all the items in the risk that you anticipate for your 12-hour duty period.
3. Check out a set of NVG's and ensure that they are set for flight:
  - a. Inspect them for any deficiencies
  - b. Fit them to your helmet and optimize
  - c. Do not leave NVG's unsecured outside of the NVG locker. You may leave them mounted to your helmet, but the helmet must be stored in the NVG locker.
4. Aircraft run up:
  - a. Conduct an engine rinse
  - b. Complete all the daily checks on the aircraft up to "Taxi Checks"
  - c. Start a new logbook page in DAW and log the engine starts
5. At the conclusion of your shift:
  - a. Sign the NVG's back in
  - b. Delete the manifest created the night prior if not used
6. The PIC is responsible for ensuring that these procedures are adhered to.



## 7.5 Ops Memo 25-05





## OPERATIONS MEMO

NO. 25-05

07 June 2025

TO: ALL PILOTS

FROM: DO

SUBJECT: **Overflying aircraft inspections**

On June 6th we overflew mandatory inspections on one of our aircraft which is a non-compliance reportable event to the FAA. Inspections are there for a reason, to ensure components are functioning properly and that they are still serviceable and not out of tolerance or limits. Fortunately, the error was discovered before any further flights were conducted.

Just reporting alone took valuable time away from myself and other management personnel.

The root cause of this occurrence was a failure to follow established company policies and procedures, as well as a lapse in attention to detail. Moving forward, if a similar situation occurs, the Pilot in Command—and any other involved crewmembers—will be subject to formal counseling, which will be documented. Repeated or egregious instances may lead to adverse employment actions in accordance with our corrective action protocol.

Our commitment to safety and regulatory compliance must be upheld without exception.

Thank you all for your continued support and for upholding the highest standards of professionalism and safety. Your dedication is essential to the success of our operations.



## 7.6 Ops Memo 25-06



### OPERATIONS MEMO

NO. 25-06

26 June 2025

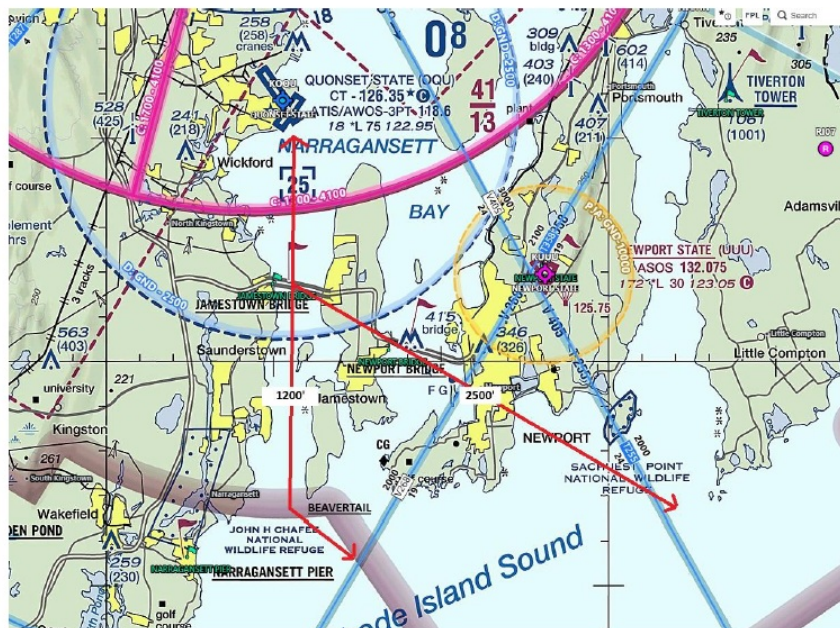
TO: ALL PILOTS

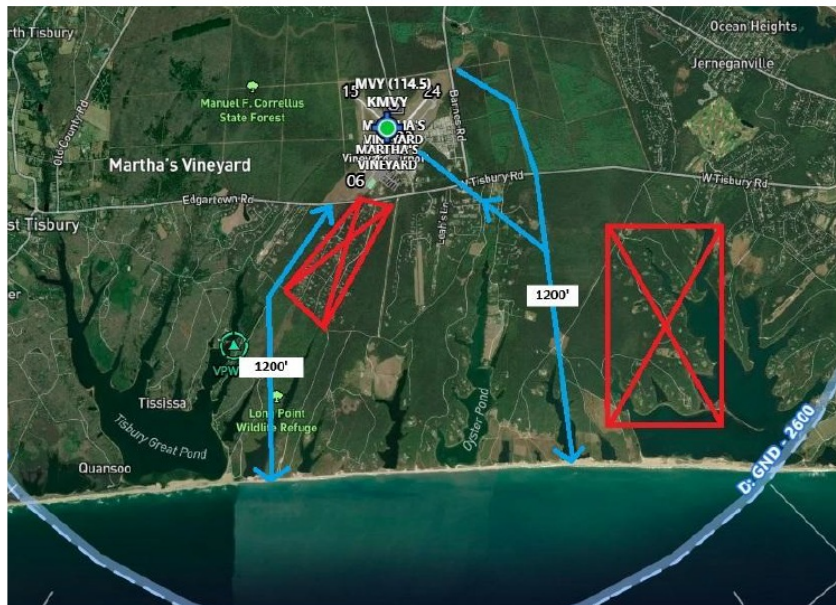
FROM: DO

SUBJECT: Course Rules for OQU and MVY

The following course rules apply for OQU and MVY flight operations. The maps below show the courses and altitudes that are to be flown to the greatest extent possible. For MVY, do not overfly the red X boxes unless it is unavoidable. As always, safety of flight, traffic avoidance, and ATC instructions take precedence. These course rules are established for neighborly flying/noise abatement and safety.

When weather conditions do not allow for the prescribed route altitudes, lower altitudes are permissible. Ensure that proper cloud clearance is maintained for the class of airspace you are in.





## 7.7 Ops Memo 25-07





## OPERATIONS MEMO

NO. 25-07

02 July 2025

TO: ALL MAINTENANCE PERSONNEL, PILOTS, DISPATCH

FROM: Director of Operations

SUBJECT: **Taking aircraft out of service for routine maintenance (New England bases only)**

Operational aircraft shall not be taken out of service for routine maintenance without the approval of the Director of Operations, Chief Pilot, or Base Mgr/Lead Pilot. There are operational elements that must be taken into consideration first to ensure we can respond to customer demands.

Operational aircraft are defined as the aircraft that is primary for that base, not the spare.



## 7.8 Ops Memo 25-08





## OPERATIONS MEMO

NO. 25-08

22 July 2025

TO: ALL PILOTS, HOIST OPERATORS, DISPATCH

FROM: DO

SUBJECT: KOQU Ramp Procedures

1. DO NOT USE SPOT 3 UNTIL FURTHER NOTICE. There is significant damage to the pavement, and it could cause damage to the aircraft.
2. In an effort to reduce the incidence of our aircraft blowing dirt/debris into hangar 157 (Merlin), while taxiing please do the following if it is safe to do so and does not impact operations negatively:
  - a. When taxiing in/out of our ramp, please use the taxi line furthest from the hangar for ingress and egress under the following conditions:
    - i. Spot 1 or 2 is clear and can be used to taxi through to reverse direction.
    - ii. The fuel truck is NOT present.
    - iii. For daylight operations only.
3. Overnight parking for the fuel truck will be moved to Spot 2 until further notice.
4. Do not ground taxi on the closed taxiway that the HHO training Conex is located on. Hover/air taxi is authorized.
5. No hoist training at the HHO training Conex while another HSUSA aircraft is operating on our ramp. Please coordinate with Dispatch on the schedule of operations for deconfliction. This is to reduce the risk of a hovering aircraft blowing FOD into an operating aircraft on the ground.



## 7.9 Ops Memo 25-09





## OPERATIONS MEMO

NO. 25-09

25 Sept 2025

TO: ALL PILOTS

FROM: Director of Operations

SUBJECT: **Passenger Briefings for Quonset Crew Change Flights**

Before taxiing, the following briefing shall be given to the passengers after they are all settled in and the cabin door is closed:

"Good (morning/afternoon/evening), welcome to HeliService USA flight to the "Vessel name", confirming that is everyone's destination, if not please speak up or tap one of us on the shoulder. Our flight time to the "Vessel name" is XX minutes. Weather en route today is \_\_\_\_\_. Please remain in your seats with your seat belts fastened until the HLO has directed you otherwise. Enjoy the flight and thank you for flying HeliService USA."

Before departing from an offshore helideck, the following briefing shall be given to the passengers after they are all settled in and the cabin door is closed:

"Good (morning/afternoon/evening), welcome to HeliService USA flight back to Quonset. Our flight time to Quonset is XX minutes. Weather en route today is \_\_\_\_\_. Please remain in your seats with your seat belts fastened until our ground staff has directed you otherwise. Enjoy the flight and thank you for flying HeliService USA."

This will be in the next revision of the OM-C.



## 7.10 Ops Memo 25-10



### OPERATIONS MEMO

NO. 25-10

27 Sept 2025

TO: ALL PILOTS

FROM: Director of Operations

SUBJECT: **AVCS and CFG's notice/recommendation**

Please see below Technical Information Letter from Leonardo regarding the AVCS and CFG's.

Please adhere to the procedures outlined here as this will reduce our incidence of AVCS issues and CFG's failing.

Apparently Phase 9 upgrade will include an automated solution to the issue.

The TIL will be available on the EFB in the AW-169 folder for reference.

If you have any questions, please reach out to Nick D. or Steven in Maintenance.

Thank you!



## 7.11 Ops Memo 25-11





## OPERATIONS MEMO

NO. 25-11

24 November 2025

TO: ALL PILOTS

FROM: Director of Operations

SUBJECT: **Pre-flight requirements**

Updates to the OM-A have been made regarding pre-flight responsibilities and are effective immediately pending FAA approval.

6.3.5 Preflight – Aircraft

### Preflight Inspection

1. HeliService USA PIC's will ensure a thorough preflight inspection of the helicopter has been completed in accordance with (IAW) the helicopter Rotorcraft Flight Manual (RFM).

### Subsequent Starts

All flights after the first flight of the day until the last flight of the day. A 360 degrees (or as close to 360 degrees as space allows) walk around check shall be performed prior to each start, consisting of, but not limited to the following:

1. Aircraft and rotor tie-downs removed.
2. General condition of the aircraft.
3. Check fluid levels.
4. Check cowling security.
5. Baggage door secured and locked.
6. Fuel cap secured.
7. Check for leaks.

**Note:** This includes anytime the rotors are stopped and on APU mode.

### 11.6.1 Pilots Pre-Flight/Post-Flight Duties

1. At the start of their shift, PICs are required to ensure their assigned aircraft is airworthy. A preflight must be completed by an AMT or the PIC to determine that. The shift AMT will



## 7.12 Ops Memo 26-01



### OPERATIONS MEMO

NO. 26-01

25 February 2026

TO: ALL PILOTS

FROM: Director of Operations

SUBJECT: AVCS and CFG's procedure

This memo was sent out last year as 25-10 and I'm resending it as this has not been adhered to for the most part.

**The procedure shall be performed as prescribed (after 4 consecutive landings).**

Please see below Technical Information Letter from Leonardo regarding the AVCS and CFG's.

Adhere to the procedures outlined here as this will reduce our incidence of AVCS issues and CFG's failing. You can also perform it more frequently.

Apparently Phase 9 upgrade will include an automated solution to the issue.

The TIL will be available on the EFB in the AW-169 folder for reference.

Thank you!

